



**Manual for
Training
Vegetation fire fighting
for fire departments**



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1 Introduction

Vegetation fires pose a serious threat to people, the environment, and infrastructure. With climate change and increasing droughts, fire departments and other emergency responders around the world are faced with the challenge of fighting increasingly frequent vegetation fires. This vegetation firefighting training manual for fire departments was developed to provide specific expertise and practical skills necessary to respond effectively to these growing challenges.

Although the phenomenon of vegetation fires is not new, their frequency, intensity, and impacts have changed significantly in recent decades. What was once considered a seasonal, localized problem can now cause widespread ecological and economic damage and increasingly endangers human life. These changes require a more comprehensive understanding of the underlying causes and dynamics of vegetation fires so that appropriate firefighting strategies can be developed.

Adapting to these new circumstances requires not only increased operational readiness and flexibility, but also in-depth training for firefighters. This is precisely where this manual comes in. It offers a comprehensive guide that includes both theoretical knowledge and practical instructions. The goal is to optimally prepare emergency personnel for the diverse challenges of fighting vegetation fires and to provide them with the necessary know-how and appropriate tools to fight the fires effectively.

Today, vegetation fires are no longer limited to remote rural areas. They increasingly threaten populated regions and urban fringe areas. This means that not only specialized emergency responders but also general firefighters must be familiar with the techniques and strategies for fighting such fires. Successful vegetation firefighting therefore requires an interdisciplinary approach that combines disciplines such as meteorology, botany, geography, and fire behavior.

This manual is aimed at both novice and experienced firefighters. It covers a wide range of topics, from basic definitions and the assessment of influencing factors to specific extinguishing techniques and tactical strategies in the field. Particular attention is paid to combining theoretical knowledge with practical exercises and examples from real-life operational practice. This is intended to ensure that the content taught can be not only understood theoretically but also applied in real-life situations.



A vital component of vegetation firefighting is the safety of emergency responders. Due to the often unpredictable and dangerous conditions under which these fires must be fought, training in safety guidelines and emergency strategies is essential. For this reason, this manual pays special attention to safety-related aspects. These include, among other things, the proper use of personal protective equipment, behavior in emergency situations, and measures to minimize health risks.

Another goal of the manual is to improve cooperation and communication during vegetation firefighting. These fires often require the involvement of multiple fire departments and organizations, making smooth communication and coordination crucial. In this context, modern technologies and tools are also presented that can increase efficiency and safety during operations.

Overall, the manual provides a comprehensive overview of the various aspects of vegetation firefighting and ensures that firefighters are equipped with the necessary knowledge and skills. It contributes to improving operational readiness and effectiveness in fighting vegetation fires and ultimately ensuring the protection of people, the environment, and property. Furthermore, this manual is intended to deepen the specialist knowledge and expand the practical skills of emergency personnel and contribute to the successful and safe conduct of operations.

1.1 Introduction to vegetation fire fighting

Fighting vegetation fires is an essential component of modern fire protection and presents numerous challenges that require specialized training and equipment. Vegetation fires, which include forest and wildfires, can be caused by natural triggers such as lightning strikes, but also by human activities. Especially in dry, vegetated areas, they pose a significant threat to people, property, and the environment.

The first step to effectively combating vegetation fires is understanding their nature and dynamics. These fires often follow specific patterns influenced by topography, vegetation type, and prevailing weather conditions. Such fires often spread elliptically, with the greatest spread occurring downwind and uphill (Handbook of Forest Fires, 2011, p. 2).

An important aspect in fighting vegetation fires is the identification and use of control lines that separate the fire from unaffected areas



Such control lines can be either dry or wet

Techniques are emerging, with dry methods often being preferred in water-scarce regions (Vegetation Fire Fighting: Operational Tactics, 2022, p. 24). The strategic use of these lines helps control and limit the spread of fire.

A crucial factor in vegetation firefighting is the rapid detection and immediate response to emerging fires. Modern technologies such as remote sensing and the use of drones facilitate the identification of new fire sources and enable rapid intervention (2022 Wildfire Season Report, p. 49). Such techniques are particularly advantageous in vast, hard-to-reach areas, as ground firefighting alone is often insufficient.

In addition to the technical and tactical aspects, the personal preparation of emergency personnel is of great importance. Personal protective equipment (PPE) plays a particularly important role in protecting emergency personnel from the extreme conditions encountered during vegetation fires. The equipment should be designed to provide optimal protection without overly compromising wearer comfort (Forest Fire Handbook, 2011, p. 6).

Another key factor in vegetation firefighting is close cooperation between different organizations and authorities. Local fire departments, forestry authorities, and disaster control units must coordinate their efforts to respond effectively. As became clear during the 2022 forest fire season in Brandenburg, cooperation between the various stakeholders is crucial for operational success (Report on the 2022 Forest Fire Season, p.

28). In addition, international cooperation is becoming increasingly important in order to learn from the experiences and techniques of other countries.

In summary, vegetation firefighting is an extremely complex and interdisciplinary task. It requires a deep understanding of fire dynamics, specialized training and equipment, and close cooperation between different stakeholders. Continuous development of the tactics and technologies used is essential to effectively address changing conditions and challenges.

1.2 Significance and relevance of vegetation fires

The relevance of vegetation fires has increased significantly worldwide in recent decades, primarily due to changing climatic conditions as a result of climate change. In Switzerland, especially in regions such as Valais and the Southern Alps, the risk of forest fires is expected to



increases significantly during the summer months, which could have serious consequences (Pezzatti et al., pp. 1-3). Climate change is lengthening dry periods while temperatures are rising, significantly increasing the risk of fires (Pezzatti et al., p. 10).

Vegetation fires have a variety of impacts on the environment and society.

From an ecological perspective, they lead to profound changes in the affected ecosystems. Not only is flora and fauna destroyed, but long-term shifts in species composition and distribution can also occur. This is particularly evident in alpine regions, where human-caused fires have severely decimated sensitive tree species (Pezzatti et al., p. 5). Furthermore, the risk of erosion and landslides increases after forest fires, especially during subsequent heavy rainfall events (Pezzatti et al., p. 2).

The consequences of vegetation fires are also serious from a socioeconomic perspective. The loss of forest areas not only affects timber production, but can also cause significant economic damage to forestry and associated industries (Pezzatti et al., p. 2). In Alaska, for example, changing climatic conditions are leading to increased fire activity, which is driving up firefighting costs (Melvin et al., pp. 1-2).

Projections indicate that the cumulative cost of wildfire suppression in Alaska by 2100 could be between \$1.1 and \$2.1 billion, depending on the climate scenario (Melvin et al., p. 1).

The human dimension of forest fires is no less significant. A significant proportion of fires are caused by human activity, whether accidental or intentional through arson (Report of the Expert Commission, p. 29).

Cities and communities near forests are constantly at risk of fires. This requires effective prevention and control measures by authorities and the public (Report of the Expert Commission, p. 11).

Another important issue is the health risks to the population and emergency responders. The smoke and pollutant particles released during vegetation fires worsen air quality and can trigger respiratory diseases. The extreme conditions and the work in the fire area pose a major health risk for emergency responders (Report of the Expert Commission, p.

41). Therefore, to minimize the risks to the people involved, strict safety protocols and appropriate protective equipment are required.

Infrastructure in vulnerable areas is also severely affected by vegetation fires. Especially in regions with high fire risk, such as the Alps, many infrastructures such as roads and settlements are located in the immediate vicinity of



Forests, which increases the risk of damage. Special protective measures are required here (Pezzatti et al., pp. 16-17).

In addition to the obvious threats to humans and nature, vegetation fires also have political and legal consequences. Legislation must be continually adapted to new challenges. For example, munitions-contaminated forest areas require special legal regulations and precautionary measures.

measures to ensure the safety of people and the environment (Report of the Expert Commission, pp. 61-63).

In summary, vegetation fires are of considerable importance on many levels. The challenges they pose require coordinated action by government institutions, science, and society. A comprehensive understanding of the risks and impacts of vegetation fires is crucial for developing effective prevention and control strategies and for mitigating the impacts on people, the environment, and the economy.

minimize.

1.3 Historical development and current challenges

The fight against vegetation fires has developed significantly over time and is now one of the central tasks of disaster protection.

Historically, vegetation fires have played a significant role in shaping landscapes and ecosystems. Even early societies faced the devastating effects of fires and developed methods to protect themselves and fight them. With the emergence of professional fire departments in the 19th and 20th centuries, vegetation firefighting also became increasingly professionalized.

In its historical development, various phases of vegetation firefighting can be identified. Initially, the focus was on extinguishing fires primarily through the use of water and simple mechanical means.

It wasn't until the mid-20th century that the integration of modern technologies and tactics began. A turning point was the introduction of air support in the 1940s and 1950s, when firefighting aircraft and helicopters were used for firefighting. These innovations significantly improved efficiency and enabled a faster response to fires.

A striking example of this historical development is the forest fire in Leuk in 2003, which is considered the most serious fire event in the Valais region and one of the largest in Switzerland (Living with Forest Fire, 2010, p. 2). Within ten days, around 300 hectares of forest, including protective forest and



Pastures were destroyed. Despite the already advanced control methods available at the time, this event highlighted the limitations of the technology available at the time and demonstrated the challenges posed by extreme weather conditions and biomass accumulation (Living with Forest Fire, 2010, p. 3).

Today's challenges in vegetation fire fighting result primarily from extreme weather conditions, which are exacerbated by climate change.

Rising temperatures and longer dry periods significantly increase the risk of fires. The report of the expert commission on the forest fires in Saxony in summer 2022 highlights that the number and intensity of fires have increased many times over in recent years due to climate change (Report of the Expert Commission on Forest Fires in Saxony Summer 2022, 2023, p. 29).

Another complicating factor is the increase in development in urban fringe areas, the so-called wildland-urban interfaces (WUI). These areas pose an increased risk because they have both high population density and are vulnerable to wildfire. The challenge is to develop appropriate prevention and emergency response measures to protect both natural vegetation and the affected population (Facing the Wildfire Spread Risk Challenge, 2023, p. 18).

Added to this is political and social resistance to the implementation of fire management strategies. The public is often skeptical of controlled burns, as they are perceived as dangerous in the short term. In the long term, however, targeted burning can help reduce the amount of combustible material and thus minimize the risk of severe fires (A System Dynamics Model Examining Alternative Wildfire Response Policies, 2019, p. 16).

The integration of modern technologies and innovative tactics is crucial to meeting today's challenges. Early warning systems, such as satellite-based monitoring and the use of drones, play a key role in this. Geographic information systems (GIS) and artificial intelligence (AI) enable better analysis and forecasting of fire risks, facilitating more precise and rapid responses to fires (Facing the Wildfire Spread Risk Challenge, 2023, p. 21).

In summary, vegetation firefighting is becoming increasingly complex and presents many new challenges. These can only be met through technological advances and interdisciplinary approaches. Historical experience shows that continuous adaptation and further development are necessary to effectively respond to the dynamic changes in the field of vegetation fire protection.



1.4 Objectives and structure of the training manual

The Vegetation Fire Fighting Training Manual for Fire Departments pursues several key objectives aimed at improving the effectiveness and safety of emergency personnel during vegetation fires. A primary objective of the manual is to impart in-depth knowledge and specialized firefighting techniques. Firefighters should not only learn the theoretical fundamentals but also acquire practical skills that they can apply in real-life situations. The training covers all levels, from basic training to specialized advanced training tailored to specific scenarios.

Another key objective of the manual is the standardization of training content and operational tactics in vegetation firefighting. This standardization is necessary to ensure a uniform approach, regardless of regional specificities. This enables firefighters from different regions and even countries to work together smoothly. Harmonization is also being sought at the international level, particularly in European civil protection (Recommendation on Training Content in Vegetation Firefighting, 2024, p. 5).

The manual is structured to provide firefighters with a logical and systematic approach. It is divided into several chapters, each covering a different topic. It begins with basic information and definitions and extends to specific tactics and strategies for vegetation firefighting. It includes analyses of influencing factors, risks, safety measures, and detailed explanations of equipment and technical resources.

The first chapter defines the basic objectives of the manual, with a clear focus on the safety of emergency responders (Forest Fire Handbook - Forest Fire Fighting, 2011, p. 1). The following chapters delve deeper into various aspects of vegetation firefighting. For example, Chapter 4 presents a hazard matrix that details potential hazards to emergency responders and the public. Chapter 6 describes the different equipment required for various operational scenarios. Finally, Chapter 8 explains tactics and strategies, including direct and indirect firefighting methods and special operational techniques.

A particularly innovative aspect of the manual is the integration of modern technologies and new techniques that are constantly being developed. This includes the use of large-scale equipment and manned aircraft (Forest Fire Manual -



Forest Fire Fighting, 2011, p. 8) as well as the use of mobile and stationary extinguishing systems. Given advancing climate change and the associated changes in the dynamics of vegetation fires, the manual must be regularly updated to keep pace with the latest scientific findings. This ensures that emergency responders always remain up to date with the latest technology and tactics (Discussion Paper Critical Vegetation Fire, 2022, p. 7).

In summary, the manual pursues the overarching goal of comprehensively and practically preparing firefighters for the complex challenges of vegetation firefighting. Through structured and standardized training, combined with the use of current technical developments and scientific findings, the aim is to increase the efficiency of firefighting and maximize the safety of emergency personnel. The implemented training content is designed to enable firefighters to fight vegetation fires quickly and effectively, minimizing damage to people, animals, and the environment.



2 Terms and basics

To effectively fight vegetation fires, a deep understanding of the relevant concepts and terminology that characterize this specialized discipline is essential. The first step in training firefighters in vegetation firefighting is therefore the precise introduction and explanation of these basic terms. Only those who confidently master these concepts can face the challenges posed by such fires.

Vegetation fires differ significantly from other types of fire, such as building fires, in many respects. Their dynamics, the fuels involved, and the specific environmental conditions require specialized knowledge about how such fires start, develop, and are ultimately fought. The "Terms and Principles" chapter not only provides a theoretical foundation but also illustrates the practical relevance of these concepts for everyday firefighting.

A solid understanding of the basic terminology in the context of vegetation firefighting lays the foundation for subsequent tactical and strategic considerations. It is crucial to precisely define terms such as flame front, fire triangle, and operational zone. A common language significantly facilitates collaboration between different emergency services and organizations, thus contributing to the efficiency and safety of operations.

Another important aspect of basic fire training is the differentiation of different fire types. Vegetation fires encompass numerous scenarios, characterized primarily by the destruction of plant material and can affect natural and built-up environments. The diversity of fire types –

Such as ground fires, crown fires, or peat fires – is highlighted in this introduction. This differentiation is of great importance, as each fire type poses specific risks and challenges that require targeted firefighting strategies.

Inextricably linked to these fundamentals is the communication of the physical and chemical processes that influence vegetation fires. Firefighters must understand precisely the conditions under which such fires start and spread. This includes factors such as wind, temperature, humidity, as well as the properties of the fuel and topographical conditions. Only by considering these factors holistically can an effective firefighting strategy be developed.

Another key element of the training is understanding the scientific principles that explain fire behavior. Topics such as fire behavior models and the various phases of a vegetation fire are covered in detail. These models are essential for emergency responders to predict fire progression.



to be able to better forecast and initiate appropriate measures at an early stage.

Equally important is knowledge of fire ecology and the interactions between fire, vegetation and the environment, which contributes to a deeper understanding of the phenomenon of vegetation fires.

The legal and organizational environment in which vegetation firefighting takes place should not be overlooked. This includes knowledge of relevant laws, regulations, and guidelines, as well as the organizational structures of the fire department and other stakeholders. A good understanding of this area enables emergency responders to carry out their measures in accordance with legal requirements and to consider the responsibilities of the parties involved.

In summary, this chapter aims to provide firefighters with a comprehensive introduction to the terminology and basic concepts of vegetation firefighting. This foundation is essential for preparing for the more advanced topics and practical applications covered later in the manual. Through a common understanding of the terms and concepts, emergency responders can increase their efficiency and significantly enhance the safety of their operations.

2.1 Basic concepts

Understanding basic terminology is essential for effective training in vegetation firefighting. A common vocabulary facilitates smooth communication and collaboration between emergency responders. It is crucial that all participants use the same terminology and concepts to avoid misunderstandings and ensure effective coordination of operations.

A central term in this context is the so-called “fire triangle”.

It describes the three essential elements required for the start and spread of a fire: fuel, oxygen, and heat. These three factors must be present simultaneously for a fire to start (State School and Technical Institution for Fire and Disaster Protection, 2024, p. 4).

The combustion triangle, also known as the fire behavior triangle, illustrates the close interplay between these two factors. Heat sources often come from external ignition sources such as lightning, human activity, or technical defects (State School and Technical Facility for Fire and Disaster Protection, 2024, p. 4).



räumlich- temporäre Vorwärmung: Sonneneinstrahlung (absorbierende Oberfläche)
initialer Energieeintrag: Brandstiftung, technischer Defekt, selten Selbstentzündung (Verrottung, Kampfmittel ...) & Blitzeinschlag
fortlaufende (endliche) Wärmefreisetzung: Leitung, Konvektion & Strahlung als Grundlage eines Ausbreitungsprozesses

Pflanzen: vital & Totholz
Berücksichtigung des augenblicklichen Zustandes
 Bei Übergang auf **Siedlungsflächen: weitere brennbare Materialien** (Feuerbrücken)



Gegenwärtige Windvariation:
 durch meteorologische Luftdruckänderung +/- brandbedingte Strömung
Geometrische/ Größe des Brennmaterials
 (Reisig vs. Stamm)

Räumlich- zeitlich Zusammentreffen der Bedingungen
 Entstehung einer selbsterhaltenden, chemischen Kettenreaktion

Figure 1. Lindon Pronto and Alexander Held. 2021. WKR training module on fire behavior.

Another key concept in vegetation firefighting is the differentiation between the different heat transfer processes: conduction, convection, and radiation. Conduction is the direct transfer of heat within a material or between two touching objects. In contrast, convection describes the transfer of heat through flowing liquids or gases. Radiation, on the other hand, occurs through electromagnetic waves and plays an important role in preheating the surrounding vegetation (State School and Technical Institution for Fire and Disaster Protection, 2024, p. 5).

Also of central importance for vegetation firefighting are the terms that describe the behavior and spread of a fire. The term "fire fringe" encompasses the entire burning area of a fire, while the "fire front" is the zone in which the fire spreads most rapidly and intensely (State School and Technical Institution for Fire and Disaster Protection, 2024, pp. 28-29). The "flanks" of a fire refer to the lateral areas that burn less intensely and spread more slowly than the fire front (State School and Technical Institution for Fire and Disaster Protection, 2024, p. 30).

Equally important are terms such as "fingers" and "pockets." Fingers are narrow bands of fire extending from the main fire line into the surrounding area, while pockets are depressions in the fire edge that are formed due to the



varying fuel density in the vegetation (State School and Technical Facility for Fire and Disaster Protection, 2024, pp. 28-29). These terms are crucial for accurately understanding the behavior and dynamics of a vegetation fire and responding appropriately.

Another important concept is the "control line." It can be either natural or artificial and serves to stop or at least slow the spread of fire (State School and Technical Facility for Fire and Disaster Protection, 2024, p. 31). A related term is the "anchor point," which marks a safe starting point for firefighting, from which emergency personnel can control the spread of the fire (State School and Technical Facility for Fire and Disaster Protection, 2024, p. 28).

Particularly dangerous are so-called "spot fires." These are caused by flying sparks or embers and can form beyond the actual fire wall (State School and Technical Facility for Fire and Disaster Protection, 2024, p.

31). They pose a significant threat because they can cause new fires in different directions, making firefighting even more difficult.

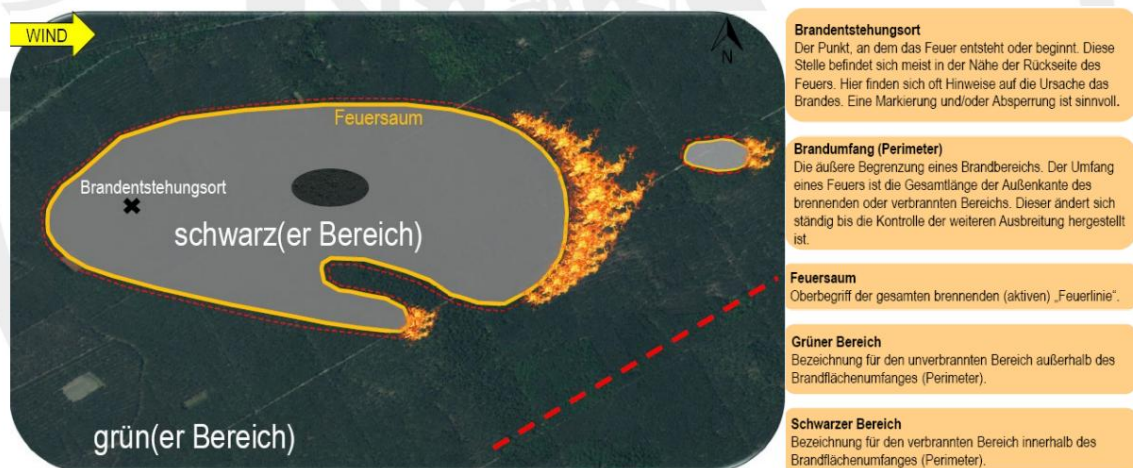


Figure 2 LG No. 612 Veg.-Bbk-Leihren1+Ausbildung intensive I AE Basic concepts/influencing factors - Fundamentals I Barth (LSTE)

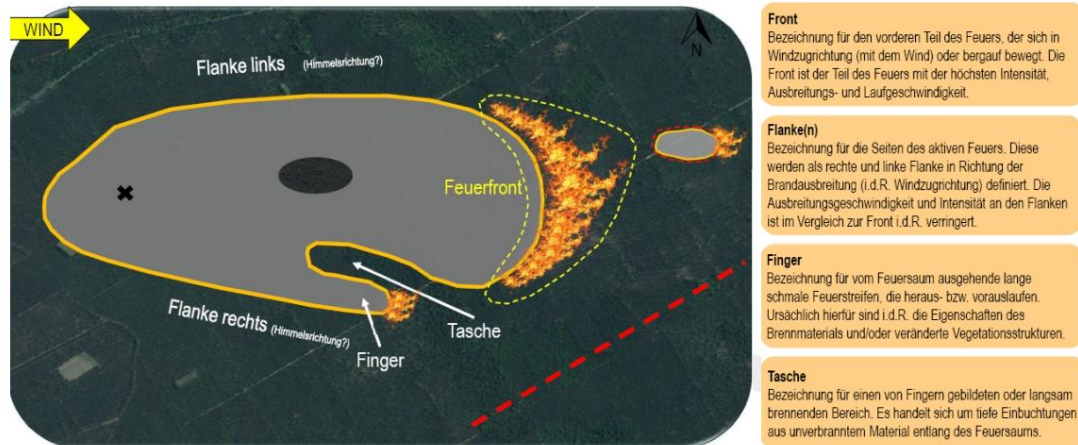


Figure 3 LG No. 612 Veg.-Bbk-Leihren1+Ausbildung intensive I AE Basic concepts/influencing factors - Fundamentals I Barth (LSTE)

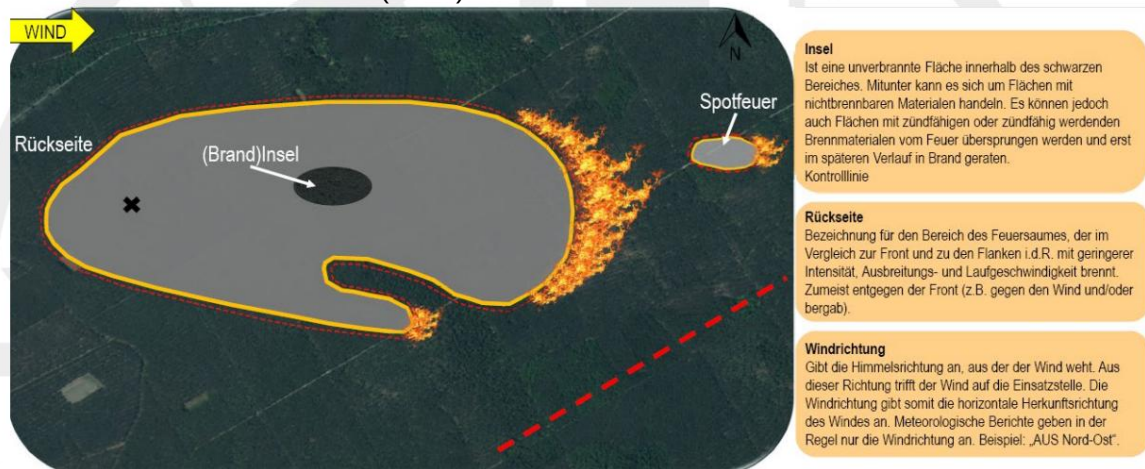


Figure 4 LG No. 612 Veg.-Bbk-Leihren1+Ausbildung intensive I AE Basic concepts/influencing factors - Fundamentals I Barth (LSTE)

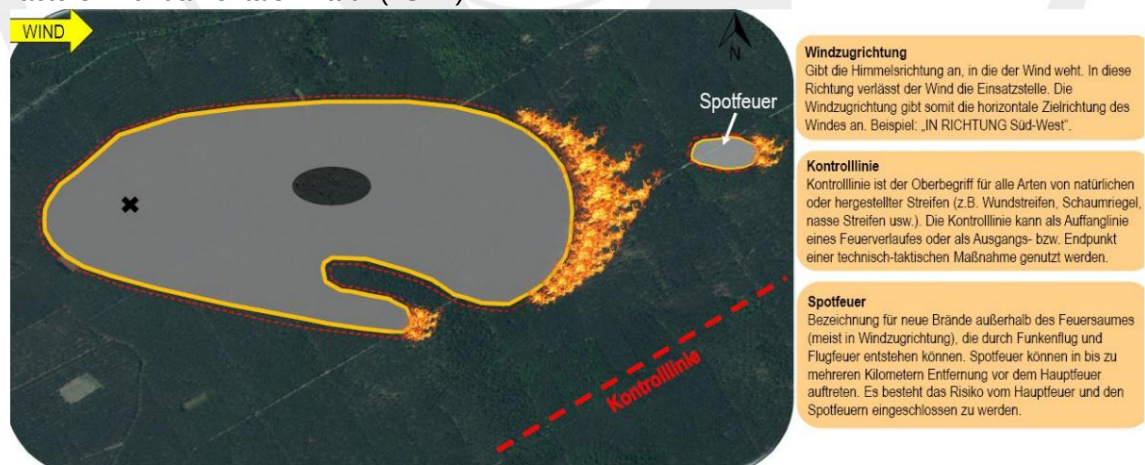


Figure 5 LG No. 612 Veg.-Bbk-Leihren1+Ausbildung intensive I AE Basic concepts/influencing factors - Fundamentals I Barth (LSTE)



In summary, understanding and correctly applying these basic concepts is crucial for successful vegetation firefighting. These terminologies not only support the analysis of the current fire situation but also play a central role in the planning and implementation of appropriate measures. Only those who fully understand these concepts can act efficiently and safely during operations.

2.2 Types of fire

The different fire types play a central role in vegetation firefighting, as they form the basis for selecting appropriate tactics and strategies. Different types of fires occur in vegetation, each presenting specific challenges and requirements for emergency responders. Vegetation fires can essentially be divided into four main categories: subsurface fires, ground fires, crown fires, and aerial fires (Barth, 2024, p. 21).

Subterranean fires, also known as earth fires, are particularly insidious because they can spread unnoticed in deeper soil layers, often beneath humus, peat, or coal layers. These fires can smolder for weeks or even months and pose a persistent threat. Fighting such fires is difficult because they can penetrate deep into the soil and retain heat, requiring lengthy and sustained firefighting efforts (Barth, 2024, p. 22).



Figure 6 LG No. 612 Veg.-Bbk-Leihren1+Ausbildung intensive I AE Basic concepts/influencing factors - Fundamentals I Barth (LSTE)

Ground fires primarily affect the uppermost layers of vegetation, such as herbaceous plants, grasses, and low shrubs. These fires typically spread rapidly with open flames, which is particularly noticeable in spring when the vegetation is still dry and highly flammable after winter (Barth, 2024, p.



23). Due to wind influences, ground fires can quickly spread to neighboring areas and thus pose a constant threat to settlements and adjacent forests. They often act as a precursor to more intense and difficult-to-control fires.
fires such as crown fires.

Crown fires, also called treetop fires, are particularly dangerous and intense. They capture the crowns of trees and spread at high speed.

Crown fires typically develop from ground fires, requiring dense tree canopies and strong winds to spread effectively (Barth, 2024, p. 25). Although such fires have so far been rare in Germany, their destructive impact remains a real threat for which preparation is essential. The transition from a ground fire to a crown fire is particularly critical and requires the utmost attention from emergency responders.

Flying sparks, also called spot fires, are caused by flying sparks. Burning particles are carried by the wind over long distances and ignite
new fires. This type of fire is difficult to predict and even more difficult

It is important to control the fires, as flying sparks can cause fires even on developed properties. Flying sparks often contribute to the creation of a widespread fire area characterized by several separate fire sources (Barth, 2024, p. 26).

The spread and intensity of different fire types are significantly influenced by factors such as wind, vegetation type and density, and weather conditions. Strong winds or extremely dry vegetation can cause fires to spread uncontrollably and more rapidly. Therefore, the tactics used must always be flexible and adapted to the current conditions. The role of early warning systems, which provide emergency responders with early information about meteorological conditions and potential hazards, is particularly important here (Barth, 2024, p. 16).

Fighting different types of fires requires specific tactics and techniques. Subterranean fires often require extensive irrigation and the removal of combustible material from deeper soil layers. Ground fires, on the other hand, are usually contained by the use of direct extinguishing agents and the creation of firebreaks to prevent further spread. Heavy equipment and firefighting aircraft are often used for crown fires, while airborne fires require rapid response times and comprehensive monitoring to detect and control new fires in a timely manner.

In summary, the fire types described differ significantly in their complexity and the demands placed on firefighting strategies. Understanding the specific fire type is crucial for effective planning and the targeted deployment of forces and resources (Barth, 2024, p. 24).



Nevertheless, any type of vegetation fire remains a challenge that requires both a detailed analysis of the prevailing conditions and flexible adaptation strategies.





3 Influencing factors

Successful measures and strategies for fighting vegetation fires depend heavily on a thorough understanding of the various factors that determine fire behavior in vegetation-rich areas. In this context, it is essential to carefully analyze factors such as fire behavior, weather conditions, the type and distribution of combustible material, and topography, as well as to tactically assess these aspects. Each of these factors contributes in its own way to the dynamics and spread of vegetation fires and thus has a significant impact on the selection of appropriate firefighting methods and on the safety of emergency responders.



Figure 7 Fire behavior triangle ©EFI/WKR

The factors that influence fire behavior encompass a range of physical and chemical processes that determine the ignition, rate of spread, and intensity of a fire. These include the ignition temperature of the fuel, thermal radiation, heat conduction, and the convection currents that drive the fire. A thorough understanding of these processes is essential for accurately predicting fire behavior and effectively planning and implementing firefighting measures.

Another crucial factor in the spread and behavior of vegetation fires is weather conditions. Temperature, humidity, wind speed, and direction play a central role in fire dynamics.

Particularly strong winds can significantly accelerate the fire front and even cause sudden changes in direction, which poses considerable challenges for emergency services. Therefore, continuous monitoring and analysis



meteorological data is crucial for the planning and implementation of operations.

The type of vegetation used as fuel is also a critical factor.

Different vegetation types—be it grassland, shrubland, or forest—have different burning characteristics. These include combustibility, moisture content, density, and structure of the material. These properties must be considered when selecting control strategies to maximize the effectiveness of the measures.

Topography, or the nature of the terrain in which the fire spreads, also significantly influences fire behavior. Slopes and gradients can accelerate or slow the fire, as fire generally spreads faster uphill than downhill. Additionally, ravines and other geographical features such as valleys or hills can lead to unpredictable air currents, making the fire unpredictable. For this reason, considering the terrain is essential for successful firefighting.

Ultimately, the tactical assessment of these influencing factors is an integrative task in which all of the above elements are brought together and placed in the operational context. A precise assessment of the overall situation, including fire behavior factors, weather conditions, vegetation types, and topography, enables sound tactical decisions. This requires not only theoretical knowledge, but also practical experience and intuition –

Both are sharpened through comprehensive education and continuous training.

Overall, it is crucial to understand the various influencing factors in their entirety and to continuously analyze them, as they form the basis for all vegetation firefighting measures. The interaction of these factors determines the effectiveness of the methods used and the safety of the emergency responders. Only through a holistic approach and regular reassessment of the relevant influencing factors can effective and safe firefighting be ensured. This chapter is therefore dedicated to the in-depth investigation and explanation of the key influencing factors that must be considered in vegetation fires.

3.1 Fire behavior factors

Fire behavior factors play a crucial role in vegetation firefighting, as they significantly influence how a fire develops and spreads. These factors not only determine the speed with which



The factors that affect not only the progression of a fire, but also its intensity and the options for successful suppression. These can be roughly divided into three main categories: fuel, weather conditions, and the topography of the terrain.

The fuel is of central importance for the behavior of the fire.

Properties such as shape, size, density and moisture content influence how easily it is flammable and how strongly it burns (Schloßmacher et al., 2021, p. 3).

Light material such as grass or dry leaves ignites very quickly and

Burns vigorously, but only for a short time. Heavy materials such as branches or tree trunks take longer to ignite, but burn more intensely and produce more heat. Whether the fuel is living or dead also plays a role: Living material contains more moisture and therefore burns more slowly and less intensely (Schloßmacher et al., 2021, p. 13).

Weather conditions are another crucial factor. Factors such as temperature, humidity, wind speed, and precipitation determine how easily vegetation ignites and how the fire behaves (Ru, 2024, p. 1). High temperatures and low humidity increase the risk of ignition and accelerate the spread of fire. Wind is particularly critical because it not only carries flames and sparks but also increases the oxygen content in the air, which further fuels the fire (Ru, 2024, p. 25). In contrast, precipitation acts as a natural fire retardant by saturating vegetation, making it more difficult to ignite (Ru, 2024, p. 42).

The topography of the area also has a significant influence on fire behavior. Steep slopes can significantly accelerate fire spread, as the fuel above the flames by convective and radiant heat

The fire is heated more quickly by the fire propagation (Schloßmacher et al., 2021, p. 10). Conversely, a fire burning downhill spreads more slowly because the fuel is not heated as quickly. Furthermore, narrow valleys and deep ravines can alter wind patterns, which can lead to unexpected fire behavior (Schloßmacher et al., 2021, p. 21).

In addition to these physical and meteorological factors, emergency responders must also consider tactical considerations based on the analysis of influencing factors. This includes assessing fire behavior patterns and choosing the appropriate operational strategy for fire containment and suppression (Ru, 2024, p. 66). In strong winds and steep terrain, direct firefighting may be too risky, and preventive measures such as creating firebreaks or controlled burns should be considered.

Another important aspect is the constant observation and prediction of fire behavior. Emergency services must be able to detect changes in the



Recognizing weather conditions early and adapting tactics accordingly (Ru, 2024, p. 28). Practical training and experience are essential to safely and efficiently apply theoretical knowledge in practice.

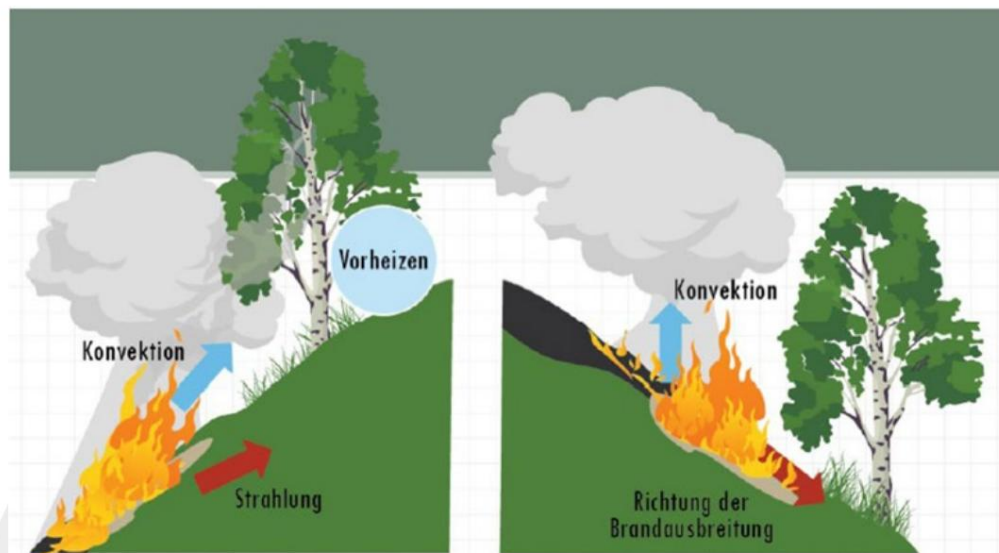


Figure 8 Effect of terrain on fire spread ©EFI/WKR

Overall, the factors described illustrate the complexity of vegetation firefighting and underscore the need for sound training for emergency responders. Careful consideration of fuel, weather conditions, and topographical features is essential to effectively manage the risks and challenges and ensure the safety of both emergency responders and the affected population. This requires continuous review and adaptation of the techniques and strategies employed to successfully counter the dynamic and often unpredictable natural forces involved in vegetation fires.

3.2 Weather

Weather plays a central role in vegetation firefighting, as it influences fire behavior both directly and indirectly. Various me-

Theoretical elements such as wind, temperature, humidity, and precipitation significantly influence the dynamics and intensity of vegetation fires. A thorough understanding of these weather factors is therefore essential for planning and implementing effective firefighting measures.

Wind is one of the most important factors, as it strongly influences the speed and direction of fire spread. A strong, constant wind can not only drive the fire front rapidly, but also



Sparks and embers can spread over long distances, igniting new fires, so-called spot fires. Reliable wind forecasts are therefore crucial for strategic operational planning. The German Weather Service (DWD) uses various models with different grid resolutions to enable precise wind forecasts (DWD Meteorology for Vegetation Fire Fighting, 2024, p. 6).

Temperature also plays a significant role. High temperatures increase the flammability of vegetation by reducing the moisture content of the plants and causing the soil to dry out more quickly. Combined with low humidity, this creates ideal conditions for the initiation and spread of vegetation fires. This interaction is also reflected in the so-called "30-

The "30-30 rule" states that at temperatures above 30°C, wind speeds above 30 km/h and humidity below 30%, the risk of severe fire spread increases significantly (Deu Wett Meteorology for Vegetation Fire Fighting, 2024, p. 69).



Figure 9 30-30-30 rule ©@fire + ©EFI/WKR

Humidity is another crucial factor related to vegetation fires. Low relative humidity causes vegetation to dry out more quickly and become more flammable. In contrast, an increase in humidity can significantly slow the spread of a fire, as more energy is required to evaporate the water contained in the plants before the material can catch fire (Deu Wett Meteorology for Vegetation Fire Fighting, 2024, p. 40).

Precipitation also influences fire behavior in various ways. On the one hand, it introduces moisture into the vegetation, which soaks the fuel and slows the spread of the fire or, in the best case, even extinguishes it. On the other hand, sudden, violent thunderstorms with lightning strikes can ignite new fires. Moderate, continuous rain is most effective.



because it has enough time to penetrate the soil and completely moisten the vegetation. Short, heavy showers, on the other hand, often only run off the surface and cannot sufficiently moisten the fuel (Deu Wett Meteorology for Vegetation Fire Fighting, 2024, p. 67).

Another important meteorological aspect is atmospheric stability. An unstable atmosphere promotes vertical air movements, which can drive fires upwards and lead to the formation of pyrocumulus clouds. These clouds are formed by intense heat development over large fires and can even cause lightning, which in turn can ignite new fires (Deu Wett Meteorology for Vegetation Fire Fighting, 2024, p. 60).

In addition to these basic weather parameters, regional and local wind systems should also be considered. Local wind systems arise from differential heating of land and water surfaces or from specific topography, such as mountain-valley winds, which must be particularly considered when fires spread in mountainous regions (Deu Wett Meteorology for Vegetation Fire Fighting, 2024, p. 30).

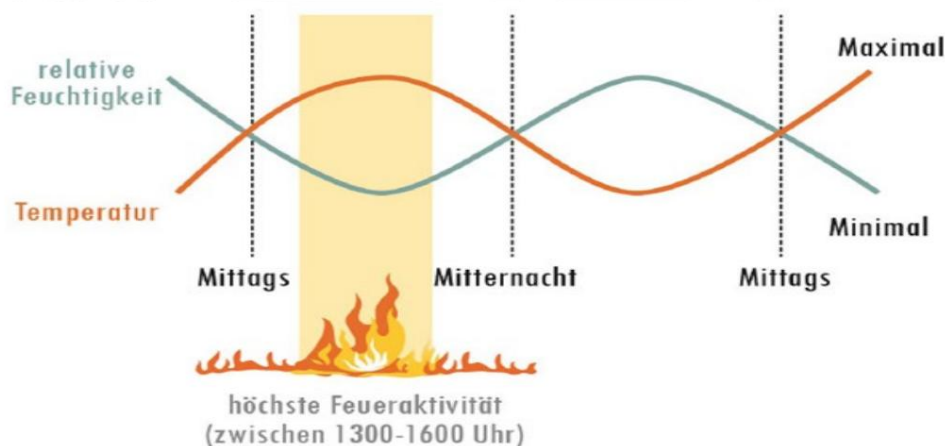


Figure 10 Humidity ©EFI/WKR

In summary, it is clear that a thorough understanding of meteorological factors and their consideration are crucial for success in vegetation firefighting. Modern technical aids and precise weather forecasts are proving to be indispensable tools for optimally preparing emergency personnel and making informed tactical decisions. Continuous observation and analysis of these weather parameters enable fires to be fought more efficiently and potential damage to be minimized.



3.3 Fuel (vegetation)

The composition and availability of fuel, i.e., vegetation, are key factors in the behavior of vegetation fires. Various aspects such as the type, moisture content, quantity, and distribution of fuel significantly influence the intensity, rate of spread, and extent of a fire.

The type of fuel encompasses various vegetation types, including grasses, shrubs, trees, and living and dead plant material. Finer fuel, such as grasses or small branches, ignites and burns more quickly. Coarser material, such as thicker branches or tree trunks, requires more energy to ignite but burns longer (Lindon Pronto, Held & Barth, 2022, p. 19). Especially in forests, fallen leaves and needle litter on the forest floor play an important role, as they can promote the rapid spread of flames.

The moisture content of the fuel is also very important. Especially in hot and dry seasons, dry material increases the risk and the speed of vegetation fire spread significantly. Although deadwood plays an important role in forest biodiversity, it can also pose a significant fire hazard when moisture content is low (Frey & Adelman, 2024, pp. 1-2). While smaller twigs and branches dry out more quickly, thicker tree trunks retain moisture longer and only pose an increased fire risk after extended dry periods.

The quantity and arrangement of fuel also play an important role in fire behavior. Dense, continuous fuel promotes the rapid and intense spread of a fire, while discontinuous fuel layers –

Barriers—such as roads, rivers, or open spaces—can slow or even stop the spread of fire (Lindon Pronto, Held & Barth, 2022, p. 19). Strategically placed firebreaks or protection strips are therefore effective measures to control the spread of fire.

However, in montane beech forests, which are normally considered less prone to fire due to their dense canopy closures and high soil moisture, prolonged drought can also lead to the drying out of litter and humus layers, increasing the fire risk (Maringer et al., 2020, pp. 2-3). After severe fires, such forests can suffer long-term damage, and the ability to regenerate depends on factors such as the presence of seed trees and the type of deadwood produced.

These specific properties of the fuel and the fact that even forests less prone to fire, such as montane beech forests, are affected by unusually



Long dry periods can cause fire, highlight the need for detailed knowledge of local vegetation and its fire behavior. Plant species, moisture content, distribution, and the size and location of deadwood must be constantly monitored and analyzed in order to develop preventive firefighting measures.

to be able to develop.

To effectively manage vegetation fires, it is crucial not only to understand the fire-ecological properties of the existing vegetation, but also to be sensitive to possible changes due to climate change and human influences

A balance should always be struck between the conservation of biodiversity and fire protection must be maintained. A differentiated approach to deadwood management and targeted silvicultural measures can help promote biodiversity and minimize the risk of vegetation fires (Frey & Adelman, 2024, pp. 3-4).

Table 1 Fuel moisture content

Brennmaterialeigenschaften	Feuchtigkeitsgehalt
Leichtes Brennmaterial	Schnellerer Feuchtigkeitsverlust, daher leichter entflammbar
Schweres Brennmaterial	Langsamere Feuchtigkeitsverlust, längere Hitzeexposition erforderlich, bevor es brennbar wird
Brennmaterial mit niedrigem Feuchtigkeitsgehalt	Verliert schnell Feuchtigkeit; brennt schnell mit hoher Intensität
Brennmaterial mit hohem Feuchtigkeitsgehalt	Verliert langsam Feuchtigkeit, zum Entzünden ist mehr Energie erforderlich; brennt mit geringerer Intensität



Figure 11 Different structure and quantity of fuel ©EFI/WKR



In summary, the properties, moisture content, quantity, and distribution of fuel decisively influence the dynamics of vegetation fires. A thorough understanding and continuous monitoring of these factors enables targeted preparation and control of firefighting. It is also essential for developing effective preventive measures and ensuring the long-term protection of forests and the species living there.

3.4 Topography/Terrain

Topography and terrain play a central role in vegetation firefighting, influencing both the behavior of the fire and the operational capabilities of the fire department. Various terrain characteristics, such as the slope, elevation, and terrain variations, have a significant impact on the direction and speed of a fire's spread.

Steep slopes accelerate fire spread because flames and heat travel upward, igniting fuel more quickly. This is due to convection and thermal radiation, which are more pronounced along slopes. The steeper the slope, the faster the fire can rise, making firefighting more difficult and requiring special caution from firefighters (Deu Wett Meteorology for Vegetation Fire Fighting, 2024, p. 66). In such situations, special tactics must be employed, such as creating firebreaks above the fire line, to prevent the flames from advancing unhindered.

Altitude also influences fire behavior. At higher altitudes, fire intensity may be reduced due to lower air density and oxygen levels. However, wind currents, which can be more pronounced at such altitudes, pose a particular challenge (Forest Road

and Operational Wildfire Response Planning, 2021, p. 10). Local wind systems are particularly important because they influence both the horizontal and vertical spread of fire. Anabatic winds, which blow up slopes during the day, and katabatic winds, which flow down slopes at night, are examples of such localized wind systems that must be considered in firefighting (Deu Wett Meteorology for Vegetation Firefighting, 2024, p. 30).

However, terrain features can also provide natural barriers that slow or stop the spread of a fire. Rivers, lakes, wide roads, or fields can serve as natural fire barriers. Such barriers are



are often included in strategic operational planning to stop fires or at least hinder their spread (Forest Road and Operational Wildfire Response Planning, 2021, p. 5). However, access to these areas may be limited by difficult terrain, requiring the use of all-terrain vehicles or even airborne units.

In flat or slightly hilly regions, where fires spread more slowly and evenly, more direct firefighting methods can be used. Fire extinguisher backpacks and hand tools are particularly effective and flexible in difficult terrain (State School and Technical Facility for Fire and Disaster Protection, 2024, pp. 15-16). The improved accessibility of fuel allows for targeted fire control in such areas.

Another challenge is the water supply in difficult terrain. Here, water intake points must be installed gradually, and the use of shuttle transport is crucial to ensure a continuous water supply (State School and Technical Facility for Fire and Disaster Protection, 2024, pp. 28-29). The terrain therefore influences not only tactical planning but also the logistical implementation of firefighting.

Finally, operational tactics must always take the topographical conditions of the terrain into account to ensure effective firefighting. This includes a precise analysis of the terrain prior to deployment, the strategic positioning of vehicles and extinguishing agents, and the use of natural barriers.

It is also important to train emergency responders in specific tactics tailored to the specific challenges of the terrain. Modern terrain analysis technologies, such as GIS systems and drones, can further optimize these strategies, significantly increasing firefighting success (Forest Roads and Operational Wildfire Response Planning, 2021, p. 9).

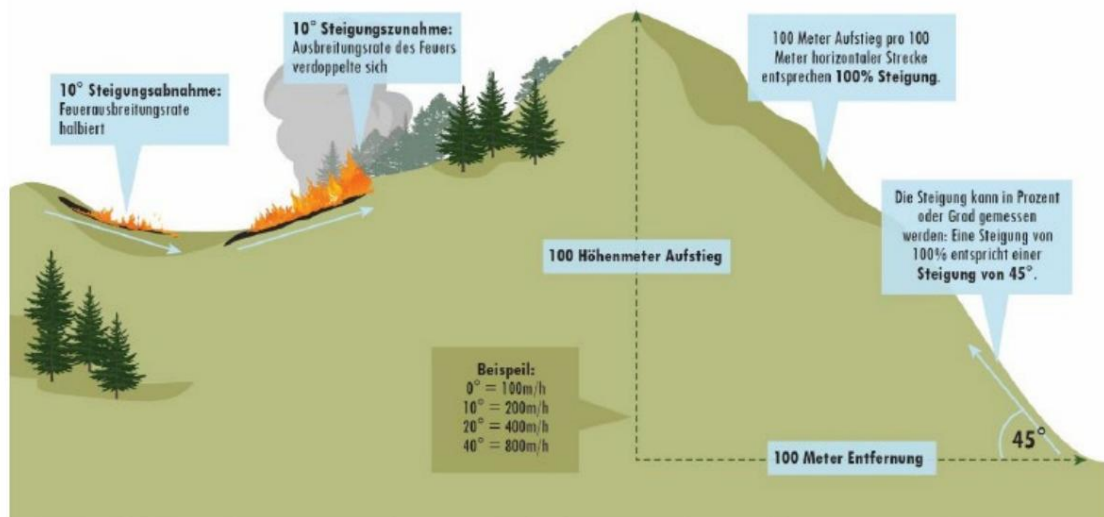


Figure 12 Fire behavior with changes in slope ©EFI/WKR

The interplay of fire behavior, weather conditions, vegetation, and terrain requires an interdisciplinary approach in which all these factors must be carefully analyzed and considered. Only then can emergency responders be equipped with the necessary knowledge and appropriate tools to effectively fight vegetation fires.

3.5 Tactical assessment of influencing factors

Assessing the tactical factors influencing vegetation firefighting requires a thorough understanding not only of the physical and meteorological conditions, but also of the organizational and logistical aspects. Precise situational assessment and sound decision-making are crucial for leading emergency personnel safely and effectively.

Fire behavior factors, weather conditions, the nature of the fuel, and the topography of the terrain are key factors that significantly influence the development and spread of vegetation fires. These parameters must be continuously monitored and analyzed to initiate effective measures. The duration and intensity of a fire depend heavily on the specific vegetation and local climatic conditions. Detailed knowledge of these factors enables emergency responders to predict how a fire will spread and which areas are particularly at risk (Maurer, p. 2).

The weather is a dynamic and often unpredictable factor that directly influences fire behavior. Wind speed and direction, temperature, and humidity are the key parameters that influence tactical



Assessment must be considered. Unexpected weather changes can significantly impact operational planning and require rapid and flexible adaptation of strategies (Barth, LSTE, p. 7).

Topography also plays a crucial role in fire development. Steep slopes can increase the speed and direction of fire spread, while fires spread more slowly on flatter terrain. Furthermore, difficult terrain, such as steep slopes or dense vegetation, can pose logistical challenges and impair the efficiency of deployed personnel (Barth, LSTE, p. 14).

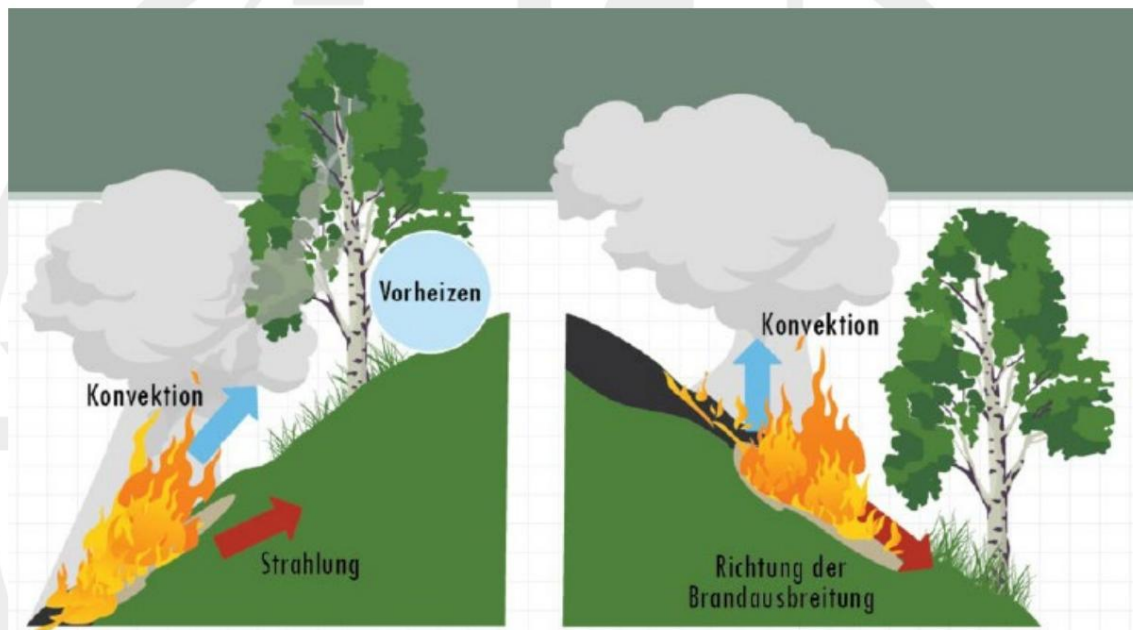


Figure 13 Effect of slope on fire spread ©EFI/WKR

The vegetation itself, i.e. the fuel, varies greatly in its flammability and burning time. Different vegetation types – from dry grassy areas to dense forests – require specific extinguishing strategies and appropriate equipment. Seasonal changes also play a crucial role, as they strongly influence the moisture content of the vegetation and thus its flammability (Barth, LSTE, p. 6).

For a well-founded tactical assessment, the systematic recording and analysis of these influencing factors is essential. The operational command must not only have comprehensive knowledge and experience, but also utilize modern technologies and current data to make a precise situation assessment (Maurer, p. 4). Digital tools, such as support parameter calculation systems and digital



Terrain models are used here. These help analyze the elevation profile of the area of operation and better plan water supply and extraction (Barth, LSTE, p. 26).

Equally important is the consistent application of a systematic decision-making process. This usually consists of a thorough assessment of the situation, the planning and evaluation of possible measures, decision-making and issuing of orders, and the constant monitoring of implemented measures (Maurer, p. 8). This process resembles a control loop in which decisions must be continuously adjusted based on current data.

A vivid example of such strategic approaches is the introduction of a dual shuttle service to the water supply in Brandenburg. Large tank fire engines were deployed at various transfer points to overcome the logistical challenges in vast and difficult-to-access areas (Barth, LSTE, p. 15).

Collaboration with other organizations, such as agricultural contractors, also plays a crucial role. This cooperation makes it possible to use tractors with water tanks as mobile water supply units, which can significantly increase the efficiency of water supply and pumping (Barth, LSTE, p. 16).

Ultimately, it is crucial that the fire command is able to integrate various influencing factors and respond flexibly to changing conditions. This requires a combination of expertise, practical experience, and the use of modern technologies. Only in this way can it be ensured that vegetation firefighting is carried out both effectively and safely, protecting the health and safety of the emergency responders and the affected population (Barth, LSTE, p. 7).

Overall, it is clear that a comprehensive tactical assessment of the influencing factors in vegetation fire fighting requires not only a high level of expertise, but also systematic and forward-looking planning and decision-making.

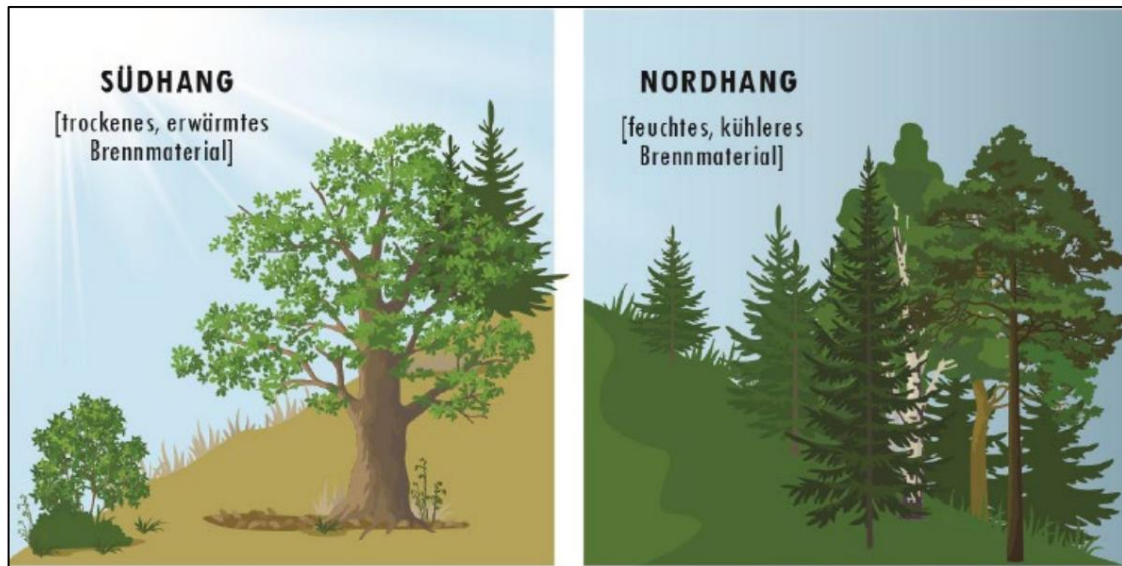


Figure 14 Effect of slope orientation on fuel ©EFI/WKR



4 Risks and dangers of vegetation fires

Vegetation fires pose a significant challenge for fire departments and society. The associated risks and hazards are diverse and complex. Due to climate change, both the frequency and intensity of these fires have increased worldwide, resulting in a growing threat to people and the environment. Fighting such fires requires not only specialized knowledge and skills, but also a deep understanding of the diverse risks associated with vegetation fires.

A thorough analysis and comprehensive understanding of the risks and hazards posed by vegetation fires are crucial for developing effective firefighting strategies and ensuring the safety of emergency responders and the affected population. Fire departments must be able to correctly assess and evaluate fire behavior, potential hazard sources, and the impacts of the fire on various ecological and social systems. Comprehensive hazard management plays a key role in this.

A key aspect of vegetation firefighting is careful planning and proactive risk analysis. Emergency responders must identify and assess potential hazards early and systematically in order to initiate appropriate preventive and reactive measures. A hazard matrix that systematically records the various risk factors and their potential impacts serves as a valuable tool to support decision-makers in selecting appropriate tactics and strategies and to ensure safety during operations.

Vegetation fires pose not only immediate hazards, such as the direct impact of flames and smoke on people and infrastructure, but also medium- and long-term risks. These include, for example, health impairments from smoke inhalation and the strain on the public health system due to potential long-term consequences. Furthermore, the destruction of habitats can lead to ecological and economic damage, which must also be considered in the risk analysis.

Another critical aspect is the safety of the emergency responders themselves. The often unpredictable dynamics of vegetation fires place high demands on tactical procedures and personal protective equipment. Emergency responders must always be vigilant and able to respond quickly to changing conditions. This requires not only solid training and regular continuing education, but also the proper use of modern equipment and technology.



In addition to protecting emergency personnel and firefighting teams, it is important to keep an eye on the potential dangers to the affected population. This includes planning and executing evacuations and providing emergency shelter. Timely and clear communication with the population is crucial to avoid panic and ensure an orderly evacuation. Information systems and warning methods must be efficient and reliable to enable rapid response in an emergency.

In summary, understanding and managing risks and hazards is a central component of vegetation firefighting. A comprehensive approach that includes systematic hazard analysis and assessment, the application of preventive and reactive measures, and the continuous optimization of operational methods is essential. Through targeted training, ongoing professional development, and the consistent use of modern technologies and strategies, fire departments can effectively address the challenges of vegetation fires and sustainably ensure the safety of emergency personnel and the public.

4.1 Overview of the hazard matrix

Vegetation fires pose a multifaceted threat that endangers not only the environment but also human health and infrastructure. The hazard matrix provides a systematic method for identifying and categorizing these hazards in order to develop preventive measures and ensure the protection of emergency responders and the affected population.

One of the central components of the hazard matrix is the so-called fire behavior triangle, which relates the three key propagation factors – wind, fuel, and topography – to each other (Barth, 2024, p. 6). These three elements influence both the behavior and the intensity of the fire and

The potential risks are therefore appropriate. Wind-driven fires tend to spread rapidly and unpredictably, while fuel-driven fires are strongly influenced by the type and density of vegetation. Topography also plays a significant role, especially on slopes, where fire can spread much faster uphill.

Another important element of the hazard matrix is the effective recognition and communication of hazards by all emergency services. Even the smallest potential hazards that are overlooked can lead to serious accidents. Therefore, a high degree of situational awareness and consistent communication are crucial (Barth, 2024, p. 5). Since vegetation fires often cover large areas



and can develop dynamically, identified risks must be communicated to management immediately.

Particular attention must also be paid to the respiratory toxins released by incomplete combustion processes. These include carbon monoxide and polycyclic aromatic hydrocarbons, which pose health risks to both emergency responders and the affected population (Barth, 2024, p. 12). To minimize these risks, emergency responders should always take appropriate protective measures, including wearing personal protective equipment and using CO detectors.

In addition to the immediate dangers of fire, there are also secondary risks, such as the danger of explosions at former battlefields. Particularly in areas suspected of containing explosive ordnance, the heat generated by fire can increase the likelihood of uncontrolled explosions. In such cases, special safety precautions, such as extensive cordons, are required (Barth, 2024, p. 19).

Another significant danger is the risk of entrapment due to uncontrolled

In complex and large-scale fire areas, emergency personnel may be unexpectedly trapped by the flames. This danger can be partially minimized through precise operational planning and the continuous updating of escape routes and safety zones (Barth, 2024, p. 11).

The hazard matrix also takes into account the physical stress and health risks associated with vegetation fire fighting.

These include, among other things, the risk of dehydration and overheating, which can be caused by prolonged deployments under extreme conditions and inadequate fluid intake (Barth, 2024, p. 14). Such conditions can have serious health consequences, such as heatstroke or circulatory problems.



Gefahren durch:	A	A	A	A	C	E	E	E	E
für:	Atemgift	Angstreaktion	Ausbreitung	Atomare Strahlung	Chemische Stoffe	Erkrankung/ Verletzung	Explosion	Elektrizität	Einsturz
Welche Gefahren müssen bekämpft werden?									
Menschen									
Tiere									
Umwelt									
Sachwerte									
Vor welchen Gefahren müssen sich Einsatzkräfte schützen?									
Mannschaft									
Gerät									

Figure 15 Hazard matrix in the context of vegetation fires LG No. 612 Veg.-

Bbk-Leading 1+Training intensive I AE Hazards / Safety - Basics I Barth (LSTE)

In summary, the hazard matrix is an indispensable tool in vegetation firefighting. It provides a systematic method for identifying, assessing, and communicating the numerous risks that can arise during such operations. The integration of these factors, combined with preventive measures, intensive training, and ongoing hazard monitoring, is crucial to ensuring the safety of emergency responders and the affected population. Consistent application of this matrix allows immediate hazards to be addressed quickly and effectively, and long-term risks to be reduced through forward-looking planning and coordinated operations.

4.2 Dangers to emergency services and the public

Fighting vegetation fires poses a significant risk to both emergency responders and the affected population. These risks arise from a variety of factors that can affect both physical and mental health and significantly impede the effectiveness of the operation.

For emergency personnel, responding to vegetation fires places a particularly high physical strain on them, which is exacerbated by high temperatures, dense smoke, and difficult terrain. A key risk is the danger of overheating and dehydration, as the protective clothing (PPE) that must be worn further increases the heat load. If PPE is not regularly cleaned or replaced, the risk of health damage from harmful particles and chemicals also increases. It is therefore crucial that emergency personnel



Taking regular breaks and drinking sufficient fluids is essential to reduce the risk of heat stroke (Technical Committee of the German Fire Brigades, 2021, pp. 3-5). Regular cleaning and changing of protective clothing are also essential to prevent long-term health problems (Technical Committee of the German Fire Brigades, 2021, p. 19).

Another serious risk for emergency responders is smoke development during vegetation fires. The smoke contains a variety of toxic substances that can be absorbed through the respiratory tract and cause both acute respiratory irritation and long-term health damage. Carbon monoxide, in particular, poses an immediate threat that must be reduced through the use of respiratory masks and carbon monoxide detectors (Technical Committee of the German Fire Brigades, 2021, pp. 17-18).

The often impassable and unfamiliar terrain in which vegetation fires occur presents an additional challenge. Dense vegetation, hard-to-reach areas, and steep slopes make it difficult for emergency services to progress and increase the risk of injuries from falls or from thorns and branches.

In addition, so-called "heat stress" – the overloading of the body due to the combination of physical exertion and extreme environmental conditions – can pose a serious problem (Technical Committee of the German Fire Brigades, 2021, pp. 4-6).

The population is also exposed to significant dangers in areas adjacent to forests or vegetation. There is a risk that buildings will be engulfed in flames, so evacuation measures must be carried out quickly and efficiently to protect residents. The psychological strain associated with the impending danger and evacuations can lead to considerable stress and result in both short-term and long-term health problems (Technical Committee of the German Fire Services, 2021, p. 4).

In addition to the direct dangers posed by fires, the public also faces the risk of smoke and toxic gases. These can irritate the eyes and respiratory tract and are particularly dangerous for older people or those with pre-existing medical conditions.

It is crucial to evacuate affected areas promptly and inform residents about appropriate protective measures, such as closing windows and doors (2022 Wildfire Season Report, p. 13).

Vegetation fires also pose indirect risks, such as the panicked flight of wild animals or the spread of pollutants caused by burning plants. To minimize these risks, thorough



Protective measures are necessary both during and after the fires (2022 Forest Fire Season Report, p. 25).

In summary, fighting vegetation fires poses significant risks for the emergency services and the affected communities. These risks require a careful preparation, appropriate equipment, and continuous monitoring of the health of all involved. This includes measures to protect the respiratory tract, regulate fluid intake, and provide safety training for both emergency personnel and the public (Technical Committee of the German Fire Brigades, 2021, pp. 15-16). It is essential that fire departments and other organizations develop and regularly update strategic plans to ensure the safety and health of everyone.





5 Security

Safety is a central element in the fight against vegetation fires and plays a prominent role in the training manual for fire brigades.

Given the diverse and often unpredictable challenges that arise during such operations, careful planning and consistent implementation of security measures are crucial to protect both the emergency responders and the affected population. This chapter presents basic security principles as well as specific approaches aimed at minimizing risks and maximizing the effectiveness of emergency responders.

Vegetation fires are characterized by their rapid spread, extreme temperatures, unpredictable wind changes, and the nature of the fuel. These dynamics require firefighters to have a deep understanding of the factors that influence fire behavior—including fuel types, weather conditions, and topography. This knowledge is critical for making informed decisions during an incident and ensuring the safety of all involved.

One of the main tasks of the fire department is therefore not only direct firefighting, but also the continuous education and training of emergency personnel in safety issues. Both preventive and reactive measures must be observed. The preventive aspect includes preparing for possible scenarios and imparting knowledge of safety protocols and how to deal with potential hazards. The reactive aspect refers to behavior during an operation, particularly the ability to recognize hazards and respond appropriately to avoid injuries or worse consequences.

A fundamental component of incident preparation is understanding safety principles, which provide a structured basis for conduct during an incident. These principles help ensure an orderly and coordinated response to the challenges of a vegetation fire. A safety-oriented approach requires clearly defined roles and responsibilities within

of the response teams to ensure effective coordination and that both preventive and emergency measures are implemented consistently.

Another important aspect is promoting a strong safety culture within the fire department. This is based on mutual trust and a willingness to adhere to safety protocols and share knowledge. A culture of openness, in which lessons are learned from past incidents, promotes continuous improvement and increases awareness of potential hazards. Regular



Training and practical exercises are essential to apply theoretical knowledge in practice and to provide emergency personnel with the necessary tools to deal with stressful situations.

The handling and maintenance of equipment also plays a central role in safety planning. From personal protective equipment (PPE) to technical aids, everything must be kept in optimal condition and used efficiently. All equipment should be regularly inspected and maintained to ensure it functions reliably in an emergency. A comprehensive knowledge of the available technologies and technical aids enables the targeted use of these resources to effectively fight the fire while reducing the risk.

to minimize.

Special attention is also paid to protection against contamination and the associated health risks. Vegetation fires can release a variety of toxic substances that can cause short-term and long-term health problems. Protection against such risks requires specialized knowledge and clear protocols that must be integrated into the safety concept to protect the health of emergency personnel.

Overall, safety forms the foundation of successful and sustainable vegetation firefighting. A comprehensive safety concept combines preventive measures with reactive strategies and promotes a culture of vigilance and learning. Through careful planning, thorough training, and adherence to proven safety protocols, the risks to emergency responders and the public can be significantly reduced.

5.1 Safety principles

When fighting vegetation fires, safety principles are crucial to both ensuring the safety of emergency personnel and increasing the effectiveness of the measures. Systematic adherence to these principles contributes significantly to reducing risks and making operations more efficient. The safety principles encompass several key elements that must be considered both individually and as a team.

A fundamental aspect is early hazard detection and assessment. All emergency personnel should maintain a high level of situational awareness in order to identify potential threats as quickly as possible. This hazard detection begins even before the deployment through comprehensive situation analyses and risk assessments.

During the operation itself, it is necessary to constantly monitor changes in the



environment or the behavior of the fire and react accordingly (PP LG-No. 612 Veg.-Bbk Fü1+Ausbildungen intensiv AE-Gefahr-Sicherheit, pp. 1-5).

Another essential element of the safety principles is teamwork so-
such as ongoing communication among emergency services. Security is created
not in isolation, but through the cooperative action of all involved. It is crucial that information about hazards is shared
immediately within the team so that a joint response can be achieved. Standardized communication tools play a key
role in avoiding misunderstandings or loss of information (PP LG No. 612 Veg.-Bbk Fü1+Ausbildung intensiv AE-Gefahren-Sicherheit, p. 28).

Maintaining safety zones and defining escape routes are also of utmost importance. These measures serve to protect
emergency personnel from dangerous situations from which they would have difficulty escaping.

Safety zones are designed to be free of flammable vegetation and at a sufficient distance from the fire front. Likewise,
escape routes are clearly defined so that emergency personnel can escape the danger zone quickly and safely in an
emergency (PP LG No. 612 Veg.-Bbk Fü1+Ausbildung intensiv AE-Ge-Sicherheit, pp. 29-30).

Another fundamental principle is ensuring personal protective equipment (PPE). Emergency personnel must wear
appropriate protective clothing at all times to protect themselves from extreme weather conditions, smoke, and heat.

These include protective suits, helmets with visors, gloves, and breathing apparatus. However, the effectiveness of
this equipment depends heavily on its proper use and maintenance. Regular training is necessary to ensure that
emergency personnel know how to use the equipment correctly in an emergency (PP LG No. 612 Veg.-Bbk
Fü1+Ausbildung intensiv AE-Gefahren-Sicherheit, pp. 14-15).

Consideration of weather and terrain conditions is also essential. Sudden changes in weather conditions, such as
changes in wind direction or drops in temperature, can significantly influence the behavior of vegetation fires. Therefore,
current weather data must be constantly monitored and correctly interpreted. Wind direction and speed, in particular,
play a crucial role, as they significantly influence the spread of the fire (PP LG No. 612 Veg.-Bbk Fü1+Ausbildungen
intensiv AE-Gefahren-Sicherheit, p. 10).

Ultimately, it is necessary to continuously evaluate security strategies and adapt them as needed. Since the conditions
of an operation often change dynamically, risk assessments and deployment strategies must be regularly reviewed.
Only through this flexible approach can security be ensured in the long term.



of the emergency services must be ensured (PP LG-No. 612 Veg.-Bbk Fü1+Ausbildungen in-tensiv AE-Gefahr-Sicherheit, p. 6-7).

In summary, safety principles are essential for safe and effective vegetation firefighting. The combination of hazard recognition, effective communication, the use of PPE, and consideration of weather and environmental factors creates a solid foundation that enables emergency responders to operate safely and successfully.

5.2 LACES

The LACES acronym is an essential component of vegetation firefighting safety protocols. Its five core components—Lookouts, Awareness (Anchor Points), Communications, Escape Route, and Safety Zones—provide a comprehensive safety system tailored to the unique challenges of wildland firefighting operations.

The first component, "lookouts," refers to specially assigned observers who continuously monitor the fire's development and assess potential hazards. This role requires experienced individuals who not only understand the terrain and fire behavior but are also able to detect changes early. Lookouts must also competently use communication systems to provide timely warnings and support decision-making to emergency responders. This minimizes the risk of emergency responders being caught unprepared by sudden changes in the fire (Lindon Pronto et al., 2022, p. 30).

Another important component of the LACES principle is the so-called anchor point.

These are safe locations that emergency personnel can use as a base from which to launch operations or retreat if necessary. Anchor points enable structured and safe firefighting because they provide a fixed reference point for operational planning. These points are essential, especially in dynamic, rapidly changing environments.

The communication aspect also plays a central role.

Effective coordination is virtually impossible without clear and constant communication between team members and managers. It's all about communicating status reports, warnings, and tactical instructions in a timely manner. In addition, everyone involved must know how to respond correctly in emergency situations and be familiar with the communication tools used (Vegetation Fire Fighting, 2022, p. 34).

Escape routes are another critical component of LACES.

These escape routes must be clearly defined before each operation, and



It should be ensured that they remain accessible during the operation. The strategic selection of escape routes must also be continuously reviewed to ensure that they can be used quickly and safely even under changing fire conditions (Vegetation Fire Fighting, 2022, p. 33).

The safety zones are the final pillar of the concept. These zones serve as refuges for emergency personnel and should be free of flammable materials. It is important that these zones not only provide sufficient space but also be regularly adjusted to ensure maximum protection. Continuous evaluation of the safety zones is essential.

The integration of the LACES system into vegetation firefighting training and practice is absolutely essential. It requires not only technical knowledge from emergency responders, but also the ability to make appropriate decisions in pressure situations and respond to sudden changes. A central component of this training is regular practice. Through continuous exercises, emergency responders internalize the principles and routinely apply them in an emergency. It is difficult to overemphasize the importance of these trainings to ensure that safety standards are effectively implemented (Lindon Pronto et al., 2022, p. 39).

It is also important that LACES is integrated into tactical operational planning. Leaders must ensure that these safety principles are not only understood theoretically but also applied in practice. This means that safety standards should be continuously monitored and documented to enable optimization in future operations (Notes on Practical AE, 2024, p. 1).

Overall, LACES is an indispensable tool in vegetation firefighting. Applying these principles reduces risks and significantly increases the safety of emergency responders. It is the responsibility of all involved to recognize these principles and integrate them into their operational strategies.

5.3 Principles of use

The operational principles of vegetation firefighting are crucial to ensuring both effectiveness and safety during firefighting operations. These principles are based on a blend of many years of practical experience and sound scientific knowledge. A core element is strategic planning, which must always be based on a comprehensive assessment of the situation.



An essential component of these principles is the continuous assessment of the fire situation. Firefighters must constantly monitor the behavior of the fire, the current weather conditions, the topography of the area of operation, and the nature of the burning material. These factors are crucial for the

Effectiveness of the measures implemented and for the safety of the crew (Schloßmacher et al., 2021, pp. 1-3). Since weather phenomena such as wind, humidity, and temperature can significantly influence the behavior of the fire, it is essential that the fire command always has access to up-to-date meteorological data (ibid., p. 4).

Another key principle is the selection of the appropriate operational tactics, which must be adapted to the prevailing conditions. These tactics can range from direct extinguishing, used for smaller and less intense fires, to indirect methods, which are effective for larger and more difficult-to-access fires. While the direct approach involves working directly at the fire's edge, the indirect approach involves establishing control lines to keep the fire at a distance (Pronto et al., 2022, pp. 1-4).

Particular attention should be paid to the selection and securing of anchor points. These anchor points serve as strategic positions from which emergency responders can launch their control measures. They must be selected to contain as little combustible material as possible to minimize the risk of fire spreading to these locations (ibid., pp. 6-7).

The incident command is responsible for ensuring that all operational personnel clearly understand the tactical concepts and their respective roles. The LACES principle plays a crucial role here. It encompasses the aspects of lookouts, awareness, communication, escape routes, and safety zones. These elements are essential to ensuring the safety of the crew during the operation (ibid., p. 39). Regular briefings and debriefings are also important to ensure the flow of information and to adapt tactics as needed.

The use of heavy equipment and technical aids must be carefully planned and monitored. Machines such as forestry mulchers or remote-controlled devices can help effectively clear large areas of vegetation and contain the spread of fire (Schloßmacher et al., 2021, p. 18). However, handling such equipment requires specially trained personnel and strict adherence to safety protocols to minimize risks.

Emergency plans and preparatory training are also of great importance.

These measures ensure that the emergency services can act efficiently and safely in complex and dynamic situations.

Regular exercises and



Simulations help to internalize processes and optimize teamwork (ibid., p. 42).

In emergency situations, it is important that the behavior of the emergency services is clearly defined and trained. Quick reactions and clear communication are the key

in order to identify dangers early and act accordingly.

Special concepts such as the Security and Support Unit (SUE) additionally ensure the safety and support of the emergency services on site (Pronto et al., 2022, p. 19).

In summary, the operational principles of vegetation firefighting require thorough planning, constant situational assessment, adaptable operational tactics, and comprehensive safety measures. These principles form the foundation for successful and safe firefighting operations and are crucial for controlling the spread of vegetation fires and

how to protect people and property.

5.4 Behavior in emergency situations

Emergency situations in the fight against vegetation fires pose enormous challenges for emergency services, as they can often occur suddenly and unexpectedly. Despite all preventive measures, it is essential that the emergency services

are prepared for such events at all times. Comprehensive training and targeted preparation are crucial here in order to be able to act quickly and effectively in critical moments (PP LG No. 612 Veg.-Bbk Fü1+Ausbilden intensiv AE-Gefahren -Sicherheit-G 1-24.pdf, p. 31).

A key aspect of behavior in such situations is the careful planning of escape routes. These should be considered during the deployment preparation phase and adapted to the specific on-site conditions. Efficient escape routes must be clearly defined, regularly reviewed, and adapted as needed (PP LG-No. 612 Veg.-Bbk Fü1+Ausbilden intensiv AE-Gefahren -Sicherheit-G 1-24.pdf, p. 29). These routes should always lead to safe areas that serve as safety zones. These zones must be sufficiently far from the fire front and located in areas that are difficult or impossible to ignite, such as damp meadows or vegetation-free areas (PP LG-No. 612 Veg.-Bbk Fü1+Ausbilden intensiv AE-Gefahren -Sicherheit-G 1-24.pdf, p. 30).

In an acute emergency situation, it is essential that the emergency services respond quickly and in a coordinated manner. Warning of danger is usually given by specially assigned observers, but any other person in the team can also issue the warning if necessary. Once the warning is given, the withdrawal must be



into the previously designated safety zones. All emergency personnel should be familiar with these routes and use them immediately to reach safety. If retreat is no longer possible, shelters must be sought at short notice that offer at least temporary protection (PP LG No. 612 Veg.-Bbk FÜ1+Ausbil-den intensiv AE-Gefahren -Sicherheit-G 1-24.pdf, p. 32).

Another crucial aspect in emergency situations is the reorganization of units. This tactical reorganization serves to protect the emergency personnel and secure the scene. Intensive training and regular exercises are essential to be able to react quickly and efficiently in such situations (PP LG-No. 612 Veg.-Bbk FÜ1+Ausbildung intensiv AE-Gehnen_ -Sicherheit-G 1-

24.pdf, p. 32).

Particular attention must be paid to adapting tactics to the specific emergency situation. For fires with flames less than 1.5 meters long, direct extinguishing measures can still be used. However, if the flames exceed this limit, a flanking attack is advisable. For flames over 3.5 meters long, direct extinguishing measures should be avoided, and indirect methods, such as the use of heavy equipment or air support, must be used (PP LG No. 612 Veg.-

Bbk FÜ1+Training intensive AE-Hazards -Safety-G 1-24.pdf, p. 33).

In addition to purely tactical measures, the physical and psychological resilience of the emergency personnel also plays a central role in such emergency situations.

Stress management and self-awareness skills are essential components of training and help emergency personnel make the right decisions in critical moments. Leaders must be able to recognize their team's personal limitations and initiate self-rescue measures in a timely manner (Decision-Making in Uncertainty – A Daily Challenge and Solution Strategies in the Fire Service, p. 12).

In summary, vegetation fire emergencies require well-thought-out retreat planning, clear and efficient communication channels, and a high degree of tactical flexibility. Through intensive training and practical exercises, firefighters can be prepared to respond quickly and decisively to ensure the safety of emergency personnel in such critical moments.

5.5 Self-protection measures

Self-protection measures are an essential part of vegetation firefighting training, as the physical and psychological stresses to which the firefighters are exposed can be considerable. In order to protect the health and



To protect the operational capability of firefighters, special strategies are therefore required.

A central element of these measures is the selection and correct use of personal protective equipment (PPE). The specifics of the area of operation, the weather conditions, and the type of fire largely determine which PPE is used. Important not only is protection from external hazards but also supporting the physical well-being of the emergency responders. For example, it is crucial that firefighters' clothing is cleaned after each operation to prevent overheating and dehydration (Personal Protective Equipment for Fire Departments Against the Background of New Challenges, 2021, pp. 1-3).

Risk assessments also play an important role in minimizing the individual risks of emergency personnel. Factors such as the

The nature of the site, the weather, the topography, the season, and the vegetation are taken into account, as these can significantly influence physical strain (Personal Protective Equipment for Fire Departments Against the Background of New Challenges, 2021, p. 3). Fire department leaders have a special responsibility here by ensuring regular rotations, adequate breaks, and refreshments.

Preventing overheating and dehydration is particularly important. Emergency personnel must have access to sufficient fluids throughout the entire operation. Since the route from the emergency vehicles to the fire scene is often long, providing drinks can be difficult. Therefore, it is important to ensure that emergency personnel are supplied with fluids in a timely and regular manner (Personal Protective Equipment for Fire Departments Against the Background of New Challenges, 2021, p. 13).

In addition to the physical stress, protective measures against harmful substances such as smoke and particles are essential. In many situations,

Respiratory protection equipment such as compressed air breathing apparatus or filter masks are essential to protect emergency personnel from toxic substances in the air (Selection of personal protective equipment for fire service operations, 2016, p. 17).

The psychological strain should also not be underestimated. The farther the escape route from the scene, the greater the mental stress caused by the uncertainties and dangers during the operation (Personal Protective Equipment for Fire Departments Against the Background of New Challenges, 2021, p. 4). It is therefore essential that emergency personnel know clear escape routes and are regularly briefed on escape strategies.



The working environment, especially during forest and vegetation fires, presents an additional challenge. The physical strain is further increased by dense vegetation, rugged terrain, and extreme temperatures (Personal Protective Equipment for Firefighters Against the Background of New Challenges, 2021, p. 6). Special training is required to prepare emergency responders for these conditions and ensure they adhere to the necessary safety measures.

In addition to PPE, organizational and tactical concepts such as the LACES principle (Lookouts, Awareness, Communication, Escape Route, Safety Zones) are also essential. These concepts support safe withdrawal and continuous communication for situation assessment (Notes on Practical AE Object Defense/Follow-up Extinguishing (Fundamentals), 2024, p. 3).

In summary, self-protection measures in vegetation firefighting consist of a variety of preventive and reactive measures. From the selection of suitable PPE and the provision of necessary resources to training and consideration of psychological stress –

All of these aspects contribute to ensuring that emergency personnel can carry out their tasks safely and effectively. Continuous evaluation and adaptation of these measures are particularly important in order to be able to respond to the constantly changing challenges of operations.

5.6 Control thresholds

Control thresholds are critical parameters in vegetation firefighting that enable safe and effective fire management. They define specific conditions under which the tactics employed or phases of the operation must be adapted to ensure both the safety of the firefighters and maximize the efficiency of firefighting.

The definition of these control thresholds is based on a variety of factors, such as weather conditions, flame height, fire spread rate, fuel type and distribution, and the topography of the affected area. These variables are subject to constant fluctuation, requiring a flexible and situation-adapted interpretation of the control thresholds.

A key aspect of control thresholds is the assessment of flame height and its influence on the selected response methods. If flames exceed a critical height, tactics must be quickly adjusted to ensure the safety of the emergency responders. This may mean switching from a direct



attack to an indirect or more defensive approach to minimize the danger to firefighters (Barth, 2024, p. 7). These thresholds are often based on empirical values and scientific studies that address the thermal loads and physical properties of vegetation.

deal with fires.

Weather conditions also play a key role in determining control thresholds. Wind speed and direction, humidity, and temperature significantly influence the behavior and intensity of a vegetation fire. In the event of sudden changes in these weather factors, incident commanders must be able to respond quickly and effectively to ensure both the safety of the emergency responders and the containment of the fire (Training Module on Fire Behavior, 2021, p. 9).

The topography of the terrain is another crucial factor in establishing control thresholds. Steep slopes and narrow valleys can accelerate fire spread and significantly increase the risk for emergency responders. In such areas, it is especially important to define control thresholds for escape routes to ensure the safety of firefighters. The strategic positioning of escape points and anchor points therefore becomes an essential part of operational planning (Fire Behavior Training Module, 2021, p. 10).

Another aspect that influences the definition of control thresholds is the type and distribution of the fuel. Different vegetation types exhibit different burning behaviors, which directly influences the choice of firefighting strategies. If a high concentration of highly combustible material is present,

If the fire is dangerous, the emergency services must be particularly vigilant and, if necessary, adapt the extinguishing methods used, for example by increasing the use of extinguishing foam or other extinguishing agents to control the spread of the fire (Training Module Fire Behavior, 2021, p. 12).

Overall, these factors demonstrate that control thresholds act as dynamic landmarks in vegetation firefighting, enabling continuous adaptation of tactics and thus ensuring the safety of emergency responders in a constantly changing environment.



Flammenlänge	Feuerverhalten	Empfohlene Taktik und Technik
≤ 0,5 m	Schmelbrand oder Bodenfeuer mit sehr geringer Intensität. Feuer kann von selbst erlöschen.	Angriff (Handwerkzeuge, Löschangriff) oder keine Aktion aber Überwachung je nach Standort und Situation
Achtung: Ein Angriff auf die Front bei Flammenlängen von mehr als 0,5 m ist gefährlich!		
0,5 – 1,5 m	Bodenfeuer geringer Intensität	Angriff mit Handwerkzeugen (Feuerpatsche, Rucksackspritze, Spatenschaufel und Laubbläser) und/oder D-Schlauch, Pump-and-Roll, etc.
Achtung: Ein Angriff auf die Front bei Flammenlängen über 1,5 m ist zu unterlassen.		
1,5 – 2,5 m	Mittlere Feuerintensität; zu intensiv, als dass die Einsatzkräfte nahe an den Flammen arbeiten können	Löschangriff über die Flanken (Pump-and-Roll, D-Schlauch), Flankierender/paralleler Angriff mit schwerem Gerät (Forstmulcher, Räumschild, Pflug) und oder taktischen Feueinsatz ⁴ empfohlen.
2,5 – 3,5 m	Hohe Feuerintensität; das Feuer ist so intensiv, dass die Einsatzkräfte nicht in der Nähe der Flammen arbeiten können	Löschangriff über die Flanken nur mit ausreichender Wasserversorgung Verteidigung mit indirekten Taktiken. Einsatz von schwerem Gerät empfohlen. Unterstützung aus der Luft empfohlen.
Achtung: Bei Flammenlängen über 3,5 m sollten umfassende Verteidigungsmaßnahmen zur Brandbekämpfung eingeplant werden.		
3,5 – 8 m	Sehr hohe Feuerintensität; Einsatzkräfte müssen aus großer Entfernung arbeiten	Verteidigung mit indirekten Taktiken. Einsatz von schweren Geräten notwendig. Unterstützung aus der Luft empfohlen.
Achtung: Bei Flammenlängen über 8 m sind ausschließlich Verteidigungsmaßnahmen möglich! Extremes Feuerverhalten und schnelle Brandausbreitung sehr wahrscheinlich → Rückzug der Einsatzkräfte auf sichere Positionen veranlassen!		
≥ 8 m	Extreme Feuerintensität und -verhalten; Einsatzkräfte können nicht sicher angreifen	Verteidigung mit indirekten Taktiken. Schutz von Infrastruktur und Siedlungsflächen. Einsatz von schweren Geräten notwendig. Unterstützung aus der Luft empfohlen. Rückzug und Evakuierung in sichere Bereiche.

Figure 16 Control thresholds LG No. 612 Veg.-Bbk-Führen1+Ausbildung intensive I AE Overview of technical-tactical procedures - Basics I Barth (LSTE)

The integration of control thresholds into the tactical and strategic approach to vegetation fires plays a central role, not only in immediate firefighting but also in preventive operational planning. Through the systematic analysis of past operations and the continuous adjustment of these thresholds, fire departments can optimize their operational strategies. This contributes to increasing the safety and efficiency of future operations. A crucial aspect of this is also the training and targeted training of emergency personnel in the recognition and application of these control thresholds. Such measures are essential to optimally prepare personnel for real-life operations (Training Module on Fire Behavior, 2021, p. 23).

A practical example of the application of such control thresholds is the AFFEN action plan. This plan provides a structured approach to vegetation fires and integrates clearly defined control thresholds into every step of the process. It supports emergency responders in proceeding systematically and safely by enabling them to continuously ensure control of the fire and the safety of the crews (Barth, 2024, p. 9).

The application of such models underscores the importance of a flexible yet well-structured approach to vegetation fires. By regularly training emergency responders in the application of these control thresholds, they are able to make quick and informed decisions in dangerous and dynamic environments.

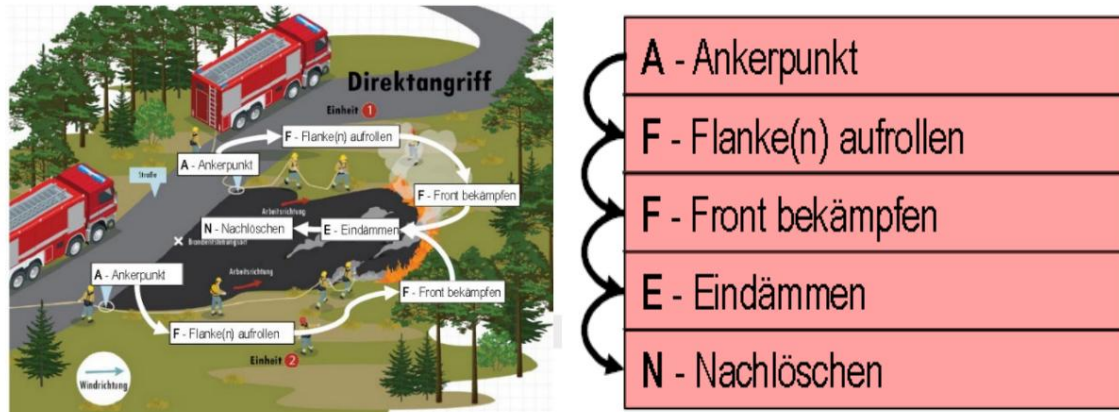


Figure 17 Action scheme MONKEYS LG No. 612 Veg.-Bbk-Leihren1+Ausbildung in-tensiv I AE Overview of technical-tactical action procedures - Basics I Barth (LSTE)

In summary, the definition and application of control thresholds is an indispensable tool in vegetation firefighting. They enable flexible, situation-appropriate adaptation of operational strategies, contribute significantly to the safety of emergency personnel, and increase the efficiency of firefighting measures. Continuous development and adaptation of these thresholds, based on the latest scientific findings and practical experience from past operations, is therefore crucial for the successful and safe conduct of future operations.

5.7 Safety concept – Security and support unit (SUE)

A central element of safety strategies in vegetation firefighting is the establishment of a security and support unit (SUE). This unit consists of specially trained and equipped emergency personnel whose primary task is to support the work of the regular units while simultaneously ensuring their safety. The SUE plays a crucial role, especially in complex and dynamic operations where the hazard situation can change rapidly.

The critical tasks of the SUE include the establishment and monitoring of safe anchor points and escape routes. These anchor points provide emergency personnel with safe starting points for their work and serve as escape routes in the event of an emergency. They are established in areas with little or no vegetation to minimize risk and provide emergency personnel with a safe working space (Barth, 2024, p. 26). These anchor points are crucial for the safe



Carry out fire-fighting measures and act as assembly points in the event of a retreat.

Another core element of the SUE's safety concept is the continuous monitoring and analysis of fire development. By establishing a systematic lookout system, dangers can be identified early and communicated to emergency personnel. This enables timely adjustment of operational strategies and the initiation of protective measures before the situation escalates (Barth, 2024, p. 24).

The SUE also assumes responsibility for communication and coordination between the various emergency services. Since communication at the scene is crucial, the SUE ensures that all important information, such as warnings of danger or instructions on tactical measures, is communicated quickly and clearly (Barth, 2024, p. 28).

Another key task of the SUE is providing emergency personnel with the necessary equipment. This includes water, breathing apparatus, and other important materials required during operations. The constant availability and maintenance of this equipment ensures that emergency personnel can work effectively and safely (Barth, 2024, p. 22). The SUE also monitors the prescribed escape routes, which are regularly inspected to ensure they remain clear of obstacles and safely passable (Barth, 2024, p. 29).

Another key element of the SUE's security measures is the provision of tactical reserves. The creation of tactical reserves. These serve to quickly provide additional forces in unforeseen and difficult-to-control situations to relieve the burden on emergency services and ensure safety (Barth, 2024, p. 22). This is particularly important when fire behavior changes suddenly and an immediate response is required.

Another safety-relevant aspect covered by the SUE is the prevention of vehicle fires. Emergency vehicles are parked in the direction of escape and secured during the operation. Closing windows and equipment compartment doors minimizes the risk of flying sparks igniting the vehicles (Barth, 2024, p. 22).

Overall, the security and support unit contributes significantly to the safety of vegetation firefighting through its comprehensive security measures. It not only ensures the immediate safety of the emergency services but also supports them through strategic planning and the logistical provision of the necessary resources.

The implementation of a



The implementation of a safety concept by the SUE is therefore essential to ensure effectiveness and safety in dangerous and demanding operational situations (Barth, 2024, p. 23).

5.8 Contamination

Fighting vegetation fires presents a number of challenges, including the risk of contamination, which requires special protective measures and clear protocols. Contamination can occur through the use of

occur in contaminated areas and pose a significant threat to health.
the safety of the emergency services.

Firefighters are exposed to a variety of chemical, biological, and physical contamination risks, including airborne contaminants and solid particles. Personal protective equipment (PPE) plays a crucial role in reducing the risk of contamination events. A sound risk assessment is essential to ensure that the level of protection is appropriate and that the equipment is tailored to the specific operational conditions (DGUV, 2016, p. 9).

The use of respiratory protective equipment is an important component of protection against vegetation fires. These devices must be carefully selected and regularly inspected to ensure maximum protection against potentially hazardous substances (DGUV, 2021, p. 55). Regular training is also required to ensure that emergency personnel are familiar with the use of these devices.

Contamination prevention measures also include the correct handling and maintenance of PPE. In particular, cleaning and disinfection of all equipment, such as helmets and gloves, is important to minimize the risk of secondary contamination (DGUV, 2016, p. 79).

Another critical issue is the management of potentially contaminated water sources used in firefighting. Contaminated water can not only harm the environment but also pose a health threat to emergency responders. Therefore, the risk of contamination must be considered during deployment.

The choice of water sources should be taken into account. The use of filter and treatment systems can help ensure water purity (DGUV, 2016, p. 59).

Radioactive contamination also poses a particular risk. In such cases
the use of special protective clothing and respiratory protection equipment is required,



provide sufficient protection against ionizing radiation without excessively restricting the freedom of movement and operational capability of firefighters (DGUV, 2016, p. 13).

In addition to equipment, the behavior of emergency personnel is crucial in contamination management. Regular training and exercises help prepare emergency personnel for potential contamination scenarios and train them in the use of contamination-safe equipment. Systematic documentation and debriefing of operations contribute to deepening knowledge and continuously improving strategies (DGUV, 2021, p. 69).

A key component of contamination management is post-emergency decontamination. This includes the proper removal and disposal of contaminated equipment. Adherence to strict decontamination protocols is necessary to completely remove residues and prevent re-exposure of emergency personnel (DGUV, 2021, p. 123).

It is also important to plan emergency response measures for dealing with contamination in advance. This includes providing decontamination kits and training emergency personnel in emergency procedures. A quick and efficient response can minimize the impact of contamination and ensure the safety of emergency personnel (DGUV, 2021, p. 97).

In conclusion, a comprehensive understanding and consistent application of protective measures are essential to minimize the risks of contamination during vegetation fires. This requires not only the right protective equipment, but also regular training, careful maintenance, and thoughtful planning before each deployment. Only in this way can the health and safety of emergency personnel be protected in the long term.



6 Equipment and resources for vegetation fire fighting

Effectively fighting vegetation fires is a complex and demanding challenge that requires a thorough understanding of the equipment and resources used. In practice, the success of these operations depends largely on the correct selection and appropriate use of these resources. This chapter is dedicated to the basic and specialized equipment, as well as the various resources specifically developed and used for vegetation fire fighting.

Vegetation fires are particularly dangerous due to their highly dynamic nature and unpredictable development. The equipment used by firefighters plays a key role in protecting lives, the environment, and property.

The requirements for equipment arise from the specific conditions and hazards associated with such fires, such as rapid fire growth, high flames, and widespread spread due to wind and weather conditions. Specialized equipment is therefore essential to meet these challenges.

First, personal protective equipment (PPE) forms the foundation for the safety of emergency responders. This protects firefighters from extreme temperatures, smoke, sharp objects, and other immediate hazards that can arise during vegetation fires. Well-equipped PPE consists of flame-resistant clothing, helmets with visors and face shields, breathing apparatus, and robust gloves and boots to provide essential basic protection.

In addition to personal protective equipment, specialized vehicles and equipment are used to meet the operational requirements in difficult terrain. These include special all-wheel drive fire engines with large water tanks that can operate even in difficult-to-access areas. These vehicles are often equipped with modern technology, such as fire monitors and mobile water pumping equipment, to ensure efficient firefighting under difficult conditions.

Modern technologies and technical tools have contributed significantly to the advancement of vegetation firefighting in recent years. These include thermal cameras that localize fire sources through heat detection, and drones that enable real-time aerial surveillance. These technologies provide valuable information for strategic planning and the execution of operations. Mobile communication devices and GPS



Systems play a crucial role in ensuring the coordination of emergency services and keeping an eye on the current situation at all times.

This chapter also discusses the use of large equipment such as tracked vehicles and firefighting helicopters. These devices are particularly crucial for large-scale and difficult-to-access fires. They can distribute significant amounts of water or extinguishing agents, creating large fire barriers to control the spread of the fire.

Ground-based equipment such as hand-held fire pumps, fire beaters, and chainsaws complement the range of equipment and are particularly important in the initial phase of a fire and during post-fire extinguishing operations. These devices enable precise firefighting in hard-to-reach areas and help secure control lines to prevent the fire from spreading.

Another important component of vegetation firefighting equipment is the use of firefighting aircraft and helicopters. These aircraft enable rapid and targeted attacks on fires, even in remote areas. With their capacity to drop large quantities of water or extinguishing agents, they provide indispensable aerial support and help bring fires under control quickly.

Overall, this chapter highlights the diversity and strategic importance of equipment and resources in fighting vegetation fires. A thorough understanding of the respective advantages and disadvantages of these resources, as well as their optimal application, is crucial for successful firefighting. Only in this way can the risks to people and the environment be effectively minimized.

6.1 Personal protective equipment (PPE)

Personal protective equipment (PPE) is an essential component of vegetation firefighting, as it provides the necessary protection for emergency personnel against the diverse hazards posed by such fires. Vegetation fires pose specific challenges, such as extreme heat, difficult terrain, and long deployment times, which is why PPE must be specifically tailored to these conditions (Technical Committee of the German Fire Brigades, 2021, p. 1).

The essential components of PPE include heat-resistant protective clothing, helmets, gloves, goggles, and breathing apparatus. The protective clothing must not only protect against heat and flames, but also provide comfort and freedom of movement to avoid additional strain on the emergency responders. A thorough risk assessment is necessary to select the appropriate PPE for



to select different fire operations and conditions, as emphasized in DGUV Information 205-014 (German Social Accident Insurance, 2016, pp. 1-3).

A particularly important aspect is protection against smoke gases and respiratory toxins, which are produced in large quantities during vegetation fires. The use of respiratory protective equipment such as filter masks or self-contained breathing apparatus is therefore essential. Filter masks, such as FFP3 masks, protect against particles and pathogens, for example (Technical Committee of the German Fire Brigades, 2021, p. 18). However, they do not offer protection against toxic gases such as carbon monoxide, which is why constant monitoring of environmental conditions is necessary.

The choice of PPE must also be adapted to the specific hazards and the respective area of operation. In dense vegetation or uneven terrain, additional protective measures may be necessary to prevent injuries from thorns, falls, or animal contact (Technical Committee of the German Fire Brigades, 2021, pp. 4-5). Reinforced protective suits and sturdy footwear are particularly important here to ensure safety without restricting mobility.

Another key issue is operational hygiene and the prevention of contamination. Contaminated PPE can contain dangerous residues that can endanger the health of emergency personnel and impair operational capability. Therefore, thorough cleaning and maintenance of equipment after each operation is essential (Technical Committee of the German Fire Brigades, 2021, p. 1).

In addition to the correct selection and maintenance of PPE, regular training and exercises are crucial for emergency personnel. This training includes the correct handling of the equipment, as well as behavior in emergency situations and the safe conduct of rescue measures (German Statutory Accident Insurance, 2021, pp. 69-71).

PPE is also divided into groups depending on the area of application. For example, there is PPE 11 for external firefighting and PPE 12 for internal firefighting. For vegetation fires, special forest fire PPE is recommended, which is particularly lightweight and tailored to the specific requirements of such operations (Technical Committee of the German Fire Brigades, 2021, p. 10).

In summary, PPE used in vegetation firefighting must be adapted to the specific conditions and hazards. Careful selection of equipment, regular maintenance, and comprehensive training of firefighters are crucial to ensuring effective protection and safety during firefighting. Only through these measures can



the challenges of vegetation fires are mastered efficiently and safely.

6.2 Vehicles and equipment knowledge

Vegetation firefighting requires specialized equipment, particularly in the area of vehicles and equipment, to ensure an effective and coordinated response to wildfires. Various vehicles are used.

tools and hand tools are used, which vary depending on the area of application and the specific requirements.

First, fire engines specifically equipped for vegetation firefighting deserve mention. These vehicles often require high off-road capability, as they are used in difficult-to-access areas.

Fire engines and tank fire trucks are equipped with powerful pumps and large water tanks to ensure a continuous water supply. They can also carry portable firefighting equipment such as fire backpacks or motor-driven sprayers, which can be used flexibly and directly at the fire scene (Südmersen, 2024, p. 10).

A central element in vegetation fire fighting are also the vehicles used to create control lines, which are intended to prevent fires from spreading uncontrollably.

Both manual and mechanical methods are used.

Tools such as brush cutters, saws, and hand saws are used for manual work, while machines such as mulchers and forestry equipment are used for the mechanical construction of control lines. To ensure safety at the entry

To ensure safety at the scene, these machines are often accompanied by security guards and water-carrying fire engines (Südmersen, 2024, p. 15).

In addition to the vehicles, equipment knowledge plays a crucial role. Hand tools such as the Pulaski axe, a combination of an axe and a hoe head, are particularly suitable for working rooted or rocky soils and for exposing embers. The McLeod tool, which combines a raking and hoeing function, is particularly useful for removing combustible material from sandy ground (Südmersen, 2024, p. 18). Such tools are indispensable for direct and indirect firefighting measures and must be carried by emergency personnel.

be served more safely and efficiently.

Technical aids such as GPS devices and thermal imaging cameras are also of great importance. GPS systems help with the precise marking and zoning of fire areas, as well as with navigation in complex terrain. Thermal imaging cameras enable embers to be quickly located, thus enabling a thorough



To ensure control over the progression of the fire (Südmersen, 2024, p. 8). These modern technologies contribute significantly to strategic planning of operations and increased safety for emergency personnel.

In hard-to-reach areas, special equipment and techniques are used. Fire slappers and sand throws are often used to directly combat flames, while fire backpacks with water tanks and double-action pumps allow emergency responders to fight the fire at its root. These devices are particularly useful in areas where fire engines cannot be deployed (Südmersen, 2024, p. 10).

Another important tool for firefighting operations is the firefighting backpack, which allows for targeted water delivery in difficult terrain. Firefighting backpacks are essential for operations that extend over many hours or even days.

During these phases, tools such as picks and fire hoses are also used to expose and extinguish embers (Südmersen, 2024, p. 8).

In summary, vehicles and equipment knowledge play a central role in the success of vegetation firefighting. From powerful fire engines to specialized hand tools and modern technologies, the right combination of these elements significantly improves the efficiency and safety of emergency responders. A thorough understanding and the ability to correctly use this equipment are essential components of training and contribute significantly to effectively fighting vegetation fires and minimizing the impact on people and the environment.

6.3 Technical aids and modern technologies

Vegetation fire fighting has made significant progress in recent years through the use of technical tools and modern technologies.

These technologies enable fires to be detected early, localized precisely and fought in a targeted manner, which significantly improves the efficiency and safety of emergency services.

A key technological advance is the integration of cyber-physical systems (CPS), which combine various technologies such as the Internet of Things (IoT), unmanned aerial vehicles (UAVs), and unmanned ground vehicles (UGVs). IoT sensors continuously monitor environmental conditions and thus detect vegetation fires at an early stage. The collected data is transmitted in real time to drones and ground vehicles, which conduct detailed situation reconnaissance and initiate targeted extinguishing measures based on this (Battistoni et al., 2023, p. 1).



UAVs are particularly useful in firefighting because they can quickly reach hard-to-reach areas and enable comprehensive aerial surveillance. Equipped with sensors and infrared cameras, UAVs can identify fire sources and track the direction of fire spread. Using advanced algorithms such as YOLOv5 for object detection, UAVs can precisely locate fires and transmit this information in real time to ground vehicles or control stations (Battistoni et al., 2023, p. 9).

Unmanned ground vehicles (UGVs) are another technical tool used autonomously for firefighting. These vehicles can be equipped with water or foam and operate in difficult terrain. They have the advantage of being able to remain continuously at the source of the fire to prevent it from reigniting (Battistoni et al., 2023, p. 12).

The combination of UAVs and UGVs creates an effective synergy. UAVs provide critical data from the air, while UGVs perform targeted firefighting operations on the ground. This collaboration is supported by advanced communication protocols and control systems such as the Robot Operating System (ROS). Technologies such as the ROS Bridge enable real-time communication between the various systems, ensuring coordinated and efficient firefighting (Battistoni et al., 2023, p. 10).

Data processing and analysis also play a central role. Machine learning can analyze weather data, vegetation information, and topographical data to predict fire danger. These forecast models provide valuable information for strategic operational planning and help utilize resources more efficiently (Umunnakwe & Davis, 2023, p. 3).

Satellite monitoring and remote sensing are other technologies that enable large-scale monitoring. Convolutional neural networks (CNNs) can analyze satellite images and automatically detect fires.

Specially adapted models such as MobileNetV2 offer high accuracy and efficiency, even on devices with limited computing power (James et al., 2023, p. 3).

A central component of modern vegetation fire fighting are the ground stations, which monitor and control the resources.

These stations provide real-time information on temperature, humidity and wind speed, enabling the operational command to make quick and informed decisions (Battistoni et al., 2023, p. 14).

In summary, technical aids and modern technologies offer significant improvements in vegetation fire fighting. They enable



more efficient firefighting, minimize risks for emergency responders, and reduce damage to the environment. However, the continuous development and adaptation of these technologies to the specific requirements of firefighting operations remains an ongoing challenge that requires further research and development.

6.4 Use of large equipment

The use of large-scale firefighting equipment is crucial, especially for large-scale and difficult-to-access fires. These equipment offers significant advantages, as they are capable of delivering large quantities of extinguishing agents and performing tactically decisive actions that would not be possible with smaller equipment (Thompson, Gannon, & Caggiano, 2021, p. 1).

A key advantage of large equipment such as bulldozers and heavy firefighting vehicles is their ability to create vegetation barriers. Such barriers serve as buffer zones that can slow or stop the progression of fire.

Combining these barriers with other techniques, such as controlled burning, significantly increases effectiveness and helps contain the spread of fire (Thompson et al., 2021, p. 5).

Heavy-duty firefighting vehicles offer the ability to pump large quantities of water or foam directly into the fire. Equipped with high-performance pumps and water cannons, these vehicles can operate even in difficult terrain where conventional firefighting equipment reaches its limits. They also play an important role in supporting smaller units by providing water for portable firefighting equipment, thus ensuring a continuous supply of extinguishing agents (Thompson et al., 2021, p. 7).

Aircraft such as helicopters and firefighting aircraft complement the use of large equipment on the ground by distributing large amounts of water or firefighting foam over burning vegetation. These aircraft are particularly useful in inaccessible or steep terrain, as they can extinguish fires precisely from the air while increasing the safety of ground responders. However, the efficiency of these aircraft depends heavily on coordination with ground forces to ensure optimal implementation of operational plans (Thompson et al., 2021, p. 3).

Another important application of large equipment is the transport of personnel and equipment over long distances and difficult terrain. Modified off-road vehicles and trucks enable firefighting units to quickly reach the



Such vehicles are not only crucial for logistics but also contribute to the safety of emergency personnel by ensuring quick and safe access to incident sites (Thompson et al., 2021, p. 11).

The efficiency of large-scale equipment has been further increased through the use of modern technologies. GPS-based navigation systems enable emergency personnel to make more precise decisions and identify and avoid danger zones more quickly. Modern monitoring systems also allow real-time observation of fire development and the movements of deployed vehicles, enabling optimal coordination of resources (Thompson et al., 2021, p. 9).

In addition to their tactical advantages, large-scale firefighting equipment also contributes to the sustainability of firefighting efforts. The use of heavy machinery allows for the removal of combustible materials, such as dead wood, reducing the risk of future fires. These preventive measures promote ecological stability in the affected regions and contribute to long-term fire risk reduction (Thompson et al., 2021, p. 10).

In summary, large firefighting equipment offers decisive advantages. Their ability to combat fires both preventively and actively not only improves the efficiency and safety of emergency responders but also makes them an indispensable component of modern firefighting methods.

Integration into modern technological systems and close coordination with other firefighting techniques further contribute to large-scale firefighting equipment playing a key role in fighting vegetation fires.

6.5 Use of ground-based equipment

The chapter "Use of Ground-Based Equipment" highlights the various tools and techniques used in ground-based vegetation firefighting. These tools are particularly important because they can be used in hard-to-reach and remote areas where the use of fire engines and hose lines is not possible.

Ground-based equipment offers a wide range of applications for direct extinguishing measures, exposing embers, or establishing control lines. The use of these devices requires physical effort and technical skills, which is why correct handling and safety are paramount.



One of the basic tools is the firefighting backpack, which holds between 16 and 25 liters of water and is equipped with a double-action pump. These backpacks allow emergency personnel to extinguish fires directly at the root of the flames and are particularly effective for smaller fires (Südmersen, 2024, p. 10). Fire beaters and fire swatters are also used to extinguish and extinguish flames. These devices are particularly popular due to their ease of use and high efficiency.

Hand-held tools such as shovels, spades, and various rakes and axes also play a central role. Shovels and spades are used in particular for sand throwing, in which sand is thrown onto the fire to extinguish it. This technique is particularly useful when only limited amounts of water are available (Südmersen, 2024, p. 10).

More specialized tools such as the McLeod tool and the Gorgui tool combine cutting and raking functions and are suitable for different soil types and vegetation structures. The McLeod tool is particularly suitable for sandy soils and is often used to remove combustible material and expose burn strips (Südmersen, 2024, p. 18). The Gorgui tool is versatile and offers a wide range of applications in vegetation firefighting (Südmersen, 2024, p. 19).

In practice, these tools are often used in combination to maximize firefighting efficiency. For example, one fire crew can extinguish the flames, while another crew extinguishes the remaining flames, and a third crew carries out follow-up extinguishing operations with fire backpacks (Südmersen, 2024, p. 11). This coordinated approach minimizes the risk of reignition.

Ergonomics and safety of the emergency services are also of utmost importance. The physical work requires good fitness, and emergency personnel must wear appropriate personal protective equipment (PPE). In addition to helmets, gloves, and flame-resistant clothing, safety goggles and respirators are essential to protect emergency personnel from heat, smoke, and flying particles (Südmersen, 2024, p. 8).

Another important aspect is adapting techniques and tools to local conditions. For large burn areas, additional measures such as establishing control lines may be necessary. These lines serve to stop the spread of the fire and create a safe barrier. Techniques include removing vegetation down to the mineral soil or creating fire-free strips (Südmersen, 2024, pp. 13, 15).



Continuous monitoring and maintenance of these control lines are essential to ensure their effectiveness.

Modern technologies, such as GPS, are also used to precisely locate and document fire sites and embers. These technologies increase the efficiency and safety of emergency responders by enabling more precise planning and tracking of operations (Südmersen, 2024, p. 8).

In summary, the use of ground-based equipment in vegetation firefighting requires a variety of specialized tools, each with its own specific advantages and disadvantages. The correct selection and combination of these tools, as well as thorough training and strict safety guidelines, are crucial to the success of operations.

6.6 Use of manned aircraft

The use of manned aircraft plays a crucial role in vegetation firefighting, especially in large-scale fires where ground-based measures often reach their limits. These aircraft are capable of reaching difficult and remote areas, conducting rapid initial reconnaissance, and delivering targeted extinguishing agents. They thus provide valuable support in firefighting strategies.

A key advantage of manned aircraft is their ability to deliver extinguishing agents precisely and effectively to hard-to-reach fires. Helicopters and aircraft are often equipped with special external firewater tanks, such as the flexible Bambi Bucket, which allows water to be collected from open bodies of water and dropped over the fire area (State School and Technical Facility for Fire and Disaster Protection, 2024, p. 23). These tanks have a capacity of up to 10,000 liters and ensure a flexible water supply.

The constant supply of extinguishing agents over long distances relieves the burden on ground troops and improves the efficiency of firefighting.

However, the mobilization of these aircraft requires extensive logistical planning and close coordination between the forces on the ground and in the air.

Communication between pilots and operations control, for example via air traffic control and digital radio, is crucial for precisely coordinating drop zones and implementing safety precautions (State School and Technical Facility for Fire and Disaster Protection, 2024, p. 22). It is essential to monitor the takeoffs and landings of rotorcraft and coordinate the movements of emergency personnel in the drop zone to avoid accidents caused by falling branches or the strong downwash generated by the rotors.



The suitability of the aircraft used must also be carefully checked.

While rigid external load containers such as the Smokey are unsuitable for direct withdrawal from firewater tanks, flexible containers offer a more promising application (State School and Technical Institution for Fire and Disaster Protection, 2024, pp. 23-24). The choice of the appropriate aerial device and equipment depends on various factors, including the type of fire, the availability of water sources, and the topographical conditions of the area of operation.

In addition to firefighting, aircraft can also be used for important tasks such as aerial reconnaissance. These reconnaissance missions enable better situational assessment and more precise planning of responses. They can also conduct evacuations and emergency transport, especially in areas cut off from supply lines by fire.

Despite the numerous advantages, risk mitigation remains a top priority. Risks such as contact with falling branches or the weakening of tree stability due to massive shedding should not be underestimated. Furthermore, the possibility of so-called downwash or sidewash must be considered, in which the powerful airflow from the rotors can unpredictably fan the fire (State School and Technical Institution for Fire and Disaster Protection, 2024, p. 24).

Overall, manned aircraft offer significant advantages in vegetation firefighting, but also pose complex challenges for emergency services.

Careful planning, coordination, and training are essential to minimize risks and maximize the benefits of these high-tech tools. Compliance with safety protocols is also crucial to ensure the protection of emergency responders and civilians. The strategic use of technology and air support can significantly increase the efficiency of firefighting operations, ultimately contributing significantly to the containment and control of vegetation fires.



7 Water supply and pumping

Water supply and pumping is a fundamental aspect of vegetation firefighting. Water is not only the primary extinguishing agent, but also a critical resource whose availability and efficient use can significantly determine the success of an operation. Given increasing climatic changes and extreme weather phenomena, the challenge of providing adequate quantities of water quickly and effectively in affected areas is becoming ever greater. Therefore, it is essential that fire departments and other relief workers are comprehensively informed about the various aspects of water supply and pumping.

promotion training.

Vegetation fires differ from other fire types in many respects, particularly in the often large burned areas and the varying availability of water in the affected regions. In-depth knowledge of local conditions as well as seasonal and meteorological influences is necessary to effectively plan and implement water supply. Since forest fires and other vegetation fires can quickly reach threatening intensity, ensuring a constant water supply is of immense importance to firefighting teams.

The use of water as a blanket resource in fighting vegetation fires presents particular challenges. For many large-scale fires in forest and grassland regions, conventional water supplies are often insufficient or even nonexistent. In such cases, alternative water sources must be developed or innovative extraction techniques must be used to achieve the necessary extinguishing effect. Geographical conditions and available technical equipment play a key role.

Another essential factor in water supply for vegetation fires is the efficiency of water pumping. This includes not only the quantitative provision of extinguishing water, but also the logistical planning and implementation of water pumping from the source to the fire sites. Techniques such as the construction of hose lines, the use of shuttle water pumping, and the implementation of water transfer points must be precisely coordinated and coordinated. This requires a high degree of tactical understanding and organizational skills on the part of the emergency services.

The long-term security of firefighting water supplies, especially in areas repeatedly affected by vegetation fires, must also be considered and planned. Strategies for the creation and maintenance of water reservoirs, the expansion of water abstraction and pumping infrastructure, and the creation of redundancies are essential in this regard.



Given changing environmental conditions, such as increasing drought periods, a proactive and forward-looking approach to resource security is required.

The importance of a well-organized and well-thought-out water supply and

The importance of water is further emphasized by the fact that it not only serves as an immediate extinguishing agent, but can also fulfill preventive functions. Creating firebreaks through targeted water use or wetting vegetation to minimize the fire load are just a few examples of how water can be used strategically in firefighting.

In summary, water supply and pumping in vegetation firefighting is a complex and multifaceted issue that requires comprehensive knowledge and skills. Mastering this topic is crucial for successful firefighting and the protection of people and nature.

Continuous training and further education of emergency personnel, the development and adaptation of new techniques, and the regular review and improvement of existing procedures are essential in order to adequately meet the increasing challenges in the field of vegetation fire fighting.

7.1 Water as a scarce resource

Water supply is a central issue in vegetation fire fighting.

Water shortages represent a significant limitation in the practice of firefighters, particularly in regions affected by water scarcity or during periods of drought. This problem occurs not only in arid regions but increasingly also in temperate climates, primarily due to the effects of climate change and associated climatic anomalies such as summer droughts.

and heat waves.

A thorough understanding of the importance of water as an extinguishing agent and the challenges of its availability is essential for the optimal preparation and execution of firefighting operations. Water not only serves as a means of containing and suppressing fires, but is also crucial for cooling and suppressing pockets of fire that may persist after the main extinguishing operations (Module 4, 2022, pp. 1-20).

The specific nature of vegetation fires poses particular challenges for water supplies. These fires often occur in rural or remote areas where water infrastructure is often poorly developed or nonexistent. These geographical conditions require



Mobile and flexible supply solutions, such as mobile water tanks, shuttle service of tank fire engines, and the use of water from natural sources such as lakes, rivers, or ponds, are available. However, access to these water sources can be hampered by topography and prevailing fire conditions (Module 4, 2022, p. 3).

Effective water management also involves optimizing existing resources. This includes establishing and maintaining water intake points in densely forested areas to ensure quick and efficient water withdrawal in the event of an emergency. Advance planning and regular inspection of these intake points are crucial. Permanent water reservoirs, which are installed in many communities, should be regularly maintained and expanded as needed to provide sufficient water supply (Module 4, 2022, p. 6).

Given the water shortage, it is also important to consider alternative firefighting methods and technologies. The use of wetting agents, which increase the wetting capacity of water, or firefighting foam can be useful in many cases. These agents enable more efficient use of available water quantities by better incorporating the extinguishing water into the burning material and enhancing the cooling effect (Module 4, 2022, p. 10). However, the ecological impacts must be considered, and the use of such agents must be coordinated with local environmental authorities.

Another important aspect is the sustainable security of firefighting water supplies. Long-term planning and investments in infrastructure are particularly necessary in times of climate change. Innovative water extraction technologies and the integrating alternative water sources, such as rainwater harvesting systems and the reuse of treated water, can address the challenges posed by increasing water scarcity. Targeted storage of water during wetter seasons for use in the dry months also represents a practical solution (Module 4, 2022, p. 18).

Collaboration between various stakeholders—fire departments, local authorities, forestry administrations, and landowners—also plays a crucial role. This cooperation facilitates access to necessary information about available water resources or the use of shared facilities. By sharing experiences and jointly developing strategies, operations can be made more efficient and water-related bottlenecks minimized.

The measures and findings presented here demonstrate that meeting the challenge of water supply requires continuous adaptation and innovation. Against the backdrop of changing climatic



Given the current conditions and the increasing frequency of vegetation fires, this remains a key focus in the further development of vegetation fire fighting.

7.2 Water intake point

Water supply is a central element in vegetation fire fighting.

The availability of water is crucial for the success of operations, especially in regions affected by water scarcity or during periods of drought. In Brandenburg, a distinction is made between dependent and independent firefighting water supplies (Barth, 2024, p. 3).

Dependent firefighting water supplies draw their water from the public drinking water network. However, this network is often exhaustible, especially under the challenging conditions of a major fire. In such cases, independent firefighting water supplies represent an important alternative, as they utilize water from natural or artificial sources such as bodies of water, ponds, wells, and other reservoirs. These sources often provide an inexhaustible supply of water, which is needed for extended firefighting operations (Barth, 2024, p. 3).

A key aspect is the accessibility of these water intake points, especially in densely forested or remote areas where the density of conventional water intake points is lower. These points can vary in access depending on the season and can sometimes be difficult to reach, complicating efficient water supply (Barth, 2024, p. 6). Technical and tactical considerations are crucial to optimize flow rates and minimize distances to the site.

In practical implementation, it is essential that water intake points are clearly marked and easily accessible. This requires coordinated planning and regular monitoring of available water sources. Investments in the infrastructure of these water intake points are therefore necessary for the long-term supply of firefighting water in rural and densely forested areas. It is recommended that firefighting water reserves be strategically distributed and maintained in the form of tanks or ponds (Barth, 2024, p. 4).

A pragmatic approach to increasing firefighting water capacity is the integration of agricultural resources, such as slurry tanks, which can be used as mobile firefighting water reserves (Barth, 2024, p. 16). However, such systems must be thoroughly cleaned and maintained to ensure both technical functionality and hygienic safety.



Another important aspect is the differentiation in the planning and provision of water intake points. In development areas without increased property or personal risk, responsibility lies with the local fire protection authorities. For specialized properties and in forest areas, the installation and maintenance of water intake points is the responsibility of the property owners and forest owners (Barth, 2024, p. 4).

This differentiated allocation contributes to the efficiency and sustainability of fire water supply and requires close cooperation between public and private actors.

Furthermore, innovative solutions and modern technologies should be integrated to increase the effectiveness of water intake points. Digital tools can provide valuable support by enabling the measurement of distances and elevation profiles, as well as the display of hydrant locations (Barth, 2024, p. 26). These technologies improve the accessibility and usability of water intake points by providing accurate and up-to-date information.

For a comprehensive safety concept, it is essential that firefighters are trained not only in the technical handling of water-carrying hoses and pumps, but also in the efficient use of water extraction points. Experience from Brandenburg shows that knowledge of the various circuit series, consisting of open and closed systems, and their application in different topographical and infrastructural conditions is of central importance (Barth, 2024, p. 20).

In summary, the effectiveness of water intake points plays a crucial role in vegetation firefighting. Technical and organizational measures can ensure the availability of extinguishing water even under difficult conditions. The use of natural resources, modern technologies, and strategic planning of water intake points form the cornerstones of successful firefighting and prevention.

These integrative approaches must be continuously developed and adapted to the growing requirements of vegetation fire protection.

7.3 Water pumping

Water extraction is an essential component of vegetation firefighting, as it directly influences the efficiency and effectiveness of firefighting efforts. Given the often limited water resources in rural and densely forested regions, it is important to develop and implement sustainable water extraction methods.



When it comes to water supply, a fundamental distinction is made between dependent and independent water supplies. Dependent water supplies use the public drinking water network, which, however, is exhaustible and therefore only partially suitable for long-term operations. In contrast, independent water supplies obtain their water from natural sources such as rivers, lakes, or artificial firefighting ponds. These sources are generally inexhaustible and can provide larger quantities of firefighting water (Firefighting Water Supply, 2024, p. 3).

To efficiently transport water from remote water intake points to the fire scene, powerful pumps and hose lines are used. The length of the hose lines and the associated pressure loss play a crucial role here. The hose lines must be designed and operated in such a way that a constant and sufficient flow rate is guaranteed. This often requires the use of booster pumps to maintain a stable pressure and compensate for hose line defects, such as bursts (Fire Water Supply, 2024, p. 19).

A common method of water delivery in vegetation firefighting is shuttle transport with tank fire engines. These vehicles shuttle between the water intake point and the fire scene to continuously deliver water.

However, this method is time-consuming and complex, which is why it is usually only suitable for initial use. In the long term, shuttle transport aims to temporarily ensure the firefighting water supply until more stable systems can be established (Firefighting Water Supply, 2024, p. 10).

To maximize the efficiency of shuttle traffic, water transfer points can be used. These allow for strategic transfer of water from tank fire engines to smaller vehicles or directly to the scene. The selection and equipment of the water transfer points must be carefully planned to ensure smooth operations (Fire Water Supply, 2024, p. 14).

In rural areas, agricultural machinery, such as water-carrying trailers, can also be used to pump water. However, this requires careful planning and coordination with local farmers to ensure their availability and the technical requirements. In Brandenburg, cooperation with the agricultural sector is firmly integrated into the operational strategy (Fire Water Supply, 2024, p. 16).

In addition to the technical implementation, tactical considerations in water supply are of great importance. This includes the creation of operational sections depending on the size of the fire area and the planning of booster pump stations along the hose lines to ensure a continuous water supply.



(Fire Water Supply, 2024, p. 14). Ensuring the fire water supply permanently also requires the establishment of long-term strategies, such as the construction and maintenance of permanent water tanks and reservoirs.
in areas close to forests.

Modern digital tools significantly support water extraction. These technologies make it possible to measure distances and elevation profiles, determine hydrant locations and calculate delivery parameters such as inlet and outlet pressure. These functions improve the precise planning and execution of on-site operations (Fire Water Supply, 2024, p. 26).

Overall, it is clear that successful water supply in vegetation firefighting requires a combination of technical expertise, tactical skill, and innovative solutions. Only by integrating all of these aspects can effective and sustainable firefighting be ensured.

7.4 Commuter traffic

Shuttle transport in vegetation firefighting is an important method for ensuring water supplies in remote or poorly developed fire areas. Given the often limited natural water supply, shuttle transport is often the first option for fighting fires. Fire engines with water tanks continuously travel between a water intake point and the scene to transport the required extinguishing water.

The effectiveness of shuttle traffic depends on several factors, including the number and capacity of the tank fire engines deployed, the distances covered, and the availability and yield of water intake points (State School and Technical Facility for Fire and Disaster Protection, 2024, p. 27). Due to the dynamic nature of shuttle traffic and the numerous variable influencing factors, planning and organizing this method is challenging. Nevertheless, its flexible application makes it an indispensable technique in vegetation firefighting.

However, after the first firefighting operations, in which shuttle service was initiated due to its rapid availability, significant weaknesses of this method also became apparent. Continuous water supply is often difficult to plan, as it depends on the availability of vehicles, the timing of their movements, and possible failures (State School and Technical Facility for Fire and Disaster Protection, 2024, p. 27). For longer distances, shuttle service requires comprehensive management organization and the implementation of



technical-tactical measures, such as the use of stationary water pumping systems or additional firefighting water tanks. This often leads to increased coordination and resource requirements.

Although shuttle transport is primarily a temporary solution, it can be optimized using various approaches depending on the situation. For example, deploying a large number of small tank fire engines can accelerate water delivery, while larger vehicles can transport a greater volume of water and reduce the amount of water required per trip. The flexibility of this solution also allows for spontaneous response to changing conditions and operational requirements.

Technically, there are various types of shuttle transport that can be used depending on the deployment scenario and resource availability. A basic distinction is made between single, double, and multiple shuttle transport with a water intake point, as well as closed, open, and combined shuttle transport (State School and Technical Facility for Fire and Disaster Protection, 2024, p. 26). These variants offer different advantages and disadvantages depending on the deployment conditions.

Additional measures can significantly increase the efficiency of commuting.

These include well-planned water transfer points that enable faster and more efficient water withdrawal, as well as increased transport capacity through the deployment of additional fire engines. Close cooperation and coordination among firefighters, as well as precise operational management, are crucial for optimizing shuttle traffic.

In the long term, however, shuttle service should be replaced or at least supplemented by more stable and reliable methods. These include stationary water pipes installed along the emergency route to ensure a consistent water supply in the long term (State School and Technical Facility for Fire and Disaster Protection, 2024, p. 26). While these solutions require greater logistical and technical effort, they offer a safer and continuous water supply compared to shuttle service.



Figure 18 Double commuter traffic ©IdF NRW

In summary, shuttle transport is an important, albeit temporary and interim, method for supplying water during vegetation fires. Its flexibility and rapid deployment make it particularly

This makes it an important option, especially in the initial phase of an operation. However, the long-term goal should be to implement more reliable systems to ensure a continuous supply of extinguishing water and sustainably safeguard the effectiveness of firefighting.

7.5 Water transfer points

Water transfer points play a central role in the logistical organization of water supply in vegetation firefighting. These points act as interfaces between various modes of transport and the end users of the water at the fire scene, enabling efficient and continuous water supply. Given the often limited availability of water in rural or remote areas, they are crucial (Fire Water Supply/Promotion, 2024, p. 14).

A key element of water transfer points is their strategic location. They should be positioned so that they are easily accessible, both for supply vehicles and for the fire engines that transport the water further.

Paved access roads and sufficient space for maneuvering vehicles are essential (Fire Water Supply/Supply, 2024, p. 15). Well-planned location of water transfer points can significantly increase the efficiency of fire water supply and minimize downtime.

The water transfer points should be equipped with sufficiently large containers to ensure a continuous supply, even if the supply vehicles are temporarily missing (Fire Water Supply/Promotion, 2024, p. 14).

For this purpose, folding containers or other improvised water reservoirs can be used, which can be set up quickly and have sufficient capacity for the duration



of the operation. The selection and equipment of these water transfer points must be based on the specific requirements of the emergency situation, such as the size of the fire and the availability of extinguishing water nearby (Extinguishing Water Supply/Supply, 2024, p. 15).

The technical equipment at the water transfer points is also crucial. Firefighting centrifugal pumps and other fittings must be present to ensure the smooth transfer of water (Fire Water Supply/Conveyance, 2024, p. 15). These devices should be regularly maintained and checked for functionality to ensure immediate deployment in the event of an emergency.

be.

Another important aspect of water transfer points is the safety of emergency responders. This includes ensuring a clear and safe operating environment free from potential hazards such as loose soil or unstable structures. Emergency responders should receive clear instructions on the handling and operation of the water transfer points, observing all safety precautions (Module 4 Vegetation Fire - Post-Fighting Work, 2022, p. 16).

Challenges can arise from topography and terrain. In hilly or difficult-to-access areas, special solutions must be developed to ensure that water can be distributed efficiently without exposing vehicles to unnecessary risks. This may include the use of specially designed off-road vehicles or additional pumping stations for pumping water over longer distances.

The integration of agricultural water transport components, such as tractors with attached slurry tankers, can also represent a useful extension of water transfer points. These vehicles are often already adapted to local conditions and can be deployed flexibly and quickly to transport large quantities of water (Fire Water Supply/Promotion, 2024, p. 16).

In addition to the physical infrastructure, communication and coordination between the various units using the water transfer points is crucial. A comprehensive communication network ensures that all parties involved are informed about the status of the water supply at all times and that adjustments can be made quickly (firefighting water supply/

funding, 2024, p. 14).

A future-oriented approach to water transfer points could include the integration of modern technologies and digital tools that improve the management and overview of water availability and use. Through the use of GPS technologies, real-time

from digital terrain models and



Data communication enables emergency services to plan and respond more effectively (Fire Water Supply/Promotion, 2024, p. 26).

Finally, regular training and practice of all emergency personnel in the use of water transfer points is an essential component of vegetation fire preparedness. Only through continuous training and practical exercises can it be ensured that all procedures are followed in an emergency and that water is supplied without delay. Such training measures should include both theoretical background knowledge and practical exercises in the field to be prepared for all eventualities (Module 4 VEGETATION FIRE - FOLLOW-UP FIRE EXTINGUISHING WORK, 2022, p. 14).

Overall, water transfer points are a critical element in vegetation fire fighting, and their planning and operation require utmost attention.

Through well-thought-out strategic positioning, appropriate technical equipment, comprehensive security measures and continuous training of employees, emergency services can maximize their effectiveness and minimize the risk for the emergency services. be nimated.

7.6 Long-term security of fire water supply

Ensuring the long-term supply of firefighting water represents a significant challenge in light of climate change and the associated increase in extreme weather events such as droughts. Strategic measures and coordinated efforts are necessary to ensure the availability of firefighting water in the future. A key basis for this is the systematic planning and continuous review of existing water withdrawals.

positions and infrastructures.

In recent years, it has become apparent that the availability of natural water sources is becoming increasingly unreliable. Therefore, alternative water supply strategies, such as the installation and regular maintenance of artificial firefighting water reservoirs, are of key importance. These measures can help maintain the water supply even during dry periods (Firefighting Water Supply/Supply, p. 3). The expert commission from Saxony recommends establishing and regularly monitoring both fixed and mobile firefighting water reservoirs in order to be able to act quickly in the event of a vegetation fire (Report of the Expert Commission, 2023, p. 53).

A strategic element for ensuring the long-term supply of firefighting water is the planning of double shuttle services and long water supply routes. These approaches make it possible to supply water over long distances and longer



To provide firefighting water for extended periods of time. In Brandenburg, for example, the use of feeder tank fire engines and temporary water intake points is recommended to ensure a continuous water supply (Firefighting Water Supply/Supply, pp. 8-9). Effectively managing shuttle traffic and optimizing these processes are therefore crucial measures to meet water needs during long-term operations.

The development of a water transfer system that incorporates large tank fire engines and delivers water at strategically important points is another option for optimizing firefighting water supply. These points must be selected according to specific criteria and regularly adjusted to function efficiently (Firefighting Water Supply/Supply, pp. 14-15). Such systems are particularly essential in rural and densely forested regions where direct water sources are often difficult to access and the construction of long-distance supply lines is necessary.

In the long term, the integration of modern technologies and digital tools should also be sought to support water supply. Digital tools can assist in the evaluation and modeling of scenarios, identify optimal locations for water transfer points, and improve the efficiency of water supply. Smartphone applications and web platforms have already contributed to improving operational planning in various regions (Firefighting Water Supply/Support, pp. 26-27).

In addition, the use of alternative water sources should be further investigated and promoted. This includes, for example, the use of agricultural infrastructure, such as slurry tanks, which, after appropriate cleaning measures, can be used for firefighting water. Close cooperation with farms can tap into additional water resources.

and ensure the supply of fire-fighting water, particularly during longer fire-fighting operations (Fire-fighting water supply/supply, p. 16).

Another important aspect is the legal and organizational anchoring of such measures. It is essential that regulations are in place that support the expansion and operation of such infrastructures and provide financial incentives.

In Saxony, for example, it is proposed to legally enshrine fire prevention inspections and the regular revision of forest fire protection concepts in order to ensure a high level of water supply security in the long term (Report of the Expert Commission, 2023, p. 47).

In summary, ensuring the long-term supply of firefighting water for vegetation fires requires a combination of strategic planning, technical innovation, legal framework, and practical implementation.



A stable and reliable water supply can be ensured through coordinated efforts of all actors involved and continuous adaptation to changing climatic conditions.





8 Tactics and strategies in combating vegetation fires

Fighting vegetation fires is one of the most complex and demanding tasks for firefighters. Unlike building fires, where the causes and spread of the fire can usually be quickly identified, fighting fires in vegetation-rich areas requires a differentiated approach and comprehensive knowledge of various natural, weather, and landscape factors. The tactics and strategies used are also more diverse and less standardized, making comprehensive training and continuous professional development of emergency personnel essential.

Vegetation fires can be triggered by numerous factors, including natural causes such as lightning strikes or human activities such as campfires and negligence. Fighting them requires not only specific technical skills and equipment, but also a deep understanding of the nature of fire itself. Therefore, developing tactics and strategies for fighting these fires efficiently and safely is a central component of the training and preparation of firefighting units.

A fundamental principle in fighting vegetation fires is the use of both direct and indirect methods. Direct methods refer to immediate intervention at the scene of the fire to extinguish the fire or prevent its spread. This includes, for example, directly controlling the flames with water or other extinguishing agents. Indirect methods, on the other hand, are based on preventive measures aimed at slowing or stopping the spread of the fire by creating fire barriers or burning vegetation in a controlled manner. The choice of method depends on various factors, such as the intensity of the fire, the topography of the terrain, weather conditions, and the available personnel and materials.

In addition to these basic approaches, there are a variety of specialized operational techniques that can be applied in practice. These include, among others, the targeted burning of vegetation to create firebreaks or the targeted use of natural barriers such as rivers or roads. These techniques require precise planning and coordination to maximize their effectiveness and minimize the risk to emergency responders and residents.

The simple tactical wildfire forecast also plays a crucial role in the development of tactics and strategies. It enables emergency responders to predict the development of a vegetation fire and take appropriate measures. This forecasting tool is based on various



Parameters such as weather conditions, vegetation types and terrain characteristics to estimate the behavior of the fire and determine the most effective tactics.

men.

The components of assessment and decision-making in vegetation fire scenarios are complex and require in-depth knowledge and experience. Incident commanders must not only consider the immediate dangers to people and the environment,

but also consider long-term perspectives. This includes assessing fire dynamics, analyzing escape routes, resource availability, and the potential impacts of a fire response on the environment and the affected region.

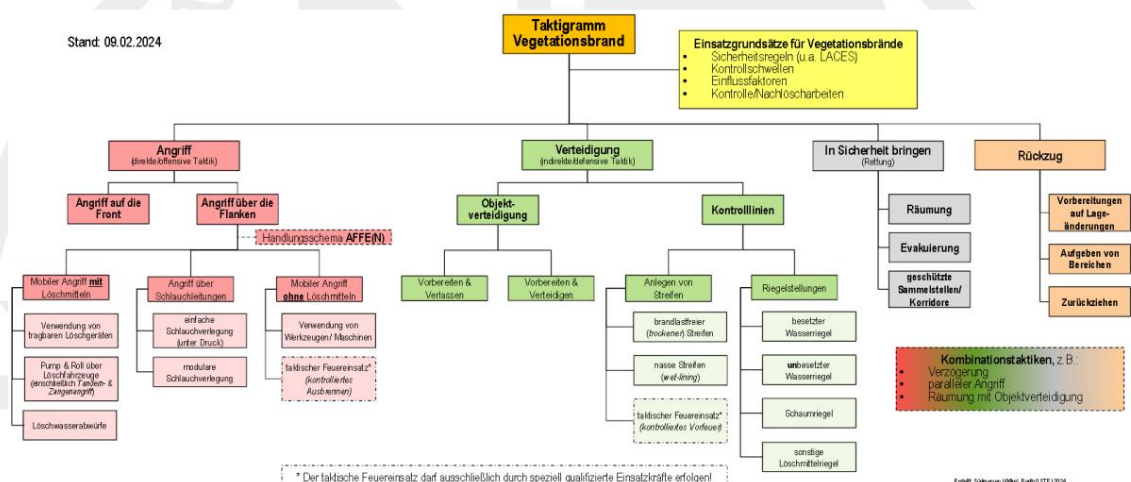


Figure 19 Vegetation fire tactigram, LG No. 612 Veg.-Bbk-Leihren1+Ausbildung intensiv I AE Overview of technical-tactical procedures - Basics I Barth (LSTE)

Overall, the tactics and strategies used to fight vegetation fires aim to bring the fire under control as quickly and efficiently as possible, while ensuring the safety of emergency responders and the affected population. Through continuous learning and adaptation

The methods allow firefighters to respond to changing circumstances and new respond to challenges arising from climate change and increasing human-nature interactions.

This chapter provides a comprehensive introduction to the various tactics and strategies for fighting vegetation fires and presents the essential concepts and considerations that are crucial for successful firefighting. It emphasizes the need to integrate both preventive and reactive measures to increase the effectiveness of firefighting.



maximize and minimize risks. Careful planning and execution of these strategies enables safe and effective firefighting.

8.1 Direct and indirect methods

Direct and indirect methods are crucial in fighting vegetation fires, as they offer different advantages and disadvantages depending on the fire progression and the specific conditions at the scene. These methods can be roughly divided into two categories: direct firefighting and indirect firefighting. Both methods have their own tactics and strategies that should be applied depending on the circumstances.

Direct firefighting focuses on fighting the fire directly at the fire line. One of the main techniques of direct firefighting is the frontal attack, in which firefighters work at the active fire front to either cool, smother, or separate the fire from unburned fuel (Pronto, Held & Barth, 2022, p. 9). This tactic is primarily used for low- to medium-intensity fires with a flame length of up to approximately 1.5 meters. The advantage of this method is that the burned area can be used as a safety zone, immediately stopping the spread of the fire.

Another direct approach is the flanking attack, in which a control line is established along the flanks of the fire to gain control of the sides of the fire. This method is preferred because it allows firefighters to retreat to the already burned areas in an emergency, increasing the efficiency of the control line (Pronto, Held & Barth, 2022, p. 12). The flanking attack allows for the spread of the fire to be managed under safe conditions, making firefighting more controlled.

In contrast, the indirect method involves establishing control lines at a safe distance from the active fire front. These control lines can represent natural or artificial barriers that prevent the fire from spreading further (Pronto, Held & Barth, 2022, p. 13). This method is often used for high-intensity, hard-to-reach fires to ensure the safety of emergency responders. Controlled burning between the fire front and the control line can keep the fire at a safe distance. However, a disadvantage of this method is that it often requires extensive destruction of vegetation to create an effective barrier.

Another indirect approach is controlled burnout or pre-fire.

the fuel is stored between the side of the control line facing away from the fire and



The actual fire front is intentionally burned. This technique helps reduce the amount of fuel that could reach the main fire, thus creating an additional barrier. However, the pre-fire must be conducted under strict supervision and with sufficient personnel to prevent uncontrolled spread (Pronto, Held & Barth, 2022, p. 37).

Both methods require a deep understanding of fire behavior and the specific conditions, including weather, terrain, and existing vegetation. Factors such as wind speed and direction, relative humidity, and topographical features play a crucial role in the choice of methodology (UAG Training of the Working Group on Natural Forest Fire Protection, 2024, p. 13). Furthermore, it is necessary to respond flexibly to changing conditions to ensure the safety of emergency personnel and maximize the success of extinguishing measures.

The choice of the appropriate method depends on the specific situation, with the primary goals always being to contain the fire and protect people, the environment, and property. In addition to the techniques described, the

Available resources, the efficiency of the methods, and the potential risks for emergency responders must be taken into account (Pronto, Held & Barth, 2022, p. 15). In many cases, the use of a combination of direct and indirect methods may be the best approach to efficiently address the complex challenges of a vegetation fire.

In summary, the choice between direct and indirect methods is based not only on current fire conditions, but also on a careful assessment of safety aspects and the optimal use of available resources. A well-thought-out strategy that takes into account the changing conditions of fire propagation can make a decisive contribution to successfully controlling vegetation fires and minimizing potential damage.

8.2 Control lines and prevention

Control lines are a key element in vegetation firefighting, as they serve to prevent or at least delay the spread of a fire. They are established before, during, and after an operation and can include both natural and artificial barriers. The effectiveness of a control line

depends on various factors such as fire behavior, weather conditions, topography and vegetation (Module 4 VEGETATION FIRE - POST-EXTINGUISHING WORK, 2022, p. 7).



A key objective of establishing control lines is to physically separate the burning and unburned material. This can be achieved by removing vegetation and other combustible materials down to the mineral soil. This ensures that the fire does not overflow and spread further (Module 4 VEGETATION FIRE - FOLLOW-UP WORK, 2022, pp. 6-7).

The establishment of a control line ideally begins at an anchor point. Such a point should have as little fuel as possible and can provide a natural barrier such as a body of water or an already burned area. From this point

From then on, the line will be systematically expanded and, if necessary, reinforced with further measures (Module 4 VEGETATION FIRE - FOLLOW-UP FIRE EXTINGUISHING WORK, 2022, p. 7).

An example of this is the creation of narrow but continuous firebreaks in the terrain that interrupt the fuel continuity (Vegetation Fire Fighting: Operational Tactics, 2022, p. 6).

In addition to physical separation by control lines, "wet lines" can also be established. These are created by the targeted application of water at the edges of the fire, but usually have only a temporary effect and are dependent on weather conditions (Vegetation Fire Fighting: Operational Tactics, 2022, p.

7). Especially in large-scale and intense fires, the use of water is often insufficient to effectively contain the fire. Therefore, a combination of dry and wet techniques is often used (Vegetation Fire Fighting: Operational Tactics, 2022, pp. 24-25).

The preventive creation of buffer zones or protective strips can also be an effective preventive measure. Controlled burning or the removal of fuel can reduce the fire hazard in specific areas (Vegetation Fire Fighting: Operational Tactics, 2022, p. 7). Regular maintenance and management of forests also help reduce the fuel load and impede the spread of fires.

Controlled burnouts, also known as "pre-fires," represent another important prevention tool. This involves deliberately burning fuel between a control line and the actual fire front to impede the fire's advance. However, this technique requires specialized knowledge and skills from emergency responders to ensure safety and avoid uncontrolled fires (Vegetation Fire Fighting: Operational Tactics, 2022, p. 33).

Choosing the right technique and appropriate measures depends on many variables. In steep terrain, additional safety measures may be necessary.

nical, such as digging trenches to prevent burning materials from rolling away (Vegetation Fire Fighting: Operational Tactics, 2022, p. 22).



The width of the control line must be adapted to the specific conditions (Vegetation Fire Fighting: Operational Tactics, 2022, p. 18).

Techniques for establishing control lines include both manual and mechanical methods. Hand tools, machinery such as forestry mulchers or excavators, as well as the use of fire trucks and aircraft are common tools. An effective communication and leadership structure is essential to ensure the success of these measures (Vegetation Fire Fighting: Operational Tactics, 2022, pp. 18-19).

In conclusion, control lines and preventive measures are essential for the effective suppression of vegetation fires. They require careful planning, well-thought-out tactics, and experienced emergency personnel. By combining various methods and techniques, the spread of fires can be effectively contained, ultimately ensuring the protection of people, the environment, and property.

8.3 Special deployment techniques

Specialized firefighting techniques play a crucial role in fighting vegetation fires, as they can significantly improve the efficiency and safety of firefighting. The choice of appropriate techniques depends on the specific conditions of the fire, including the type of vegetation, weather conditions, and the topography of the terrain.

A frequently used technique is the "pump & roll" approach, in which a tank fire engine continuously drives with the pump running and delivers extinguishing agent via hand-held nozzles. This method enables flexible and mobile firefighting along the flanks of the fire and is particularly effective for low to medium flame lengths, in areas with less dense vegetation and relatively flat terrain (Discussion Paper Critical Vegetation Fire, 2022, p. 6).

The so-called "caterpillar method" is another method. In contrast to the mobile "pump & roll" method, the vehicle remains stationary while hose lines are laid and gradually moved to the next section.

This method is particularly useful when the vehicles are not equipped for mobile firefighting or the terrain is too impassable (Recommendation on training content in vegetation firefighting, 2024, p. 137).

Modular hose laying, also known as "modular hose deployment," is a tactical method that allows for effective firefighting in hard-to-reach or extensive fire areas.



Attack and supply lines are laid gradually to ensure a stable water supply and to fight the fire at different points simultaneously (Recommendation on training content, 2024, pp. 141-145).

Special techniques also include the use of large equipment such as bulldozers or forestry planers to create firebreaks and firebreaks. These "burnbreaks" serve to remove fuel and thus prevent the fire from spreading. Especially in dense forests or heavily overgrown areas,

These measures are crucial to facilitate access for firefighting units and reduce fire intensity (Recommendation on training content, 2024, p. 89).

Another innovative tool is the leaf blower, which is used to remove dry leaves and other combustible material from critical areas.

This technique helps to slow the spread of ground fires and make the routes safer for emergency services (Recommendation on training content, 2024, p. 81).

The technique of controlled burning, also known as "backburn," is a preventative measure in which an area is intentionally burned to deprive the main fire of fuel and prevent its spread.

However, this method requires careful planning and implementation, as it can get out of control in adverse weather conditions or insufficient knowledge of local vegetation (Recommendation on training content, 2024, p. 49).

An integral part of firefighting is aftercare, which includes systematic smoldering searches and extinguishing. Modern equipment such as thermal imaging cameras make this work much easier, as they detect hidden embers that are invisible to the naked eye. This technique is particularly important for preventing the fire from reigniting and for ultimately securing the scene (Recommendation on Training Content, 2024, p. 61).

Additionally, cooperation with supporting units, such as forestry, agriculture, or even the military, is of great importance. This cooperation enables the deployment of additional personnel and specific equipment not typically available to firefighters, such as agricultural equipment or heavy clearing equipment (Recommendation on Training Content, 2024, p. 85-

89).

The special operational techniques mentioned above illustrate that successful vegetation fire fighting requires a combination of well-thought-out tactics, modern technology, and interdisciplinary cooperation. Understanding and



The application of these techniques contributes significantly to maximising the effectiveness of the measures and ensuring the safety of the emergency services.

8.4 Simple Tactical Forest Fire Forecast

The simple tactical forest fire forecast is an essential tool in vegetation firefighting, enabling emergency responders to act in a targeted and effective manner. By predicting fire behavior depending on various influencing factors, appropriate tactics and strategies can be developed and protective measures implemented in a timely manner.

Essential components of such a forecast include analyzing current fire behavior and considering weather conditions, existing vegetation, and topography. These factors, which have been discussed in detail in several sources, have a significant impact on the spread and intensity of fires (Barth, 2024, p. 1).

The immediate weather conditions are crucial for predicting fire behavior. High temperatures, low humidity, and strong winds are indicators of rapid spread and increased intensity of vegetation fires. Wind speed and direction, in particular, significantly influence the dynamics of a fire by increasing the risk of flying sparks and flying sparks, which can lead to new fire sources (Barth, 2024, p. 10). Since the weather can be subject to short-term fluctuations, continuous monitoring and adaptation of operational tactics are necessary.

Fuel also plays a crucial role in wildfire forecasting. The combustibility and moisture content of the material, as well as its spatial arrangement and density, are important. Highly combustible materials such as grasses and small shrubs ignite more quickly and promote fire spread, while heavier materials such as tree trunks take longer to catch fire but can cause more intense and long-lasting fires (Barth, 2024, pp. 12-14). These aspects must be considered in forecast models to make precise statements about the potential intensity and extent of a fire.

Topography also influences fire behavior. Steep slopes favor the rapid spread of fires uphill through convection and radiation. Exposed south-facing slopes, which are subject to more intense solar radiation, often lead to drier conditions and thus a higher fire risk (Barth, 2024, pp. 17-18).



Another aspect of tactical wildfire forecasting is the consideration of historical data on comparable fires in the region. Modern technologies such as geographic information systems (GIS) can be used to create models that predict the behavior of future fires.

Identifying "hotspots" and critical areas that pose an increased risk to the population is also important. This requires close cooperation between emergency services and meteorological services to obtain reliable weather data. Tactical measures, such as the use of the "pump and roll" method in hard-to-reach areas or the use of thermal imaging cameras to detect embers, depend directly on the results of the wildfire forecast (Lindon Pronto et al., 2022, pp. 33-36).

The forecast must be continuously updated and adjusted to respond to changes in external conditions. Constant communication

and coordination between the emergency services on site and the control center is crucial

Only through regular updates and rapid responses can operational tactics be effectively implemented and risks minimized.

In summary, the simple tactical wildfire forecast is a dynamic and multifaceted tool. It enables emergency responders to make informed decisions and respond flexibly to changing situations. Precise consideration and analysis of weather conditions, vegetation, and topography, as well as the use of historical fire data, are fundamental components that together enable effective firefighting and damage minimization.

8.5 Components of the assessment

The assessment of vegetation fires is a complex process consisting of various aspects to develop a comprehensive understanding of the current situation and make informed decisions. A central component of this assessment is the analysis of current fire behavior. This includes the rate of fire spread, flame height and intensity, and smoke development. These factors can be captured using descriptive analytics that combine historical data and real-time information to create an accurate picture of fire dynamics (Thompson et al., 2019, p. 6).

Another important aspect is the assessment of weather conditions, which have a significant influence on fire behavior. These include temperature, humidity, wind speed and direction, and atmospheric stability.

Predictive analytics provides valuable insights by using past data



and current weather forecasts, possible developments in fire behavior are predicted (Thompson et al., 2019, p. 6).

The terrain and vegetation type are also crucial factors. Different vegetation types exhibit different burning behaviors, and the terrain can influence fire spread patterns. A topographical analysis that considers slope gradients and obstacles helps identify potential hazard zones and plan appropriate measures (Thompson et al., 2019, p. 20).

An integral part of the assessment is also the estimation of risks and potential damage. This includes the identification of vulnerable infrastructure, residential areas, and natural resources with the goal of minimizing the danger to human life, property, and the environment. Prescriptive analytics are used here, which, based on descriptive and predictive data, provide recommendations for action for optimal resource allocation and hazard mitigation (Thompson et al., 2019, p. 6).

Coordination and communication between the involved emergency services are also crucial. An efficient information system and a reliable communications network are necessary to quickly and accurately communicate current conditions to all relevant actors. This supports decision-making on the ground and ensures targeted operational tactics (Thompson et al., 2019, p. 13).

In addition to operational assessment, long-term learning and continuous improvement are also of great importance. After each deployment, data and findings should be systematically collected and evaluated to analyze the effectiveness of the measures taken and draw conclusions for future deployments (Thompson et al., 2019, p. 1, pp. 10-11). This contributes to the continuous optimization of assessment processes and the overall efficiency of vegetation firefighting.

The use of modern technologies should also receive special attention. Technologies such as GIS, drones and advanced sensor networks offer new opportunities to collect and analyze comprehensive and precise data in real time. These technological tools should be fully integrated into the assessment process to enable data-driven decision-making (Thompson et al., 2019, p. 14).

Finally, it should be emphasized that the assessment of vegetation fires is based not only on the knowledge and experience of the emergency services, but also on a robust database and the application of advanced analytical methods.



This comprehensive and detailed assessment is essential for the effective control of vegetation fires (Thompson et al., 2019, p. 4, p. 8).





9 Use of extinguishing techniques and firefighting methods in vegetation

Fighting vegetation fires is one of the most difficult and dangerous tasks facing fire departments worldwide. It requires not only specific expertise, but also the use of highly specialized technical equipment and precise coordination of tactics. Vegetation fires often occur in forests, meadows, or other vegetated areas.

They can spread rapidly and have serious consequences for the environment, human health, and infrastructure. Therefore, it is crucial to take a systematic approach to firefighting in such areas.

The role of extinguishing techniques and procedures for fighting vegetation fires cannot be overemphasized. These methods are the foundation of any operational strategy and have a significant impact on the success or failure of firefighting efforts. Vegetation fires are not just about extinguishing the fire; rather, their spread must be controlled, post-fire extinguishing efforts conducted efficiently, and safety aspects rigorously observed. Different geographical and climatic conditions require specific techniques and procedures to meet the diverse challenges. A thorough understanding of these techniques and their applications is therefore essential for every firefighter.

Water is the most commonly used extinguishing agent and plays a central role in fighting vegetation fires. However, its use must be carefully and efficiently planned to achieve the desired results.

and resources. This includes methods such as firefighting water

supply via hoses, the use of firefighting aircraft or helicopters as well as the establishment of temporary water intake points and shuttle traffic.

In addition to water, other extinguishing agents such as foam or fire retardants are also used.

mer are used, which offer specific advantages in certain situations.

Another important aspect is the use of mobile and stationary equipment in vegetation firefighting. Mobile equipment such as portable pumps, fire backpacks, and portable water tanks enable flexible use even in hard-to-reach areas. Stationary equipment, such as permanently installed fire monitors and remote-controlled firefighting systems, offers the possibility of continuous and often automated firefighting.

The post-extinguishing phase is a particularly critical step in the fight against vegetation fires. Here, embers must be detected and removed in order to



to prevent the fire from flaring up again. This requires not only experienced personnel, but also suitable technical aids such as thermal imaging cameras and special tools for working the soil.

The safety of emergency personnel is a top priority. Due to the high level of danger posed by vegetation fires, strict adherence to safety protocols is essential. A comprehensive safety concept, including both preventive measures and rules of conduct for emergency situations, is crucial for minimizing risks. Furthermore, individual protective clothing and equipment should be adapted to the requirements of the respective emergency situation.

Equally important are a formalized exit plan and handover protocol. These elements ensure that the operation is concluded in an orderly and safe manner, and that responsibility is clearly transferred. This is particularly relevant when multiple units or organizations are involved in an operation. A well-structured handover protocol ensures that all necessary information is shared and minimizes the risk of misunderstandings and errors.

Overall, this chapter provides a comprehensive introduction to the techniques and procedures of vegetation firefighting. It highlights the importance of a well-thought-out, structured, and systematic approach to effectively mastering the diverse challenges encountered during an operation. The primary goal of all these measures is to protect human life, minimize environmental and property damage, and successfully fight the fire.

9.1 Use of water and other extinguishing agents

The effective use of extinguishing agents, especially water, is a key component of vegetation firefighting. Water is often used as the primary extinguishing agent because it is widely available and relatively easy to handle. However, its use varies greatly depending on the intensity and type of fire. For low-intensity fires, i.e., with flames up to approximately 0.5 meters long, a direct attack with hand tools or targeted extinguishing is often sufficient.

However, for more intense fires, especially with flame lengths of up to 3.5 meters, more complex tactics and heavier equipment are required (Notes, 2024, p. 5).

Efficiency and economy in water delivery are particularly important, especially in regions with limited access to water resources. Studies show that short, heavy rainfall events are often less effective than longer, lighter rains in adequately moistening vegetation (Module 2, 2022,



(p. 8). This finding highlights the critical importance of using water strategically to achieve the greatest possible impact.

In addition to water, there are a number of other extinguishing agents, such as foam and special chemical additives, that can be used depending on the size and environment of the fire. The use of these agents often depends on their availability and the specific requirements of the fire site. Foam, for example, can help smother the fire and prevent embers from reigniting. This is

particularly important in rough terrain where traditional methods, such as stirring the embers, may not be as effective (Notes, 2024, p. 21).

The use of mobile and stationary fire extinguishing techniques is another crucial Aspect in the fight against vegetation fires. Mobile firefighting techniques, including all-terrain vehicles, enable the rapid distribution of water, especially in difficult-to-access areas (Notes, 2024, p. 9). A key method in this context is the "pump & roll" method, in which vehicles with a water tank remain in motion while simultaneously releasing water (Notes, 2024, p. 8). This methodology promotes dynamic and flexible firefighting, which proves particularly advantageous in difficult terrain.

Stationary equipment is often used to establish control lines, which can be either wet or dry. These lines act as barriers that support fire protection. Wet control lines are often short-term measures, while dry control lines are preferred for more sustainable firefighting (Notes, 2024, p. 22). The targeted establishment of such lines is of great importance to prevent the spread of fire and ensure that ground fires are extinguished in a controlled manner.

In addition, cooperation with other stakeholders and the use of modern technologies, such as drones, play an important role in the timely detection and suppression of fires. These technologies, especially modern sensor technologies, facilitate the rapid localization of fire sources and support effective planning of firefighting agents. Such technological advances complement traditional methods and increase safety for emergency responders.

It is crucial that the combination of different extinguishing agents and techniques is tailored to the specific fire situation. Continuous training and continuing education of emergency personnel are therefore essential. Through regular training and realistic scenarios, knowledge of the most effective extinguishing techniques can be deepened and adapted to specific operational conditions (Notes, 2024, p. 10). This strengthens the resilience of emergency personnel to different



Fire conditions and contributes to more effective and safe firefighting.

In summary, the use of water and other extinguishing agents depends heavily on the specific environmental conditions and the type of vegetation fire. Effective strategies combine the targeted use of water, the use of supplementary agents such as foam, the use of modern technologies, and the training of emergency responders in a variety of techniques. This integrated approach enables a flexible firefighting strategy that effectively addresses the diverse challenges of vegetation fires.

can.

9.2 Mobile and stationary equipment

Fighting vegetation fires requires the targeted deployment of a variety of mobile and stationary equipment to ensure effective extinguishing measures. The selection and application of this equipment varies greatly depending on the accessibility of the site, the type of fire, and the available resources.

Mobile equipment includes handheld firefighting devices such as fire backpacks and motorized devices. Fire backpacks typically have a volume of 16 to 25 liters and are equipped with double-action pumps that allow continuous water delivery directly to the root of the flame. These devices are particularly useful for vegetation fires, where larger vehicles often cannot reach (Notes for Practical AE Equipment Knowledge/Hand Tools, 2024, p. 10). Fire beaters and fire swatters are also used to extinguish flames through mechanical action. These methods are effective for shorter flame lengths and enable precise firefighting on the ground.

Additionally, motorized spray, fan, or fog devices are used, enabling efficient extinguishing of different fire materials and intensities. These devices can deliver a larger amount of extinguishing agent in a shorter time and are particularly useful during intense fire phases. The combination of fire backpacks and motorized devices can increase the efficiency of fire extinguishing.

significantly increase the effectiveness of firefighting because they are both flexible and powerful (Notes for practical AE equipment knowledge/hand tools, 2024, p. 10).

Stationary equipment, such as tank fire engines, often represents the primary source of water supply for vegetation fires. These vehicles are capable of transporting large volumes of water and delivering it via various types of nozzles or monitors. The hoses can be laid over long distances.



to reach hard-to-reach fires. In combination with mobile units, stationary equipment enables a continuous water supply and targeted firefighting, which increases the efficiency and effectiveness of the firefighting

measures improved.

Innovative technical solutions offer significant advantages, especially in densely forested and difficult-to-access areas. Modern technologies such as mobile, self-contained extinguishing systems operated by drones or unmanned vehicles can penetrate dangerous or inaccessible areas and deliver targeted extinguishing agents. Such systems complement traditional firefighting vehicles and significantly expand their operational capabilities.

The combination of modern technologies with traditional extinguishing agents plays a key role, even in difficult-to-reach areas. Portable firefighting equipment, including fire backpacks and handheld sprayers, is crucial for fighting smaller fires (Notes for Practical Firefighting Equipment/Hand Tools, 2024, p. 10). Stationary equipment such as motorized pumps or permanently installed water storage tanks also ensure a long-term water supply and facilitate replenishment for mobile units.

A well-thought-out safety concept is crucial to ensure that both mobile and stationary equipment can be used effectively without compromising the safety of emergency personnel. Clear communication channels and regular training are essential to ensure the optimal use of resources.

In summary, the effective use of mobile and stationary equipment plays a crucial role in vegetation firefighting. The combination of the flexibility of mobile equipment and the comprehensive capacity of stationary equipment enables emergency responders to respond effectively to the various challenges. Advancing technological developments will continue to play a key role in further improving the efficiency of firefighting measures and meeting the increasing demands of firefighting.

9.3 Follow-up extinguishing work and search for embers

Follow-up fire extinguishing activities, as well as the identification and extinguishing of embers, are essential measures in vegetation firefighting. They are crucial to ensuring that a fire is completely extinguished and does not reignite (Module 4 VEGETATION FIRE - FOLLOW-UP FIGHTING WORK, 2022, p. 3).



The first step in fire extinguishing is to identify and completely extinguish all embers. These embers can be hidden in various materials such as dead wood, stumps, or deep layers of ash.

Identifying and dealing with these embers requires special attention as such as the use of multiple senses – sight, hearing, smell and touch (Module 4 VEGETATION FIRE - FOLLOW-UP FIRE EXTINGUISHING WORK, 2022, p. 11).

Different techniques are used for the follow-up extinguishing work.

A proven method is to expose the embers, followed by smothering them with sand, earth, or water. Wetting agents are added if necessary to enhance the cooling effect (Module 4: Vegetation Fire - Post-Extinguishing Work, 2022, p. 8).

Post-extinguishing without water is becoming increasingly important, especially in situations of water scarcity. In these cases, hand tools such as spades, hoes, or rakes, as well as natural materials such as sand and earth, are used (Notes on Practical AE Object Defense/Post-Extinguishing (Basics), 2024, p. 13).

A key component of post-extinguishing activities is the systematic securing and monitoring of the fire site. For this purpose, the fire area is often divided into zones, which are then specifically searched for embers (Module 4 VEGETATION FIRE - FOLLOW-UP FIREFIGHTING WORK, 2022, p. 14). Emergency personnel work in groups to ensure that no areas are left unattended, significantly reducing the risk of a fire reigniting.

In difficult terrain, follow-up firefighting operations pose a particular challenge. Special safety measures are necessary to protect emergency personnel. Cooperation with other organizations, such as the mountain rescue service, may be necessary in such situations (Module 4 VEGETATION FIRE - FOLLOW-UP FIREFIGHTING WORK, 2022, p. 17).

In addition, it is crucial to monitor areas outside the control line. Rolling burning material or sparks could ignite new fires (Module 4 VEGETATION FIRE - FOLLOW-UP FIRE-EXTINGUISHING WORK, 2022, p.

7) Constant vigilance is required, and control lines may need to be improved to prevent the fire from spreading.

The fire command plays a central role in the follow-up firefighting operations, as they decide which measures are necessary and how they are implemented (Module 4 VEGETATION FIRE - FOLLOW-UP FIREFIGHTING WORK, 2022, p. 6). Numerous factors must be considered during planning and execution, such as weather conditions, the type and condition of the vegetation, and the time of day, to ensure the operation is effective (Module 4 VEGETATION FIRE - FOLLOW-UP FIREFIGHTING WORK, 2022, p. 5).



Another important aspect of post-extinguishing activities is the continuous monitoring and adaptation of measures to changing conditions. This includes observing and, if necessary, improving control lines and requesting additional emergency personnel if necessary (Module 4 VEGETATION FIRE - POST-EXTINGUISHING WORK, 2022, p. 6). It is crucial that emergency personnel continue to act prudently and with safety in mind even after active firefighting activities have been completed to minimize the risk of injury.

Overall, it is clear that post-fire extinguishing activities and the search for embers are complex tasks that require precise planning and careful execution. They are an integral part of vegetation firefighting and contribute significantly to ensuring the safety of emergency personnel and sustained control of the fire (Guidelines for Practical AE Object Defense/Post-Fire Extinguishing (Fundamentals), 2024, p. 5).

9.4 Security

Safety during vegetation firefighting plays a key role in protecting both the emergency responders and the affected population. It is crucial that emergency responders are protected from the diverse hazards in the area through systematic risk analyses and appropriate precautionary measures.

be protected.

A key safety principle in these situations is the application of the so-called LACES rules (Lookouts, Awareness, Communications, Escape Route, Safety Zones). These principles are of great importance for the safety of emergency personnel and must be continuously observed and implemented. Continuous monitoring and assessment of the area of operation and fire behavior are particularly important. Responsible incident commanders should set up suitable observation posts (lookouts) to alert the firefighters to potential hazards early on (Barth, 2024, pp. 23-24).

Clearly defined and communicated escape routes are also essential. These routes lead emergency personnel safely out of the danger zone and end in safety zones that must be pre-determined and appropriately marked. Safety zones should be free of vegetation and flammable materials and ideally located in open areas or in areas with a low fire load (Barth, 2024, p. 30).

In addition, it is necessary for emergency personnel to undergo regular safety training and exercises in order to be able to react appropriately in emergency situations. This includes knowledge and implementation of the



Guidelines for behavior in critical moments, such as timely warning and an orderly retreat to the safety zone. In the event of sudden dangers, such as a change in the direction of the fire, clear retreat orders must be given and followed immediately (Barth, 2024, pp. 31-32).

Another important element is protection against respiratory toxins and pollutants. Fire gases such as carbon monoxide, carbon dioxide, and particulate matter pose significant health risks. Therefore, it is crucial to take protective measures such as wearing appropriate personal protective equipment (PPE) and using CO detectors seriously. Emergency personnel should generally try to work in smoke-free areas and adjust their position accordingly when wind direction changes (Barth, 2024, p. 34).

Furthermore, it is important to ensure that emergency personnel do not expose themselves to excessive physical strain in the field. Risks such as dehydration, heatstroke, and exhaustion can be reduced by adequate fluid intake and regular breaks. It is recommended to drink sufficient fluids en route to the scene and to take appropriate protective and recovery measures in extreme weather conditions (Barth, 2024, p. 14).

Particular care is required when handling technical equipment and the environment, especially in areas suspected of containing munitions or with overloaded equipment and vehicles, in order to avoid explosion risks and technical failures (Barth, 2024, pp. 19-20).

Optimizing operational management and smooth communication are also key safety elements. Effective leadership and clear communication structures enable early detection of threats and the implementation of appropriate measures. This includes the use of modern communication tools and regular briefings (Barth, 2024, p. 28).

Overall, safety in vegetation firefighting is closely linked to the ability of firefighters to respond appropriately to dynamic and potentially risky changes in fire behavior and the environment. Leaders therefore have a responsibility to continuously review and comprehensively apply safety measures. Proper execution of retreat measures, full utilization of LACES principles, and continuous leadership and communication are critical to minimizing risks.

and to protect the emergency services.



9.5 Exit concept and handover protocol

A well-thought-out exit plan and a binding handover protocol are crucial components for successful vegetation firefighting. The exit plan encompasses all necessary steps to safely conclude an operation, while the handover protocol ensures that all relevant information is structured and clearly documented to enable a seamless transition to subsequent emergency responders or responsible authorities.

First, as part of the exit plan, it is essential that all personnel are thoroughly checked before leaving the scene. This serves to identify and eliminate potential hazards such as embers. Follow-up extinguishing activities and the search for embers are particularly important in this regard. For example, experience from the 2022 wildfire season showed that intensive follow-up extinguishing activities were necessary to prevent the fire from reigniting (Report on the 2022 Wildfire Season, p. 28). The risks posed by hidden embers underscore the importance of thorough follow-up care.

Another key aspect of the exit concept is the systematic decontamination of the equipment used and the personal protective equipment (PPE) of the emergency personnel. This helps minimize health risks and ensure that the equipment is ready for future deployments (post-deployment debriefing, p. 2). These measures not only maintain the operational capability of the equipment but also increase the safety of the emergency personnel.

The handover protocol serves as a binding document that records all important information about the mission. This information can include situation descriptions, measures taken, identified hazards, and remaining tasks. Structured documentation, as required in debriefings, supports a comprehensive analysis of the events (Debriefing, p. 1).

Particular attention should be paid to ensuring a smooth flow of information during the site handover. A clearly structured transfer of all relevant data minimizes risks and enables seamless follow-up measures.

Problems such as communication disruptions that occurred during the 2022 wildfire season highlight the need for optimized coordination and information transfer (2022 Wildfire Season Report, p. 15).

The implementation of specific standard procedures for the exit concept and the handover protocol can be significantly improved through modern technological tools. Digital systems for real-time documentation and communication play a key role in this. Experience from previous deployments shows



that digital radio for the Federal Office for Public Safety (BOS) is important but also susceptible to interference (Report on the 2022 Forest Fire Season, p. 15). Expanding these systems to include redundant communication channels could compensate for such weaknesses and increase the reliability of data transmission.

In addition, the handover protocol should also include information on future monitoring of the incident site and follow-up measures. In areas with potential risks, such as areas suspected of containing explosive ordnance, special instructions for monitoring and securing should be provided. The use of technological solutions, such as remote-controlled systems, could help to conduct continuous hazard monitoring (2022 Wildfire Season Report, p. 42).

In summary, a sound exit plan combined with a detailed and understandable handover protocol is essential for effective vegetation firefighting. These measures not only help prevent re-ignition but also increase the safety and efficiency of emergency responders. By integrating proven tactical concepts and modern technologies, the quality and reliability of these protocols can be further improved, ultimately leading to an optimized operational strategy and safe handling of vegetation fires.



10 Communication and coordination in action

In the dynamic and risky environment of a vegetation fire, the role of communication and coordination is crucial. These two aspects are crucial for the safety of emergency responders, the efficiency of firefighting, and the minimization of damage in affected areas. Communication and coordination should not be viewed as isolated concepts; rather, they influence each other and interact in many ways. Together, they form the foundation for successful vegetation firefighting and are crucial in all phases of the operation, from the alarm to the debriefing and analysis of the incident.

Communication is not only the transfer of information between the emergency services on the ground, but also strategic coordination at higher command levels. It is important that the exchange of information is clear, precise, and timely in order to create a common picture of the situation and keep everyone involved up to date. Various communication tools are used for this purpose.

be it radio devices, hand signals or digital platforms. The quality
The quality and reliability of these means of communication can often make the difference between
between a successful and a failed operation. A communication failure or misunderstanding can have serious
consequences, ranging from inefficient use of resources to dangerous situations for the emergency services.

Coordination refers to the structured and targeted management of the various operational components. This includes the assignment of tasks to different teams, the integration of technical resources, and cooperation with other organizations. A coordinated approach enables the efficient use of resources, avoids bottlenecks, and promotes synergies between different forces. The role of leadership is crucial here.

Managers must be able to make quick, well-founded decisions, give clear instructions and maintain an overview of the complex and often chaotic situation.

Another important aspect of coordination is situational awareness, both at the individual and command levels. Situational awareness means having a clear understanding of the current situation, available resources, potential threats, and planned actions. This requires constant review and adjustment of operational tactics based on current developments. Only with a high level of situational awareness can it be ensured that all parties involved can work together effectively and respond quickly and appropriately to unforeseen events.



The education and training of emergency personnel in the areas of communication and coordination are essential. Regular exercises and realistic scenarios help optimize communication channels, deepen understanding of the operational structure, and improve responsiveness to various situations. Such exercises can simulate realistic conditions and help identify and correct weaknesses in the communication and coordination system at an early stage.

Modern technological developments offer additional support for communication and coordination. Digital systems, such as drones for situational reconnaissance, GPS-based position monitoring, and special operations control systems, enable more accurate and faster information gathering and processing. These technologies assist leaders in decision-making and contribute to improving overall coordination. Nevertheless, the human component remains.

component is essential. Technology can facilitate communication and coordination. However, it cannot completely replace them. Direct interaction between people remains indispensable, especially in stressful and unpredictable operational situations.

Overall, it is clear that integrating communication and coordination into the operational process is an ongoing process that requires careful planning, constant adaptation, and ongoing training. Only through a harmonious and efficient combination of these two elements can vegetation fire fighting be successfully implemented and the risks to people, animals, and the environment minimized.

This places high demands on the skills and interaction of all those involved and at the same time highlights the extraordinary importance of both factors for the overall success of an operation.

10.1 Leadership and Organization

Effective leadership and organization in vegetation firefighting is essential to ensure both the safety of emergency personnel and the efficiency of the response. A clear leadership structure and precise role assignments are essential to meet the diverse challenges and dynamic risks of such operations.

A well-structured management organization begins with the formation of command and administrative staffs, which are particularly essential in large vegetation fires. These staffs are established at the state and municipal levels to ensure overall coordination and resource management (Report of the Expert Commission on Forest Fires Summer 2022 in Saxony, 2023, p.

77). Their task includes assessing the situation and ensuring the



Communication channels, with early activation necessary to maintain functionality. The guidelines for disaster management leadership must be adapted accordingly to enable smooth integration of the various administrative levels and departmental groups (Report of the Expert Commission on Forest Fires Summer 2022 in Saxony, 2023, p. 77).

The creation of a cross-resource administrative staff at the state level is particularly important. This measure enables cross-ministry action in critical situations such as forest fires and ensures consistent public relations (Report of the Expert Commission on Forest Fires in Saxony, Summer 2022, 2023, p. 79). To promote interministerial cooperation and reduce bureaucratic hurdles, regular staff exercises should be conducted to test communication channels (Report of the Expert Commission on Forest Fires in Saxony, Summer 2022, 2023, p. 95). This contributes to automating processes and enabling rapid response in an emergency.

The on-site commander is responsible for making tactical decisions and adapting operational tactics to the dynamic situation. Clear communication channels and consistent terminology are crucial to avoid misunderstandings (PP LG-No. 612 Veg.-Bbk FÜ1+Ausbilden intensiv AE-Gefahren Sicherheit-G 1-24.pdf, 2024, p. 28). Compliance with safety principles, as defined in the LACES principle, is essential. Every commander must ensure that the situation on site is accurately recorded and regularly communicated to the higher command level (PP LG-No. 612 Veg.-Bbk FÜ1+Ausbilden intensiv AE-Gefahren Sicherheit-G 1-24.pdf, 2024, p. 28).

Another crucial aspect is resource management. Large-scale vegetation fires, such as those that occurred in Saxony in the summer of 2022, demonstrated the need for central coordination of resource allocation. The Joint Reporting and Situation Center of the Federal and State Governments (GMLZ) plays a key role in this by coordinating the request and provision of forces and resources across state borders (Report of the Expert Commission on Forest Fires Summer 2022 in Saxony, 2023, p. 83). However, there is still a need for clear responsibilities and regulations to effectively manage the deployment of resources.

Modern technologies, such as drones and thermal imaging cameras, are also central to successful command and control. These devices enable the rapid identification of critical areas and the targeted control of emergency personnel. Furthermore, armored or remotely controlled vehicles can be deployed on munitions-contaminated areas to increase the safety of emergency personnel (Report of the Expert Commission on Forest Fires Summer 2022 in Saxony, 2023, p. 103).



Cooperation with air support measures is another important element of command organization, especially in difficult-to-access and topographically challenging areas. Helicopters with external firefighting water tanks and coordination with the police helicopter squadron are crucial for fighting fires as early as possible and preventing widespread spread (Report of the Expert Commission on Forest Fires Summer 2022 in Saxony, 2023, p. 93).

In summary, leadership and organization in combating vegetation fires requires a well-structured and flexible management organization that integrates modern technologies and clear communication channels. Regular exercises to improve interaction between different administrative levels and timely resource coordination are essential components.

Only in this way can efficient and safe vegetation fire fighting be ensured become.

10.2 Debriefing and learning processes

Structured debriefing of operations and exercises represents an essential aspect of the continuous improvement and professionalization of vegetation firefighting. The goal is to learn from the experiences gained and the measures implemented in order to identify weaknesses and optimize processes on this basis. A well-organized debriefing not only promotes the further development of tactical procedures but also contributes significantly to the psychological relief of emergency personnel (Schulze / HFUK Nord, 2020, p. 1).

For the successful conduct of debriefings, it is important to create a pleasant and open atmosphere in which all participants can express their observations and points of criticism, regardless of hierarchical structures.

This approach allows emergency responders to thoroughly reflect on the course of the operation. It is particularly important to pay attention to unusual behavior in individuals after stressful operations. In such cases, emergency psychosocial care may be necessary to identify and treat potential post-traumatic stress early (Schulze / HFUK Nord, 2020, p. 1).

Technical follow-up also plays a crucial role. Vehicles must be refueled, and the equipment and equipment used must be checked and cleaned. Breathing apparatus, which is often used in vegetation fires, in particular, requires a thorough inspection. Consumables must also be replenished, and damaged or contaminated equipment must be repaired. These measures are essential to ensure operational readiness.



and safety for future operations (Schulze / HFUK Nord, 2020, p. 2).

At the organizational level, follow-up includes comprehensive incident documentation, including information on personnel, injuries, hazardous substances, and near misses. This documentation is essential for deriving future improvements. Furthermore, discussions and evaluations of the incident, both with all involved forces and at the management level, are important for conducting a comprehensive analysis of the incident. Adapting the risk assessment plays a central role here, as it not only affects the immediate safety of the emergency personnel but also influences strategic decisions for future incidents (Schulze / HFUK Nord, 2020, p. 2).

Another essential component is the analysis and evaluation of problems encountered during the operation. Aspects such as access, reconnaissance, command, cooperation, radio communications, and supply should be systematically examined to identify optimization opportunities. It is advisable to develop common checklists and standard procedures to increase the efficiency and safety of future operations.

One important effect of debriefings is their role in team management and promoting cohesion. Open and constructive communication strengthens trust among emergency personnel and contributes to creating a positive working environment. Mutual understanding and the joint management of challenges can sustainably improve team dynamics.

In summary, structured debriefing and learning processes are indispensable tools in vegetation firefighting. They not only enable the continuous development of tactical and technical skills but also contribute significantly to the mental and physical health of emergency personnel. By systematically evaluating incidents and integrating feedback into training manuals, fire departments can optimize their operational strategies and better prepare for future challenges.

10.3 Collaboration and communication within the team

The cooperation and communication within a task force are crucial for the success and safety of fighting vegetation fires. Well-coordinated teamwork enables the emergency services to respond quickly and effectively to the diverse challenges and dynamic developments



to react during an operation. Both formal and informal communication structures play a central role.

A key element of collaboration is the clear allocation of roles and tasks within the team. Every firefighter should know exactly which specific tasks are assigned to them. This clear assignment not only promotes the efficiency of workflows but also minimizes the risk of errors and misunderstandings, which can have serious consequences in stressful situations. Furthermore, a precise allocation of roles increases trust and reliability within the team, which has a positive impact on overall performance.

mance affects.

Communication between team members must be direct and concise.

Information should be communicated quickly and clearly to avoid delays and misunderstandings. Various communication

Communication tools such as radios, hand signals, and mobile communication platforms are used. Radio discipline is particularly important: the ability of emergency personnel to send short, understandable radio messages.

Another important aspect is the willingness to actively give and receive feedback during the operation. This includes both suggestions for improving the operational strategy and addressing difficulties or uncertainties. An open feedback culture enables potential problems to be identified and resolved early, before they become major challenges. Post-operation debriefings support this process by bringing together all involved personnel to reflect on the course of the operation and to express criticism and suggestions for improvement. This should take place in a pleasant atmosphere (Schulze / HFUK Nord, 2020, p. 1).

The use of standardized communication protocols is also crucial. Uniform standards and codes must be used, especially when collaborating with different organizations and units, to avoid misunderstandings. Uniform communication protocols promote effective coordination between different operational forces and command levels, which in practice contributes to faster and more precise implementation of orders and instructions.

In an interdisciplinary team comprised of diverse professionals, communication across disciplines is essential. This includes integrating specialists such as meteorologists or technicians who contribute valuable knowledge and perspectives. This interdisciplinary collaboration enables the team to make more comprehensive and informed decisions, especially in complex and unexpected situations.



Another important factor is psychological support and stress management within the team. Vegetation fires are often very stressful and dangerous, which can cause additional tension and stress both individually and within the group. Regular short breaks and mindful and supportive behavior among each other can help reduce psychological stress and mentally stabilize the emergency responders.

In practical terms, this means that all team members commit to open and respectful communication. This commitment begins with the preparation for the mission, continues through the execution of the mission, and ends with the debriefing. Teamwork is thus a dynamic process characterized by continuous dialogue and mutual support.

In summary, effective teamwork and communication are essential for successfully managing vegetation fires. By establishing clear structures, fostering an open communication culture, using standardized protocols, and integrating psychological support measures, fire departments can increase their efficiency and significantly improve the safety of all involved.



11 Emergency planning and preparation for vegetation fires

Emergency planning and preparation are crucial for the effective suppression of vegetation fires. Given the unpredictability and destructive potential of such fires, thorough and systematic preparation is necessary to minimize both the risk to people and economic losses. Vegetation fires can be triggered by a variety of factors, whether natural causes such as lightning strikes or human activities. Since changing weather conditions, different vegetation types, and specific topographical features play a role in each fire, each case requires specific strategic approaches and emergency plans.

The beginning of effective emergency planning lies in a thorough analysis of potential risk areas. This includes identifying high-risk areas, as well as the location and accessibility of water resources. It is also important to consider hazard areas such as residential areas, industrial facilities, and critical infrastructure. When developing an emergency plan, all relevant information must be gathered to provide a comprehensive risk assessment. Geographic information systems (GIS) and other modern technologies are of great importance here, as they facilitate data collection and analysis and enable more precise planning.

Another fundamental component of emergency planning is the establishment of clear chains of command and communication protocols. Such structures ensure that, in an emergency, all parties involved—from on-site emergency responders to coordination centers—can act efficiently and coherently. A well-organized command center, equipped with the necessary communications tools, is essential for monitoring the fire's progression and adapting operational strategies in real time. Interoperability between the various organizations involved, such as fire departments, civil protection agencies, and police, must be ensured to enable smooth collaboration.

Preventive measures and regular training are also essential aspects of preparedness for vegetation fires. Emergency personnel must be continuously trained in specific techniques to be able to act effectively and safely in the event of a fire. This includes theoretical knowledge of fire behavior and equipment handling, as well as practical exercises that simulate realistic scenarios. Training programs should also consider the psychological aspects of dealing with extreme stress situations to promote the mental resilience of emergency personnel.

Emergency plans must be dynamic and flexible in order to respond to changes in the risk profile and new insights from past operations.



Regular reviews and updates of plans are necessary to ensure they remain up-to-date. Evaluations following real-life operations and exercises provide valuable data that can be used to identify weaknesses and make improvements. Continuous learning and adapting to new challenges are crucial to increasing the efficiency and effectiveness of vegetation firefighting.

In addition to operational planning, logistics also plays a critical role. The provision and maintenance of resources such as fire engines, hoses, pumps, personal protective equipment, and extinguishing agents must be carefully organized. Effective resource management ensures that all necessary materials and equipment are quickly available in an emergency. Both warehousing and logistical coordination during the operation are of paramount importance to ensure uninterrupted supplies.

Last but not least, public involvement in emergency planning and preparedness is essential. Awareness campaigns and training programs for citizens can increase awareness of the risks of vegetation fires and strengthen willingness to support emergency responders. Informed and prepared citizens can actively contribute to minimizing the risk of fire outbreaks and increasing the efficiency of firefighting efforts.

In summary, emergency planning and preparedness requires a complex and multifaceted approach that requires close cooperation between various stakeholders, continuous training, and flexibility. Only through comprehensive and forward-looking planning can the enormous danger potential of vegetation fires be effectively addressed and the protection of human life and property ensured.

11.1 Strategic deployment planning

Strategic operational planning is a crucial element of emergency preparedness in the context of vegetation fires. It lays the foundation for all operational measures and is essential for responding quickly and effectively to unexpected fire events. The goal of strategic planning is to identify potential fire risks early and develop appropriate firefighting and prevention measures.

A central aspect of this planning is risk assessment. This involves analyzing and evaluating the fire hazard in specific regions, taking into account various factors such as weather conditions, vegetation, and topographical features. Modern decision support systems, such as Risk Management Assistance,



(RMA) program. This program combines local knowledge with scientific analysis tools to better estimate the probabilities and impacts of fire spread (Calkin et al., 2021, pp. 2-4).

Another important component of strategic planning is the establishment and maintenance of infrastructure such as firebreaks, water intake points, and access routes. These measures are crucial for more effective fire containment and suppression. In regions with a high risk of wildfires, it is especially important to regularly inspect existing infrastructure and, if necessary, expand it to ensure an optimal supply of firefighting water in an emergency (Recommendation, 2024, p. 45).

Coordination and cooperation between different actors form a further
Another essential pillar of strategic operational planning. The involvement of local authorities, fire departments, forestry services, and, if necessary, the military or other specialized forces is crucial for rapid and coordinated action in the event of a fire. International cooperation, such as within the framework of the European Civil Protection Mechanism, can also be helpful, as they provide specialized units and equipment when a large-scale vegetation fire threatens (Recommendation, 2024, p. 105).

Effective risk management also requires continuous adaptation and optimization of operational strategies based on collected data and the evaluation of past incidents. Debriefing and analysis of previous fire incidents play a central role in this process. Through detailed investigation of fire progression and the effectiveness of measures taken, valuable insights can be gained that can be incorporated into future operational plans. This iterative learning process contributes to continuously improving the efficiency and safety of fire operations (Calkin et al., 2021, p. 16).

Targeted preparation for specific operational scenarios is equally important. This includes training and ongoing professional development of emergency personnel, as well as regular exercises to test and refine the practical implementation of theoretically developed operational plans. These exercises should be as realistic as possible and cover various scenarios to prepare emergency personnel for different challenges. Simulation tools can also be used to virtually recreate fire situations based on real data, thus improving the response capabilities of emergency personnel (Recommendation, 2024, p. 21).

In addition, strategic operational planning includes the integration of modern technologies and analysis tools. For example, emergency personnel can be trained on specific



Access fire analysis maps that indicate risk areas and potential control points. Such maps are based on historical data and current weather conditions and provide valuable support in developing and communicating containment strategies (Calkin et al., 2021, p. 10).

In summary, strategic operational planning is a complex process that requires thorough risk assessment, the establishment and maintenance of infrastructure, the coordination of various stakeholders, and the continuous training and development of emergency personnel. The use of modern technologies and the continuous optimization of strategies can significantly increase the effectiveness and safety of vegetation firefighting.

11.2 Training and exercises

Training and exercises are essential components of emergency planning and preparedness for vegetation fires. They prepare emergency responders for the specific challenges and hazards that can arise during such fires. The efficiency and effectiveness of firefighting depend critically on the quality of training and the regular conduct of realistic exercises.

A vegetation firefighting training program should, on the one hand, offer comprehensive theoretical content and, on the other hand, promote practical skills through regular exercises. The theoretical content includes basic vegetation firefighting concepts. These include concepts such as fire fringe, flanking, and spot fire, which help emergency personnel better orient themselves in the area of operation and understand the dynamics of the fire (Recommendation on Training Content in Vegetation Firefighting, p. 9).

Practical exercises must simulate realistic scenarios in which firefighters can train under conditions as authentic as possible. These exercises cover both the handling of equipment and the use of specific tactics, such as flanking or modular hose laying (Recommendation on Training Content in Vegetation Fire Fighting, p.

137). In addition, exercises on follow-up extinguishing work and ember searches should be conducted to prepare emergency services for the complete extinguishing of fires (Recommendation on training content in vegetation fire fighting, p. 61).

Another important topic in training is the safety concept. This includes the understanding and application of the "LACES" principle, which serves as a safety guideline in operations (recommendation on training content in the



Vegetation Fire Fighting, p. 177). In the exercises, safety zones and escape routes must be clearly marked and tested to ensure quick and safe escape routes in an emergency.

In addition, emergency personnel should also be trained in the use of modern technologies and technical aids. These include, for example, weather recording devices such as the Belt Weather Kit, which provides precise weather data and thus enables better planning and adaptation of operational tactics (Recommendation on Training Content in Vegetation Fire Fighting, p. 73).

Effective preparation for vegetation fires also requires regular interdisciplinary exercises. Firefighters should train together with other relevant stakeholders, such as forestry personnel and farmers, to ensure coordinated action in an emergency (Recommendation on Training Content in Vegetation Fire Fighting, p. 85).

It is also important that training and exercises also address the physical and psychological stresses of the emergency responders. Vegetation fires are often challenging and can cause enormous stress. Therefore,

Psychological support and stress management programs should be part of the training (Recommendation on training content in vegetation fire fighting, p. 109).

Overall, comprehensive training and regular exercises enable emergency personnel to consolidate their skills and expand their knowledge. This is crucial for the safety and effectiveness of fighting vegetation fires. Continuous evaluation and adaptation of training programs based on the latest scientific findings and technological developments is essential to meet current requirements.

In this way, firefighters can be optimally prepared for the diverse challenges associated with vegetation fires (Recommendation on training content in vegetation fire fighting, p. 161).

11.3 Resource management and logistics

Effective resource management is crucial in vegetation firefighting. This involves not only procuring and managing the necessary equipment, but also ensuring that available resources are used efficiently. Especially in large and long-lasting forest fires, the ability to mobilize resources in a timely and sufficient manner has a significant impact on the success of firefighting efforts.



A central aspect of resource management is the inventory and regular review of existing equipment. Procurement measures have been intensified in recent years, with the acquisition of special tank fire engines for forest fires since 2008, which have proven particularly effective (2022 Forest Fire Season Report, p. 49). The large-scale procurement of additional tank fire engines highlights the need for ongoing adaptation and expansion of equipment to meet increasing demands (2022 Forest Fire Season Report, p. 51).

Efficient provision and distribution of resources also requires well-organized logistics. This includes the timely availability of emergency resources such as tank fire engines, water pumps, and personal protective equipment. The coordination of supply and logistics is often supported by specialized units capable of responding quickly to changing operational conditions. One example of this is the establishment of temporary outposts for airborne support operations, which enable the safe conduct of firefighting operations (Report on the 2022 Wildfire Season, p. 51).

The involvement of farms and contractors has proven to be a valuable complement to traditional forest firefighting. These actors can contribute to the containment of vegetation fires through the use of water barrels and soil tillage technology (2022 Forest Fire Season Report, p.

52). It is important to have targeted planning and a better overview of agricultural resources. One way to improve coordination would be to implement an online database accessible to emergency response authorities (2022 Forest Fire Season Report, p. 52).

In addition to technical equipment, the food and accommodation of emergency personnel also play a crucial role. During long-term operations, it is important to ensure that emergency personnel are adequately supplied and protected.

Special shelter and hygiene kits could be provided to enable self-sufficient ad hoc accommodation (2022 Wildfire Season Report, p. 51).

The increasing challenges posed by vegetation fires underscore the need for flexible and well-coordinated resource management. This includes not only material resources, but also strategic planning and training for the stakeholders involved. Training measures for vegetation firefighting, such as those offered by the State Office for Road Maintenance (LSTE), aim to enable leaders to effectively coordinate both airborne and ground-based measures (Report on the 2022 Forest Fire Season, p. 43).



In summary, resource management and logistics are central components of successful operational planning in vegetation firefighting. Continuously adapting and expanding equipment, integrating external stakeholders, and ensuring food and accommodation for emergency personnel are essential elements. Well-structured and coordinated resource management contributes significantly to the effectiveness and safety of operations and helps successfully overcome the challenges posed by vegetation fires.





12 Orientation in the terrain

Terrain orientation plays a crucial role in vegetation firefighting and is fundamental to the effective and safe deployment of firefighters. Emergency personnel are often faced with the challenge of navigating in confusing and expansive areas, which is particularly important when fighting vegetation fires. This chapter therefore addresses the importance and methods of terrain analysis to ensure efficient operational planning and execution.

In the often complex and dynamic conditions in which vegetation fires occur, it is essential for emergency responders not only to understand fire dynamics but also to possess excellent navigation skills. Precise terrain orientation enables emergency responders to trace the fire's spread, identify strategic points for firefighting, and make safety-relevant decisions.

Misjudgments or disorientation can have serious consequences, both for emergency responders and for the affected environment and population. Therefore, the ability to analyze terrain is an essential skill, comprehensively covered in the vegetation firefighting training manual.

In addition to the basic ability to navigate the terrain, various techniques and aids are used. These range from traditional

Methods such as the recognition of natural terrain features, as well as modern technical aids such as GPS and digital maps, are used. This variety of orientation options is necessary because vegetation firefighting often takes place in difficult-to-access areas where conventional navigation techniques reach their limits. Especially in remote or difficult-to-see regions, the ability to orient oneself using natural landmarks can be life-saving.

tend to be.

Another important aspect of terrain analysis is precise communication and coordination among emergency responders. Identifying and utilizing common landmarks facilitates coordination and collaboration both within teams and with other organizations involved in firefighting. Efficient communication regarding positions and movements in the terrain is essential to ensure the safety of emergency responders and enable coordinated firefighting. In this context, the definition and marking of danger zones is also crucial for early risk detection and the implementation of necessary safety measures.

The chapter on terrain analysis aims to provide readers not only with theoretical knowledge, but also with practical skills and



The presentation of best-practice examples and specific case studies is intended to help apply theoretical knowledge to real-life scenarios and thus sustainably strengthen the operational skills of firefighters.

Furthermore, the importance of continuous training and education in terrain analysis is emphasized. Vegetation fires are often unpredictable and can change rapidly, requiring constant adaptation and updating of navigation skills. Firefighters must be able to

to orientate quickly and efficiently in both familiar and unfamiliar areas.
and respond flexibly to unexpected changes. Training should therefore be practical and offer emergency personnel regular opportunities to review and improve their orientation skills.

In summary, this chapter offers a comprehensive introduction to terrain analysis as a component of vegetation firefighting. Through a combination of theoretical knowledge, practical instructions, and exemplary scenarios, emergency personnel are provided with the necessary tools to operate safely and effectively in endangered areas. The goal is to increase the safety and efficiency of firefighting through targeted training measures and proven techniques, thus making a significant contribution to protecting people.

and environment.

12.1 Orientation in the terrain using terrain features

Terrain orientation is a critical skill in vegetation firefighting, as it allows emergency responders to make quick and precise decisions about their movements and positioning. This skill can give firefighters a significant advantage in the dynamic environment of a vegetation fire.

One of the most basic methods for terrain analysis is the use of terrain features. These features are visually striking and persistent objects in the landscape that serve as orientation aids. These include prominent rocks, groups of trees, hills, rivers, roads, and even structures. Because these features are generally stable and do not change rapidly, they provide reliable support for orientation. However, to use these features effectively, careful and continuous observation of the terrain is required.

necessary.

The ability to use terrain features effectively depends heavily on perception and the ability to mentally map these features.



Emergency personnel should always be alert and actively take notes of prominent Points in the field. This can be improved through regular training and targeted exercises, as highlighted in the relevant literature on vegetation firefighting (Barth, 2024, pp. 1-2).

Another important aspect of orientation is the ability to relate terrain features to one another. For example, the combination of a hill with a river and a group of trees can provide a clear orientation point. The spatial relationship between these features plays a central role.

This allows emergency services to determine their position in the terrain and movement Estimate direction and speed. This mental terrain mapping enables precise navigation, even under difficult conditions.

Additionally, an understanding of topographic maps is beneficial for comparing real-world terrain features with the maps. Emergency responders should be able to identify topographic details such as contour lines, vegetation types, and infrastructure. real elements on a map and apply this information to the real terrain. In this regard, the use of special manuals and training programs, as described in the literature, can be of great value (Barth, 2024, pp. 3-4).

A crucial aspect of using terrain features is communication within the team. Emergency personnel must be able to describe terrain features clearly and unambiguously to avoid misunderstandings.

This can be achieved through standardized terminology and training in precise communication. Knowledge of and application of such communication protocols are essential in vegetation firefighting training.

In addition, marking terrain features can aid the orientation process. Temporary markers such as colored bands or flags can be used to indicate particularly important points and serve as orientation aids. This is particularly helpful in densely vegetated or confusing areas. However, such markers should be used carefully and sparingly to avoid confusion.

It is also important to be aware of the limitations of orientation using terrain features. In dense smoke or in the dark, many visual landmarks

Identification points may disappear or be difficult to identify. Therefore, emergency personnel should also be proficient in alternative orientation techniques, such as the use of maps and electronic navigation aids, in order to be able to act operationally under all conditions.



Terrain orientation skills and techniques are essential components of the training concept for vegetation firefighters. Through continuous training and exposure to theoretical principles, emergency personnel can further develop their skills and apply them more effectively in real-life scenarios. Ultimately, sound training in terrain analysis contributes significantly to increasing the safety and efficiency of vegetation firefighting (Barth, 2024, pp. 5-6).

12.2 Orientation in the terrain using maps

The ability to navigate the terrain using maps is crucial for firefighters, especially when fighting vegetation fires. Precise map handling allows for efficient reconnaissance of operational areas, the establishment of strategic positions, and ensuring that each team member knows their location and that of other emergency personnel at all times.

knows.

First, emergency responders should be familiar with the different map types and their scales. Topographic maps are particularly valuable in this regard, as they provide detailed information on terrain features, vegetation types, water sources, and other relevant factors. These maps help emergency responders identify potential escape routes, locations for emergency vehicles, and water intake points (Barth, 2024, pp. 1-3). They also provide important terrain data that is essential for predicting fire behavior and planning firefighting efforts.

In fire service and disaster management training, great emphasis is placed on map reading. Emergency personnel must understand and be able to work with coordinates. This includes reading contour lines, recognizing potential hazards, and determining assembly points and refuge areas.

Knowledge of the use of UTM (Universal Transverse Mercator) coordinates and other coordinate systems is crucial for specifying precise positions.

Another important element in using maps is the ability to align them with the real environment—so-called "orientation." This involves rotating the map so that it aligns with the compass direction of the terrain and determining one's own location relative to prominent landmarks. This is especially important when visibility is limited or when emergency personnel are moving through unfamiliar terrain. While modern GPS technologies can be helpful, they do not replace traditional mapping techniques, as technical aids can fail (Barth, 2024, p. 4).



Communication and teamwork are essential in the fire service.

crucial. Therefore, all team members should be able to read and interpret maps correctly. Regular exercises and training are necessary to deepen these skills. Firefighters learn to record relevant information on maps, such as factors influencing fire behavior, potential danger zones, and planned tactics (Barth, 2024, p. 7).

A particularly critical element of map work is understanding the different

Different symbols and colors used on maps. This uniform symbolism enables quick and efficient interpretation of the information.

For example, firefighters need to know that green areas typically represent forests, and blue lines often indicate rivers or other water sources. Certain symbols highlight water intake points or access routes, which are crucial for water supply during a vegetation fire (Barth, 2024, pp. 9-11).

Another important aspect of map use is strategic planning and resource management. By analyzing maps, incident commanders can determine the best access routes, locations for fire engines, and possible retreat points for emergency personnel. Furthermore, map work enables better coordination with other units and more effective use of available resources. This capability minimizes risk to emergency personnel and increases safety during firefighting operations.

In summary, terrain orientation through the use of maps is a fundamental skill for vegetation firefighting. It enables precise and efficient planning and execution of operations, promotes the safety of emergency personnel, and improves coordination and communication within

of the team. Continuous training and exercises are crucial to ensure that emergency personnel are equipped with the necessary knowledge and skills to successfully navigate the terrain and effectively combat vegetation fires. The combination of traditional mapping techniques and modern GPS technologies provides comprehensive support, while the basic principles of mapping should always remain in focus.

12.3 Marking/Labeling

Landmarking and marking are crucial elements in vegetation firefighting, as they ensure both the orientation of emergency personnel and their safety at the scene. Well-thought-out marking improves coordination and helps effectively avoid potential hazards.



A key aspect of signage is the use of prominent terrain features. These can include prominent points in the landscape, such as specific groups of trees, rocks, or waterways. Such natural features act as signposts and assist emergency responders in better navigating the area (State School and Technical Facility for Fire and Disaster Protection, 2024, p. 16). Knowledge of these features is particularly valuable when planning control lines or defining safe areas.

In addition to natural features, artificial markings also play a crucial role. Various types of marking are used for this purpose.

Areas that are cordoned off are often marked with tape or other temporary markings. Colors or reflective materials can also be used to highlight paths and hazards, especially in poor visibility conditions (State School and Technical Facility for Fire and Disaster Protection, 2024, p. 16). These measures are essential to create a clear structure in the often confusing terrain and to minimize the risk for emergency responders.

The importance of marking hazard areas is immense. Vegetation fires pose numerous hazards, which can be exacerbated by differences in terrain topography and vegetation density. It is essential to clearly mark hazardous areas, such as high fire loads or fall hazards. It is helpful to use standardized symbols and colors that can be quickly and clearly understood (State School and Technical Facility for Fire and Disaster Protection, 2024, p. 24).

Tactical considerations are also crucial. Marking escape routes and assembly points is crucial for rapid response in an emergency. These areas should be positioned and marked so that they are easily accessible from any point in the area of operations and clearly distinguishable from danger zones (State School and Technical Facility for Fire and Disaster Protection, 2024, p. 18).

Modern technologies can significantly increase the effectiveness of marking and tagging systems. GPS devices and digital mapping applications enable markings to be precisely transferred to electronic devices that can be used by emergency responders. These technologies complement conventional methods and provide additional safety (State School and Technical Institution for Fire and Disaster Protection, 2024, p. 20).

Another important aspect is the consideration of local peculiarities. Especially in unknown or difficult terrain, it is crucial to take local



To identify and mark conditions and potential hazard areas.

This can be achieved through close cooperation with local experts and through early site surveys (State School and Technical Facility for Fire and Disaster Protection, 2024, p. 28).

Equally important is the continuous review and updating of markings, as conditions on site can constantly change as the fire progresses.

Continuous monitoring and adaptation of signage is necessary to address changing hazards. This not only improves the orientation of emergency responders but also contributes significantly to safety.

In summary, it can be said that marking and signage are essential in the vegetation firefighting area. Effective and well-designed signage minimizes risks, improves coordination, and ensures the safety of emergency responders. The combination of natural terrain

Detecting features, artificial markings, and modern technologies ensure that all involved forces can act quickly and safely. Continuous adjustment and testing of the markings further contribute to effectiveness and safety.

12.4 General marking of danger areas and barriers

Clearly marking hazard areas and barriers plays a key role in vegetation firefighting. Clear and consistent marking of these points allows emergency responders to navigate efficiently and safely within the area and minimizes the risk of accidents and misunderstandings.

its.

A first important aspect is that hazard areas should be clearly and repeatedly marked. Various marking methods can be used, such as colored tape, signs, or reflective materials. These markings are particularly essential in situations with limited visibility, such as dense smoke or darkness. Consistent color schemes and symbols are crucial to avoid misunderstandings.

For example, red signs could be used to mark immediate danger areas and yellow signs to indicate precautionary areas.

Secondly, it is important to conduct regular training for emergency personnel regarding the marking systems used. Emergency personnel must know the meaning of the various markings and how to respond to them. This includes understanding when and how to use new



Danger areas should be marked to ensure safety throughout the entire area of operation.

A practical example of the implementation of hazard marking can be found in the training courses on specific protective clothing and the necessary equipment. These courses provide detailed explanations of the possible uses of various handheld devices, as well as their proper handling and maintenance, to increase safety during operations (State School and Technical Facility for Fire and Disaster Protection, 2024, p. 14).

A third point concerns the dynamic nature of vegetation fires and the need for flexible adaptation of signage. Fires can change direction, and new hazards can arise. Therefore, emergency services must react quickly to changes and adjust hazard areas and barriers accordingly. Immediate communication is crucial here. The communication tools and channels provided must be ensured to ensure timely communication of changes and to keep signage up-to-date.

Fourth, barriers should be robust and durable to withstand the conditions of the fire. This means they must withstand wind, rain, and the heat generated by the fire. Furthermore, they must be designed to be clearly visible to emergency responders and potentially affected individuals and not be lost in the fire.

In some cases, a physical barrier, such as a vehicle barrier or mobile barricades, may also be useful. This is especially true for larger and unpredictable incidents. This can further increase safety at visible incident sites.

temperature, it is pointed out that vehicles for water pumping should be deployed early in order not to miss the transition from shuttle traffic to stable water supply (State School and Technical Facility for Fire and Disaster Protection, 2024, p. 12).

Finally, it should be emphasized that the marking of hazardous areas and barriers should be closely linked to the overall safety concept. This includes regular inspection and maintenance of the marking materials used, as well as training of emergency personnel in their proper use.

A systematic and structured approach to marking can make a decisive contribution to the safety of emergency services and the affected population.

These measures ensure that all parties involved are informed and made aware of the existing hazards, ultimately leading to more effective and safer firefighting. This underlines the need to



to continuously assess prevailing conditions and risks and, if necessary,
to react quickly to maximize the safety of all actors in the area of operations.

12.5 General marking for traffic control

Vegetation fires pose a significant challenge for fire departments worldwide. These types of fires often spread rapidly and uncontrollably, endangering not only natural resources but also human lives and property. In such situations, targeted traffic management is crucial to ensure efficient and safe operations. The primary goal of traffic management is to control traffic flow during an operation to ensure smooth access for emergency vehicles to the fire scene, water intake points, and other critical points.

A key aspect of traffic control during vegetation fires is clear and understandable signage. This must be clearly visible to both on-site emergency responders and the civilian population. According to the training materials of the Brandenburg State School and Technical Facility for Fire and Disaster Protection, it is of utmost importance that all markings and signs comply with local and national standards (Barth, 2024, p.

- 1). This includes both temporary markings during an operation and permanent signage in particularly vulnerable areas.

Particular attention should be paid to the operational principles and safety rules applied in firefighting (Barth, 2024, p.

- 3). This includes, among other things, ensuring a clear path for emergency vehicles while simultaneously ensuring the safety of the civilian population. Effective traffic control can help avoid panic and chaos and thus provide the best possible support to emergency responders.

The use of marking tape is a proven method for temporarily cordoning off specific areas or marking danger zones (Barth, 2024, p. 12). These tapes should be robust and highly visible to fulfill their function even under difficult conditions. It is also important that these tapes are easily understood by all involved to minimize misunderstandings and the associated hazards.

Another aspect of traffic control is the consideration of special emergency vehicles and their requirements. Different vehicles, such as tank fire engines, require different access options and turning circles (Barth, 2024, p. 9). Therefore, traffic control measures should be designed to be flexible and adaptable in order to meet the different requirements.



In addition, traffic control should be organized in such a way that the work of the emergency services is not hindered and efficient firefighting remains possible.

The topographical and infrastructural conditions of the area of operation also play a significant role. It is crucial to plan traffic routes and access points in a vegetation fire area in advance and to review them regularly (Barth, 2024, p. 6). This includes considering local characteristics such as the condition of the roads, existing bridges and their load-bearing capacity, as well as

such as the width and condition of access roads.

Traffic control during vegetation fires is therefore a multifactorial management system that takes into account both technical and human factors.

The key challenge is to find a balance between speed, efficiency, and safety. Only through careful planning and execution of traffic control can the effectiveness of emergency measures be maximized and the safety of all involved be ensured.

Overall, it is clear that traffic control is an integral part of vegetation firefighting. It requires precise planning, well-thought-out measures, and the consideration of numerous factors to ensure the smooth running of an operation. The aspects presented here, as well as the use of the scientific sources provided, illustrate the complexity and necessity of a structured approach to traffic control during vegetation fires.

12.6 Use of marking tape

The use of marker tapes in vegetation firefighting is crucial for structuring and efficient operational coordination. These tapes provide emergency responders with visual cues to clearly demarcate different operational areas, which is particularly relevant in chaotic and dynamic situations. This section examines the important role of marker tapes in the context of vegetation firefighting, particularly with regard to their various applications and advantages.

Marker tapes are simple yet highly effective tools for communicating critical information quickly and clearly. During vegetation fires, it is often necessary to mark different zones to ensure the safety of emergency responders and to implement strategic decisions precisely.

In this context, danger zones, escape routes, and assembly points can be clearly marked. This helps to minimize confusion.



and contribute to maximizing efficiency, as the emergency services can quickly identify and reliably carry out their tasks (Notes on practical AE object defense/post-extinguishing, 2024, p. 2).

The benefits of marking tape are also evident in technical and tactical operational planning. Clearly marking different areas simplifies the coordination of different teams during an operation.

The use of marking tapes allows for better control and documentation of the transitions between different measures, such as attack and defense. This clear demarcation is important not only for immediate firefighting, but also for the subsequent follow-up and evaluation of the operations, where the exact course and definition of the zones must remain traceable (Notes on Practical AE Object Defense/Follow-up Extinguishing, 2024, p. 4).

A key aspect of using marker tapes is their support during post-fire extinguishing operations. This work requires a systematic approach, in which all areas of the fire area must be carefully searched to locate hidden embers or smoldering fires. Marker tapes can be used to mark previously treated areas, avoiding duplication and increasing efficiency. They also provide emergency personnel with visual confirmation of their progress, boosting morale (Notes on Practical AE Object Defense/Post-Fire Extinguishing, 2024, p. 6).

Additionally, marker tape plays a significant role in establishing control lines and other preventive measures. These lines serve to prevent the spread of fire and can be clearly defined and monitored using marker tape. This is especially important in complex operational areas where flawless communication and coordination are critical.

Through the correct use of marker tapes, such control lines can be established more efficiently and quickly, which shortens the reaction time and supports the overall strategy (Notes on practical AE object defense/post-extinguishing, 2024, p. 7).

Another important factor in the use of marking tapes is the involvement and coordination of third parties, such as other emergency organizations or specialized units. Marking tapes offer a universal and easily deployable

standard method for communicating safety-relevant information and tactical plans. This facilitates cooperation and coordination between different actors who may have different training backgrounds and working methods, but who nevertheless pursue a common goal. The integration of these tools into the overall operational strategy can increase the effectiveness



of mixed forces and ensure a coherent approach (Notes on practical AE object defense/post-extinguishing, 2024, p. 10).

Grün	Blau	Rot	Orange	Gelb
Rettung & Wege	Wasserversorgung	Gefahren		Führung & Logistik

Figure 20 Colors for marking tape

Instead of the green markings, pink marking tapes with the inscription “Escape Routes” can also be used for the escape routes.

In summary, marking tapes are an indispensable tool in vegetation firefighting, offering numerous applications and advantages. They not only demarcate areas of operation, but also support follow-up extinguishing activities, the creation of control lines, and coordination with third parties. Thus, they contribute significantly to the efficiency and safety of firefighting operations.

The strategic use and application of marker tapes should therefore be an integral part of the training and preparation of emergency personnel. This will enable optimal preparation for future vegetation fires and enable them to effectively combat them.



13 Technology

Fighting vegetation fires represents one of the most complex challenges in firefighting. Given the increasing frequency and intensity of forest and wildfires caused by climate change and human activities, the need for in-depth and specialized training in this area is becoming increasingly urgent. The success of firefighting depends not only on the expertise and tactical knowledge of the emergency responders, but also significantly on the correct selection and effective use of technical equipment.

This chapter examines in detail the technical aspects essential for the efficient management of vegetation fires. It focuses on various devices specifically designed for this type of firefighting. Technological advances have enabled significant progress in this area, so that today a wide range of specialized tools and machines are available that can both increase the safety of emergency personnel and significantly improve the efficiency of firefighting.

Modern equipment and technologies play a crucial role in containing and fighting vegetation fires. These are often robust, versatile devices that must function reliably under extreme conditions. The requirements for this equipment are high: It must be easily transportable, user-friendly, yet effective. This requires not only innovative technical developments but also the continuous adaptation and further development of existing technologies and equipment.

A key aspect of technical equipment is the balance between efficiency and safety. Equipment used directly at the source of a fire must be both effective in firefighting and meet the safety requirements of emergency responders. These include ergonomics, durability, ease of use, and special safety features that ensure user protection.

Another important aspect of vegetation firefighting is the mobility of the deployed personnel and their equipment. In severely affected areas, emergency personnel often have to enter difficult terrain that is inaccessible to conventional vehicles. Therefore, specialized fire engines and transport vehicles tailored to the specific requirements of vegetation firefighting are of great importance. These vehicles must be able to transport large quantities of water to be used effectively in fighting large-scale fires.



Water supply is a critical issue in vegetation fire fighting.

Water remains one of the most effective firefighting tools, but in remote or hard-to-reach areas, providing sufficient quantities can be challenging. Innovative methods and specialized equipment for water extraction and distribution are therefore essential to adequately support emergency responders on the ground. These include mobile water tanks, portable pumps, and specialized spraying devices that enable the targeted application of the required amount of water, even in hard-to-reach areas.

The continuous development and adaptation of technical equipment is essential to meet increasing demands and constantly changing conditions. This includes not only the development of new equipment, but also the continuous training and further education of emergency personnel in the use of the latest technologies and methods. Only through the close integration of innovative technology and sound training can the complex and dynamic challenges of vegetation firefighting be effectively met.

In summary, technology plays a fundamental role in vegetation firefighting, extending far beyond the mere use of equipment. It requires a comprehensive understanding of the available options, their specific areas of application, and optimal use under the respective operational conditions. This chapter provides a detailed insight into the technical principles and equipment essential for the effective and safe suppression of vegetation fires. The following subchapters examine the individual aspects and equipment in detail, thus providing a sound information basis for training and practical use in vegetation firefighting.

13.1 Equipment for vegetation fire fighting

Vegetation firefighting equipment plays a central role in controlling and containing fires. They are designed to meet the specific challenges of such fires, which often develop in remote and difficult-to-access areas. It is therefore essential for fire departments to effectively utilize these modern tools to fight fires quickly and safely.

Important equipment in this context includes fire backpacks, bucket pumps, and fire beaters. Fire backpacks allow for flexible water application and are particularly useful in difficult terrain. These backpacks typically hold about 20 liters of water and have a manual pump that



enables a targeted water jet (Notes on Practical AE Equipment Knowledge/Hand Tools (Basics), 2024, p. 10). Bucket sprayers work similarly, but are easier to handle and are used for lower flame intensities. Fire beaters are used to knock down and extinguish flames and are often used on ground fires (Notes on Practical AE Equipment Knowledge/Hand Tools (Basics), 2024, p. 15).

In addition to these basic handheld devices, there are also specialized techniques and devices for water pumping. Mobile external firewater tanks, made of rigid or flexible materials, vary in capacity from 300 to 10,000 liters. They are particularly useful for extracting water from open bodies of water and transporting it to the fire (Overview of Technical-Tactical Procedures (Fundamentals), 2024, p. 23). Their versatility makes them an essential part of vegetation firefighting equipment.

Another important area is area preparation and defense through the generation of foam and wetting agents. This is done using special fittings and proportioners that ensure the foam has an appropriate penetration depth and duration of action. The technical specifications and procedures vary depending on the specific conditions at the scene (LG No. 612 Vegetation Fire Fighting - Leadership 1 (Lead Level A/B) + Intensive Training (Veg.-Bbk Leadership 1 + Intensive Training), 2024, p. 18).

The integration of technological innovations also plays a significant role. The use of digital tools and navigation technologies, such as GPS and digital maps, enables more precise and faster coordination of emergency services. Digital topographic maps offer numerous advantages in the planning and execution of operations, such as the precise identification of fire water extraction points and the assessment of fire area extent (Vegetation Fire Fighting Guide 1 (Fü-Level A/B) + Intensive Training (Veg.-Bbk Lead 1 + Intensive Training), 2024, p. 8).

In addition, specially equipped tank fire engines are of central importance because they can transport large quantities of water and have the necessary technology to operate efficiently in difficult terrain (Notes for practical AE equipment knowledge/hand tools (basics), 2024, p. 10).

In summary, the technical equipment for vegetation firefighting includes a variety of handheld and extinguishing devices, water and foam generation equipment, digital tools, and specialized vehicles. This equipment is crucial to equip firefighting teams with the necessary tools and technologies to extinguish vegetation fires quickly, safely, and effectively.



and fight them effectively. Given the increasing frequency and intensity of vegetation fires, it is essential that fire departments continuously invest in modern equipment and technologies to meet the new challenges.

to adapt to the requirements.

13.2 Handheld devices for vegetation fire fighting

Handheld devices are crucial tools in vegetation firefighting. They enable rapid response and efficient work in difficult-to-access areas, requiring minimal equipment and minimal extinguishing water. They are particularly effective for securing and controlling fires, as well as for follow-up extinguishing operations, especially when flame lengths are short (PP LG No. 612, p. 12).

A significant advantage of handheld devices is their resource-efficient application. They are generally designed for low-intensity vegetation fires and enable remote operation, independent of vehicle-based operations. Their transportability and flexibility in difficult terrain are crucial to their versatility (PP LG No. 612, p. 12).

Commonly used hand tools include the cultivator sickle, which is ideal for cutting and removing fuel, such as bushes, and for creating slash strips. Another essential tool is the hoe, which is used for removing, cutting, and loosening the subsoil, as well as for removing roots (PP LG No. 612, p. 13).

Also particularly important is the Swedish clearing axe (Sandvik), which is also used for cutting and removing fuels and for applying fire strips. The arsenal of hand tools is complemented by leaf blowers and sprayers, which are used for removing and extinguishing fuels and for humidifying (PP LG No. 612, p. 13).

In addition, there are specialized devices such as the drip torch, which are designed for controlled counter-fire. However, their use requires special training to ensure safety and effectiveness (PP LG No. 612, p. 13).

The effectiveness of these handheld devices depends heavily on proper handling and the physical resilience of the emergency personnel. Ergonomic working and the wearing of personal protective equipment (PPE) are essential to ensure the health and safety of the emergency personnel. Team morale also plays a crucial role. Managers should therefore always keep an eye on the work processes and safety aspects of their team and maintain a high level of



promote communication and trust within the team (PP LG No. 612, p. 15).

In summary, handheld devices are key tools in vegetation firefighting. They offer flexibility and efficiency, especially in hard-to-reach areas and with low fire intensity. The correct selection and correct use of these devices can effectively fight fires and ensure the safety of emergency personnel, which contributes significantly to the successful management of vegetation fires.

13.3 Vehicles

The selection and deployment of the right vehicles is crucial for effective vegetation firefighting. These vehicles are specifically designed to meet the challenges of such operations, especially in remote and inaccessible areas. The requirements for the vehicles include mobility, robustness, and the ability to transport necessary equipment and sufficient quantities of extinguishing agents.

Tank firefighting vehicles (TLFs) play a central role in this. They are designed to transport large quantities of water to the scene and distribute it efficiently (Vegetation Fire Fighting Guide 1, 2024, p. 9). One example is the TLF 4000, which is characterized by its high off-road capability and the ability to operate in difficult terrain (PP LG No. 612 Veg.-Bbk FÜ1+Ausbildung intensiv AE-Fahrzeug_Gerätekunde FÜ, 2024, p. 3). These vehicles are flexibly equipped to operate both offensively at the fire front and defensively by establishing firebreaks or screening.

All-wheel drive and differential locks are essential for many vehicles used in vegetation fire protection. They enable the vehicles to remain mobile even on difficult terrain, which is especially important since vegetation fires often occur in areas where conventional road vehicles cannot operate (PP LG No. 612 Veg.-Bbk FÜ1+Ausbildung intensiv AE-Fahrzeug_Gerätekunde FÜ, 2024, p. 10).

In addition to TLFs, firefighting vehicles (LFs) are also used, which are equipped with large extinguishing agent tanks. However, they are more specifically designed for supporting and coordinating emergency teams and offer additional communication devices to effectively support emergency personnel on the scene (Vegetation Fire Fighting Guide 1, 2024, p. 19). Command vehicles (KdW) and operational command vehicles (ELW) play an important role in the command and coordination of larger emergency units, although they are less focused on direct firefighting.



Special vehicles are also important, especially armored vehicles and tanks used in high-risk areas such as former military training areas or munitions-contaminated areas. These vehicles offer high levels of protection for emergency personnel and can operate under extreme conditions (PP LG No. 612 Veg.-Bbk Fü1+Ausbildung intensiv AE-Fahrzeug_Geräte-kunde Fü, 2024, p. 33).

Water supply is a critical aspect of vegetation firefighting. Mobile water tanks and portable pumps allow extinguishing water to be pumped from natural sources to the area of operation. These devices are crucial for a continuous water supply during operations and reduce dependence on local water supplies (PP LG No. 612 Veg.-Bbk Fü1+Ausbildung intensiv AE-Fahrzeug_Gerätekunde Fü, 2024, p. 22).

The demands placed on the vehicles are high. In addition to off-road capability and extinguishing agent capacity, stability and reliability in operation are also essential factors. The vehicles must be solidly constructed and equipped with specialized equipment such as high-pressure extinguishing systems and foam or wetting agent proportioning systems. For the safety of the emergency personnel, the option of "pump & roll" operation should be available, allowing the vehicles to remain in motion while discharging extinguishing water (PP LG No. 612 Veg.-Bbk Fü1+Ausbildung intensiv AE-Fahrzeug_Gerätekunde Fü, 2024, p. 10).

In summary, the fire service vehicle fleet for vegetation firefighting must meet diverse and specialized requirements. In addition to standard vehicles such as mobile fire tenders and fire engines, special vehicles such as armored vehicles and efficient water pumping vehicles are also indispensable. These ensure the protection of emergency personnel and effective firefighting under variable and often difficult conditions. Continuous training and practice are necessary to optimally prepare emergency personnel for the specific challenges of vegetation firefighting (Vegetation Fire Fighting Guide 1, 2024, p. 24).

13.4 Special equipment for water pumping

Water supply is a central aspect of vegetation firefighting and, due to the often difficult geographical and infrastructural conditions, places special demands on the equipment deployed. A reliable water supply is crucial for the success of firefighting operations as well as the protection of emergency personnel and property. Therefore, specialized equipment is required to effectively meet the challenges of vegetation firefighting.



Powerful pumping systems are at the heart of water delivery. These systems are designed to pump large volumes of water over long distances and are often used in combination with mobile tank fire engines (TLF). One proven method is shuttle transport, in which tankers travel back and forth between water intake points and fire sites to continuously deliver water (Barth, 2024, p. 8). Although this approach is considered a temporary solution, it is crucial for the rapid initial supply of water, especially in the initial phase of a vegetation fire.

For difficult-to-access fire areas, special pumping systems are required that offer high flow rates and pressure performance. One example is the high-performance pumping system (HFS), which is capable of pumping several thousand liters of water per minute over long distances (Kaulfuß & Hofmann, 2011, p. 24). These systems are modular and can be scaled as needed. Safety features such as pressure relief valves and special connections are integrated to ensure stable and efficient water pumping.

Transport containers also play an important role in water supply. Collapsible containers, such as the Husky Collapsible Container, enable flexible and rapid provision of water supplies (Barth, 2024, p. 18). These containers are easy to transport and quick to set up, making them particularly useful in difficult terrain. They serve as intermediate stations for continuous water supply directly at the scene.

Digital tools provide technical support for the planning and implementation of water supply strategies. Applications such as complete smartphone solutions and web-based maps, such as the Brandenburg Viewer, enable precise calculations and representations of route distances, elevation profiles, and other relevant parameters essential for efficient water supply (Barth, 2024, p. 26).

In practice, the use of agricultural equipment has proven advantageous. Tractor-water tank combinations can transport large quantities of water (Barth, 2024, p. 16). These systems offer high flexibility, especially in rural areas where infrastructure is often difficult for conventional firefighting vehicles.

Cooperation with the Federal Agency for Technical Relief (THW) may also be necessary. The THW has specialized units focused on pumping water from open waters. These units use robust, mobile pumps and long hose lines to pump water volumes of up to 8,000 liters per minute (Kaulfuß & Hofmann, 2011, p. 24). These systems are designed for use in hard-to-reach areas and ensure that the



The necessary amount of water is transported quickly and reliably to the fire sites.

In summary, water supply for vegetation fires requires the interaction of various specialized equipment and systems. Powerful pumps, mobile tank fire engines, flexible transport containers, and digital planning aids are crucial for a continuous and reliable water supply. Complemented by the use of agricultural equipment and cooperation with specialized units such as the Federal Agency for Technical Relief (THW), effective firefighting can be ensured even under challenging conditions.





14 Guide

Successfully managing vegetation fires requires precise and well-thought-out leadership. In extreme situations characterized by rapidly advancing fires, the efficiency of operational command is crucial to saving lives, minimizing property damage, and ensuring environmental protection. This chapter covers the fundamental aspects of leadership in vegetation firefighting and provides a comprehensive overview of the essential elements of operational preparation and

implementation.

The challenges posed by vegetation fires are diverse and require a high degree of adaptability and coordinated action by all involved.

Management strategies must be flexible while still providing clear structures and processes to enable quick and targeted decisions in an emergency. Not only the firefighters on site are responsible for management tasks; incident commanders, other public institutions, and local organizations also play an important role in coordinating and managing a vegetation fire.

An essential part of leadership during such an operation is preparation and planning, which begins before the actual firefighting.

This includes the education and training of personnel, the development of operational plans, and the conduct of realistic exercises simulating various vegetation fire scenarios. This preparatory phase lays the foundation for a successful operation and is crucial for the effectiveness of the measures taken.

As soon as the emergency forces arrive at the scene, specific command mechanisms come into effect, enabling the orderly and controlled execution of the operation. This includes the definition and adherence to spatial planning, which enables the emergency forces to clearly identify and fulfill their assigned tasks and areas of responsibility. The definition of staging areas and the establishment and operation of command posts are key elements that ensure

ensure efficient communication and coordination of emergency services.

Another critical factor is the division of the entire operational area into manageable parts and the initiation of firefighting measures in a targeted and systematic manner. This ensures better control and responsiveness to dynamic changes in the fire situation and enables incident commanders to allocate resources effectively.



Leadership in vegetation firefighting also includes aspects of care. It is important to provide necessary logistical support and ensure the physical and mental health of emergency personnel, who often have to work under extreme conditions.

Particular attention should be paid to the involvement of third parties, be it other government agencies, private organizations, or the affected population. An integrative leadership strategy that maximizes the cooperation and coordination potential of all available forces is essential for the overall success of the measures.

Finally, the planning and implementation of warnings and evacuations are critical areas of action in vegetation fire management. These measures require a precise assessment of the hazard situation and prompt communication to ensure public safety and enable efficient evacuation of endangered areas.

The chapter concludes with considerations on post-operational debriefing, emphasizing the importance of evaluating and documenting the operation. This phase allows for learning from each operation and continuous implementation of improvements in vegetation firefighting practices and strategies.

By examining these essential leadership aspects in the context of vegetation firefighting in detail, this chapter provides not only a theoretical foundation but also practical approaches and methods that are indispensable in real-world applications. It is intended to enable the reader to make informed decisions and successfully master the complex challenges of vegetation firefighting.

14.1 Operational preparation and planning

Pre-emergency preparation and planning are crucial for successful vegetation firefighting operations. This process begins well before the actual operation and encompasses a variety of strategic, technical, and logistical measures, all aimed at ensuring the safety of emergency personnel and minimizing damage as much as possible.

A key aspect of operational preparation is reconnaissance and situation control. This involves recording weather conditions, analyzing fire behavior, and examining fuel properties. In regions such as Brandenburg, where vegetation varies greatly, knowledge of these factors is crucial. Reconnaissance of the region using landmarks and maps also plays an important role. Emergency personnel must learn to use



To orient themselves in the terrain using landmarks and maps and to use various topographical elements such as towers, railway infrastructure or forest boundaries to determine their position precisely (Vegetation Fire Fighting Guide, 2024, pp. 1-5).

Another key element of preparation is the creation of detailed topographic maps. These maps contain precise information about settlements, transport networks, waterways, water protection areas, and other relevant elements.

In Brandenburg, such maps are available at various scales—from 1:10,000 to 1:50,000, with the latest maps scheduled to be converted to a scale of 1:25,000 starting in 2025 (Vegetation Fire Fighting Guide, 2024, pp. 8-12). Of particular note are the forest fire maps, which contain specific details such as firefighting water intake points and hazardous areas, thus enabling more precise operational planning.

Weather conditions and forecasts also play a crucial role in operational preparation and planning. The behavior of vegetation fires is strongly influenced by the weather. Therefore, it is essential that leaders are constantly informed about current fire behavior in order to be able to make safe and efficient tactical decisions (Vegetation Fire Fighting Guide, 2024, p. 3). In this context, the use of lookouts (observers) is also important, providing continuous information about potential hazards.

The integration of digital tools, such as GPS and topographic maps, also supports operational preparation. These technologies enable more precise location determination and allow for faster and more accurate planning of routes and operational strategies (Vegetation Fire Fighting Guide, 2024, p. 13-15).

A key part of the preparation is also the development of operational plans and strategies. This involves simulating various scenarios and practicing tactical operations to ensure flexible and rapid response in an emergency. The vegetation fire map for Brandenburg, for example, offers valuable support, providing information on firefighting equipment, topographical features, and potential hazards (Vegetation Fire Fighting Guide, 2024, p. 12).

In operational planning, it is essential to define escape routes and safety zones and communicate this information to all emergency personnel. These safety measures are crucial to protect emergency personnel from unforeseen hazards and to ensure quick and safe escape routes in an emergency (Vegetation Fire Fighting Guide, 2024, p. 3).

Complementary to the strategic and tactical preparation are the logistical Measures to ensure that all necessary resources are available



This includes planning water supply routes and ensuring the availability of firefighting water. Special water supply equipment and mobile command posts can be of great assistance in this regard (Vegetation Fire Fighting Guide, 2024, pp. 9-13).

Overall, step-by-step and detailed preparation is essential for the success of a vegetation fire response. Considering all relevant factors—from weather and terrain conditions to technical equipment and logistical planning—enables emergency responders to respond flexibly and safely to various scenarios and successfully minimize damage.

14.2 Organization of the room

Spatial organization is a key aspect for efficiency and safety in vegetation firefighting. A clear structure and division of operational areas are essential to maintain an overview and deploy resources effectively. Therefore, it is important that the operations command conducts a systematic survey and division of the fire area.

A proven methodology for spatial planning is to divide the operational area into different sectors. These sectors should be delimited based on geographical features such as roads, rivers, or topographical features. A clear sectoral division not only facilitates orientation for emergency personnel but also improves the coordination of firefighting measures. This corresponds to the practical requirements emphasized in relevant teaching materials (Barth, 2024, p. 1).

The role of digital navigation and the use of GNSS (Global Navigation Satellite System) systems is of great importance here. The ability to transmit precise coordinates and real-time data contributes significantly to efficiency. State-of-the-art navigation devices and digital maps enable rapid and precise mapping of areas and support operational command in decision-making (Vegetation Fire Fighting Guide 1, 2024, p. 13).

In addition, knowledge of the local infrastructure is of great importance. Paths, fire water intake points, and danger zones should be marked and integrated into operational planning. Considering these factors is essential to ensure effective logistics. The forest fire maps of the state of Brandenburg illustrate this by providing detailed information on fire hazards and infrastructure points (Vegetation Fire Fighting Guide 1, 2024, p. 11).



Proper room layout also requires careful marking of danger zones and other relevant areas. This can be achieved by using

This can be done through the use of color markings and information signs. Such measures are crucial to avoid misunderstandings and potentially dangerous situations. The use of consistent color codes increases the visibility and recognizability of important areas (Vegetation Fire Fighting Guide 1, 2024, p. 18).

Another important aspect of spatial planning is the creation of staging areas. These should be strategically selected so that they are both easily accessible and safe. Staging areas serve as assembly points for emergency resources and personnel and as starting points for further operations. Thoughtful positioning of these areas facilitates access to critical incident sites and shortens response times (Handbook on Forest Fires - Forest Fire Fighting, 2011, p. 3).

In addition, spatial planning includes the establishment of supply points where supplies and food can be provided. These points should be regularly reviewed and adjusted as needed. Providing emergency personnel with the necessary materials and food is crucial to maintaining operational capability over extended periods. An orderly supply infrastructure contributes significantly to the effectiveness of operations (Handbook on Forest Fires - Forest Fire Fighting, 2011, p. 5).

Finally, it is important to consider agreements and cooperation with local authorities and third parties. Good cooperation and regular exercises with local stakeholders, such as forestry authorities and forest owners, increase effectiveness and improve communication between the various organizations. This is also emphasized in the documents on vegetation fire fighting, which describe the integration and cooperation of various stakeholders as key to success (Handbuch Waldbrand - Waldbrandbekämpfung, 2011, p. 1).

Overall, spatial organization is an integral part of operational planning and management for vegetation fires. Careful planning and a structured approach ensure that available resources are used optimally and the safety of all involved is ensured.

14.3 Preparation rooms

Staging areas are an essential element in tactical operational management during vegetation fires. They serve as temporary locations where emergency personnel, vehicles, and equipment await their next orders. The strategic selection and planning of these areas is crucial for the efficiency and safety of the entire operation.



Setting up staging areas requires careful preparation and planning. First, suitable locations must be identified that are at a safe distance from the active fire while simultaneously offering rapid response capabilities. Accessibility for various vehicles and units is a key factor here (Barth, 2024, p. 2). Furthermore, the areas must offer sufficient space to ensure proper organization and maneuverability of the deployed forces.

The surrounding area and infrastructure must also be considered when selecting a location. Ideally, staging areas should be located near roads to enable rapid mobilization without unnecessarily blocking traffic routes. Traffic control, as emphasized in the literature, plays an important role here by creating clear markings and barriers for safety and smooth operations (Barth, 2020, p. 1).

The equipment of the staging areas must be designed so that all necessary supplies and operating materials, such as diesel and gasoline, can be recorded and made available in detail (Barth, 2020, p. 2). Mobile supply units specifically tailored to the needs of vegetation firefighting are used here. In addition, these rooms can also be used to store necessary equipment, such as operating materials and firefighting backpacks, to ensure rapid availability (Barth, 2024, p. 4).

The dynamic nature of vegetation fires requires flexible and adaptable strategies in the deployment areas. These strategies must be able to be relocated at any time if the fire front or weather conditions change. Continuous communication between the operational commands and the units on standby is crucial. The use of command centers and communication tools is essential to ensure efficient coordination (Barth, 2024, p. 8).

Another important aspect is the integration of external support forces and organizations. These can optimally contribute their infrastructure and human resources to the deployment areas (Barth, 2024, p. 9). Clear communication and task allocation among the various actors reduces organizational effort and increases the efficiency of the operational measures.

In addition to the technical and logistical aspects, the provision of food and recreation for emergency personnel plays a crucial role. Dispatch areas provide ideal conditions for breaks and rest periods. Considering the psychophysical stress of emergency personnel is important to prevent fatigue and stress. The provision of mobile toilets, washing facilities, and refreshment stations contributes to maintaining operational capability (Barth, 2024, p. 14).



Finally, structured follow-up of the staging areas is important.

This includes the cleaning and restoration of the used areas as well as the documentation of all measures taken and resources deployed. A detailed analysis of operational strategies supports the continuous improvement of vegetation fire fighting and the optimization of future operations (Barth, 2024, p. 10).

In summary, staging areas are a central element in vegetation firefighting. Their careful planning, equipment, and flexible use are crucial for the successful and safe deployment of firefighters. By serving as a hub for personnel, equipment, and communication flow, they contribute significantly to the effectiveness and safety of vegetation fire operations.

14.4 Command posts

The establishment of command posts is crucial for the effective leadership and coordination of vegetation fire operations. They act as the central nervous system of the operation, from which strategic and operational control is carried out. Command posts consolidate information flows, facilitate communication with the emergency services on the ground, and are responsible for strategic planning and tactical implementation.

The choice of location for the command post is of great importance. It should be set up in a location that is as secure as possible, offers adequate protection, and is easily accessible. Proximity to the affected areas can be advantageous for making quick decisions and responding promptly to changing conditions (Barth, 2024, p. 18). Infrastructure criteria such as electricity and water connections, as well as communication links, should also be considered.

A well-organized command post is divided into various functions and operational sections. These include the operational command, which is responsible for overall strategic leadership, as well as various specialist departments such as logistics, communications, and tactical command. This structure enables an efficient distribution of tasks and improves coordination between the different units (Barth, 2024, p. 19).

In practice, it has been shown that the equipment of command posts has a significant impact on the effectiveness of operational management. Modern technologies such as digital maps, GNSS systems, and communications equipment are essential for precise and up-to-date situation reconnaissance (Discussion Paper, 2022, p. 13).



Technologies should be adapted to the intended use and intuitive to use in order to support decision-making.

Additionally, specific training for command post personnel is necessary to ensure that all involved parties can efficiently utilize the available tools. Risk assessment and analysis techniques contribute significantly to improving decision-making. The Risk Management Assistance (RMA) program demonstrates how structured risk management approaches and analytical methods can support the management of complex operations such as large-scale fires (Calkin et al., 2021, p. 2).

Another important aspect is communication and cooperation with other organizations and institutions. This includes not only the fire department itself, but also other emergency services, municipal institutions, and, where appropriate, cooperation with neighboring countries in cross-border operations (Thompson et al., 2019, p. 14). Clear communication structures and procedures are necessary to avoid misunderstandings and delays.

Continuous monitoring of the course of operations and adapting strategies are also essential tasks of the command posts. Regular briefings are necessary to evaluate current information and adapt recommendations for action. Documenting operations is important for post-operational review to evaluate experiences and identify potential for improvement (Calkin et al., 2021, pp. 15-16).

Overall, it is clear that the establishment and operation of command posts is a complex and multifaceted task that requires thorough preparation and continuous adaptation to the current situation. Through the targeted use of modern technologies, structured processes, and comprehensive personnel training, command posts can make a significant contribution to the successful management of vegetation fires.

14.5 Section formation

Sector formation is a central element of operational management in vegetation firefighting. It contributes significantly to the efficient organization of emergency personnel and maintaining an overview of the various operational areas. Strategic considerations as well as geographical and tactical aspects play a crucial role.

By creating sections, larger operational areas are divided into smaller, more manageable sections. This is often done using natural barriers such as rivers, roads, or forest edges, which serve as boundaries for the individual sections.



(Thompson, Gannon, & Caggiano, 2021, p. 2). A well-structured system of sections enables emergency responders to have clearly defined areas of responsibility and to carry out effective measures. This also increases safety.

safety of the emergency services, as this ensures clear operational structures and communication channels be established.

The basic structure of the section formation is often based on administrative or geographical divisions. Each section is coordinated by a section leader, who is responsible for the tactical execution of operational tasks within their area. This section leader acts as an information interface between the emergency personnel on site and the higher-level operational command (Thompson et al., 2021, p. 3). Their role is crucial, as they must assess the situation, make decisions, and adapt measures accordingly.

An important aspect of section formation is the definition of clear and practical cable boundaries. These should be easily identifiable and easy to control in case of an emergency.

Topographical features are often used because they provide both orientation and natural barriers that can slow the spread of fires (Thompson et al., 2021, p. 4). This includes forest roads and trails, which not only serve as natural section boundaries but also facilitate access to the fire areas (Thompson et al., 2021, p. 5).

The technical equipment of the emergency services is also crucial. Modern positioning technologies, such as GNSS-based systems, enable the precise use of coordinates and geographical information for planning and monitoring sections (Barth, 2024, p. 13). Digital maps and navigation aids support section leaders in acting quickly and efficiently and coordinating their emergency services in a targeted manner.

Flexibility is particularly important in dynamic operational situations. If the situation on site changes—for example, due to sudden wind shifts or unforeseen fire spread—the section structure must be adjusted accordingly. This requires continuous monitoring of the situation and close communication between the section leaders and the incident command (Thompson et al., 2021, p. 6).

The integration of external actors and resources is another important component of section formation. For larger fire operations, various organizations, such as the police, civil defense, or emergency services, may be involved. These external forces must be integrated into the section structure to ensure smooth cooperation and increase the effectiveness of the operational measures (Thompson et al., 2021, p. 7). Clear communication and coordination channels are crucial.



Finally, the follow-up to a day's deployment is a relevant aspect of segmentation. After each deployment, experiences and insights from the individual segments should be collected and analyzed to identify potential for improvement. This can help better plan and conduct future deployments (Thompson et al., 2021, p. 8).

Overall, the division of vegetation firefighting requires careful planning, a well-thought-out structure, and continuous adaptation to local conditions. Only then can efficient and safe operational management be ensured.

14.6 Supply

Providing supplies to emergency personnel during vegetation fires is a key aspect of operational management. It includes the provision of food, drinking water, shelter, medical care, and other logistical support. Comprehensive supplies ensure the operational readiness and well-being of firefighters and ensure that emergency personnel can work effectively even over extended periods.

One of the most important resources that must be provided during an operation is sufficient drinking water. Continuous hydration is essential in extreme heat and high physical exertion. Firefighting operations require significant physical exertion, which is why fluid requirements are significantly higher than normal daily requirements. According to the Joint Technical Recommendation for Firefighting Teams, a carrying system with at least two liters of drinking water per firefighter should be part of standard equipment (Joint Technical Recommendation for Firefighting Teams, 2024, p. 7).

Equally important is the provision of balanced meals. The food must be energy- and nutrient-dense to compensate for the firefighters' high calorie consumption and maintain their physical performance. Providing both cold and hot meals is a proven approach, and if necessary, mobile kitchens or catering services can be used to deliver meals directly to the scene.

In addition to food, medical care is of central importance.

In addition to emergency care, this also includes preventive measures such as protective clothing and first aid equipment, which must be immediately available to emergency personnel. For example, firefighting teams are recommended to be equipped with first aid kits and additional medical equipment (Joint Technical Recommendation for Firefighting Teams, 2024, p. 7).



The supply also includes the necessary equipment and materials. This includes vehicles that need to be supplied with fuel and lubricants.

Efficient logistics for the procurement and provision of operating materials such as diesel, gasoline, and AdBlue are of great importance. This can be achieved through mobile tank attachments and canisters (Example S4 Table Operating Materials S4.pdf, 2020, pp. 1-2).

Providing accommodation for emergency personnel is also important, especially during longer operations. Mobile shelters or emergency camps can help in this regard. The accommodations should be designed to meet the recovery needs of emergency personnel while being close enough to the scene to enable rapid response times.

Another aspect is personal protective equipment. This includes not only standard firefighting clothing, but also specialized equipment such as flame-retardant hoods, breathing apparatus, and weatherproof clothing. This protective equipment must be regularly checked and replaced if necessary to ensure maximum protection (Packing List for Extended Operations, 2024, pp. 1-2).

Additionally, communication tools must be provided. Efficient coordination of emergency personnel requires reliable communication devices such as radios and cell phones. The supply of appropriate charging cables and spare batteries is also essential to ensure communication over extended periods (Packing List for Longer-Duration Operations, 2024, p. 3).

The logistics of the supply measures must be well planned and coordinated. Command centers play a crucial role here. They must ensure that all necessary resources are available in a timely manner and in sufficient quantities. This includes documenting and managing resource allocation and recording consumption, as documented, for example, in the supply report for flood relief operations (Example S4 Supply Report S4.pdf, p. 1).

In summary, supply is a complex and multifunctional area of operational management. It requires thoughtful planning and efficient logistics to provide the best possible support to emergency personnel and ensure the effectiveness of the operation. By considering and meeting the physical, medical, and logistical needs of firefighters, good supply contributes significantly to the success of fighting vegetation fires.



14.7 Involvement of third parties

Third-party involvement in vegetation firefighting is a multifunctional challenge that requires diverse strategies to respond efficiently to these often dynamic and difficult-to-control events. Various actors play a crucial role, including neighboring states, civil organizations, scientific institutions, and private companies.

A key component of third-party involvement is cooperation with neighboring countries. In the case of cross-border disasters, standardized communication channels, coordinated operational plans, and technical concepts are necessary.

This requires not only established alerting procedures, but also regular exercises to test and improve cooperation. Working groups led by central ministries are helpful for reviewing and adapting existing regulations. For example, links with command staffs in neighboring countries and joint operational-tactical command centers could improve the efficiency and speed of cross-border operations (Report of the Expert Commission on Forest Fires Summer 2022 in Saxony, 2023, pp. 95, 97).

Civilian organizations such as the Federal Agency for Technical Relief (THW), nature and environmental associations, and volunteer fire departments provide both personnel and technical support during forest fires. Their involvement can be supported through formalized cooperation agreements and standardized training measures.

According to the report of the expert commission on forest fires in Saxony, the training of cooperation partners is crucial, as it requires specific knowledge in firefighting. First, on-site training should be established that teaches the principles of crisis management and teamwork. In addition, firefighters should be regularly trained in specific courses on vegetation firefighting strategies (Report of the Expert Commission on Forest Fires Summer 2022 in Saxony, 2023, p. 85).

Scientific institutions play a central role in researching the causes and dynamics of vegetation fires. Their involvement can be promoted by initiating research projects and implementing new classification methods for risk assessment. For example, the proposed Competence Center for Silviculture and Forest Fire Protection at TU Dresden can translate current research findings into practical concepts and improve the technical equipment for combating vegetation fires. Such an institution would also support the ongoing training of emergency services and the systematic analysis of forest fires (Report of the Expert Commission on Forest Fires Summer 2022 in Saxony, 2023, p. 113).



Private companies, particularly those in the forestry and wood processing industries, can contribute to forest firefighting by providing specialized technology and financial support. Their role also includes preventive measures such as forest conversion to mixed forests less prone to fire and the creation of fire protection strips. The Expert Commission recommends more intensive education of private forest owners about the risks and protective measures to promote fire damage prevention (Report of the Expert Commission on Forest Fires Summer 2022 in Saxony, 2023, p. 45).

Special technical requirements, such as the use of air support for firefighting, require early and coordinated coordination and logistics. Police helicopters with firefighting water tanks have proven their efficiency in recent forest fire incidents when they were alerted promptly and without complex coordination procedures. Therefore, the integration of such resources should be further improved through standardized processes and communication systems (Report of the Expert Commission on Forest Fires Summer 2022 in Saxony, 2023, pp. 89, 91).

In summary, the involvement of third parties in vegetation firefighting is a complex process that should be supported by formalized collaborations, regular training, and close integration of scientific knowledge. This enables more comprehensive resource utilization and faster response to vegetation fires, thereby increasing both firefighting efficiency and the safety of emergency responders.

14.8 Warning and evacuation

Warning and clearance is an essential component in fighting vegetation fires to protect human life and minimize property damage. These measures are crucial for responding to sudden changes in the fire situation and ensuring the safety of the affected population and emergency responders.

The efficiency of warning systems plays a central role, as rapid and reliable information about the impending danger must be conveyed.

Various technological and organizational means are used. Modern means of communication such as mobile communications, radio announcements, and digital messaging platforms are suitable for disseminating warning messages quickly and widely (Vegetation Fire Fighting Guide 1, 2024, p. 18). These means of communication should be coordinated with pre-established emergency plans to enable a prompt and structured evacuation.



In addition to technological warning systems, physical warning mechanisms such as sirens and loudspeaker announcements are of great importance on site. In regions where these technologies are not widely available,

the traditional method of door-to-door warning by emergency services will be effective

(Notes on Practical AE Object Defense/Post-Extinguishing, 2024, p. 9). It must be ensured that the deployed forces have sufficient information and training to communicate the warnings correctly and efficiently.

The evacuation must be carefully planned and executed to avoid panic and confusion. Establishing evacuation routes and staging areas is essential. These routes should be regularly reviewed and adapted to the current fire situation (Living with Forest Fire, 2010, p. 14).

Care must be taken to ensure that evacuation routes are safe and clear of obstructions. Emergency personnel must be familiar with the routes and be able to use alternative routes if the primary routes are blocked.

A particular challenge in spatial evacuation is the identification and support of particularly vulnerable groups. These include the elderly, children, and people with disabilities. Specific resources and support must be provided for these target groups. This may include the provision of specialized vehicles or the involvement of supporting organizations such as the Red Cross to assist in the evacuation process (Vegetation Fire Fighting Guide 1, 2024, p. 22).

The involvement of local authorities and other relevant agencies is also crucial. This promotes the coordination and efficiency of evacuation measures. Relevant information regarding operational planning and evacuation procedures must be communicated clearly and precisely among all participating units (Vegetation Fire Fighting Guide 1, 2024, p. 19). Regular exercises and training sessions are essential to prepare emergency responders and the affected population for actual evacuation scenarios.

prepare.

The follow-up and analysis of evacuation measures are crucial for learning from the events and optimizing future procedures. This includes detailed documentation of the measures implemented and the evaluation of the effectiveness of the warning and evacuation processes. These analyses should be incorporated into the revision of emergency plans to identify and address weaknesses (Vegetation Fire Fighting Guide 1, 2024, p. 24).

A key element in warning and evacuation is public education. Information campaigns should be conducted in advance of vegetation fires to inform citizens about the correct behavior in an emergency and the



Explain the importance of signposted evacuation routes. For example, regular information events could be held in public buildings and schools to increase public awareness and preparedness (Vegetation Fire Fighting: Operational Tactics, 2022, p. 1).

In conclusion, the successful implementation of warning and clearance measures for vegetation fires requires a high degree of preparation, coordination, and flexibility. It is essential that all actors involved, from emergency services to local authorities and the affected population, work closely together to ensure safety and the protection of human life in such extreme situations.

14.9 Post-operation follow-up

The follow-up to vegetation fires is a crucial aspect of the overall operational management process. Its goal is to ensure the operational readiness of the forces while simultaneously gaining valuable insights for future operations. This phase can be divided into two main categories: technical and organizational measures.

First of all, technical measures are of great importance to restore the operational readiness of the vehicles and the equipment used.

This includes thoroughly checking, cleaning, and maintaining breathing apparatus, as well as other tools and vehicles. It is also important to refill all consumables, such as water and foam concentrate, and to repair any damage to equipment (Schulze, 2020, p. 1). Such technical steps are essential to ensure that emergency responders are optimally prepared for their next deployment and find their equipment in perfect working order.

An equally important area of follow-up includes organizational measures. This involves a structured debriefing with the participating emergency personnel, in which the problems and challenges of the operation are identified and discussed. This reflection is crucial for uncovering any weaknesses and planning targeted improvement measures (Schulze, 2020, p. 1). For example, an inefficient flow of communication during the operation could result in the need for training to improve these processes in the follow-up.

Another key aspect of organizational follow-up is the documentation of the incident. This includes the preparation of incident reports and entry into exposure databases. Equally important is the reporting of near misses and other relevant incidents to external bodies, such as



Accident insurance providers (Schulze, 2020, pp. 1–2). This comprehensive documentation not only supports the analysis of specific operations but also promotes long-term observation and improvement of operational tactics.

The follow-up should also consider the mental health of the emergency responders. Operations during vegetation fires can be emotionally and psychologically stressful. Therefore, it is important to monitor any unusual behavior of individual emergency responders and, if necessary, offer psychosocial emergency care (PSNV-E) (Schulze, 2020, p. 1). This contributes to the mental stability and long-term operational readiness of the emergency responders.

Additionally, the debriefing should not focus exclusively on technical and organizational aspects. An open and cooperative environment promotes exchange between the emergency responders. Even during the debriefing, the lessons learned can be used to inform future operations. A moderator could help ensure that all relevant aspects of the operation are addressed and that the feedback of all participants is taken into account.

In summary, the post-mission review is a central component of the overall operation. It not only ensures the technical operational

Not only does this affect the readiness of the equipment, but also the organizational and psychological stability of the emergency services. Through careful implementation of this follow-up, valuable insights can be gained and targeted measures can be implemented to improve operational strategy and tactics. This not only increases the efficiency of future operations but also ensures the long-term operational capability of the fire service units.



15 Environmental protection and aftercare for vegetation fires

Managing vegetation fires represents a significant challenge for fire departments worldwide. These fires are often unpredictable and difficult to control, making them a serious risk to people and the environment.

Therefore, it is essential not only to develop effective strategies to combat these fires, but also to implement comprehensive environmental protection and post-fire care measures. This chapter addresses the various aspects of environmental protection and post-fire care in the context of vegetation fires and highlights the different dimensions required to mitigate the ecological and social consequences of such events.

Vegetation fires can cause profound ecological changes as they strongly influence the soil, flora and fauna, and the atmosphere.

A key focus of this chapter is therefore on the immediate and long-term impacts of fires on the environment. Critical aspects include habitat destruction, greenhouse gas release, and soil erosion. These factors must be thoroughly investigated to gain a comprehensive understanding of the ecological damage and plan for the necessary regeneration processes.

In this context, strategies for reforestation and ecological management will also be discussed. The restoration of the affected areas requires both technical know-how in agriculture and forestry as well as also requires a deep understanding of ecological interrelationships and natural regeneration processes. A systematic approach to reforestation and biodiversity promotion is crucial to strengthening ecosystem resilience to future fires.

Another key issue is the prevention of future vegetation fires. This involves both technological and organizational measures aimed at minimizing the likelihood and intensity of future fires. The development and implementation of early warning systems, the promotion of sustainable land-use practices, and public awareness of the risks of vegetation fires are essential elements of a comprehensive prevention strategy.

It is important to emphasize that environmental protection and post-fire management should not be viewed in isolation. An integrative approach that involves the cooperation of various actors – from governmental institutions to non-governmental organizations government organizations to local communities – is essential. Only through coordinated efforts can the



effectively reduce the ecological, economic and social consequences of vegetation fires and achieve sustainable management.

Given the increasing frequency and intensity of vegetation fires as a result of climate change, the discussion on environmental protection and aftercare is particularly urgent. Global warming and changing precipitation patterns are expected to increase the risk of vegetation fires in many regions, underscoring the need for enhanced protection and prevention measures.

Adaptation strategies must be developed and continuously adapted to changing environmental conditions in order to ensure the long-term sustainability of the affected areas.

This chapter aims to raise awareness of the complexity and urgency of environmental protection and aftercare in connection with vegetation fires. It provides a sound basis for the development and implementation of measures that reduce the ecological consequences of fires and support the regeneration capacity of affected areas. By comprehensively addressing the various aspects of vegetation fires and their impacts, this chapter will make a valuable contribution to the training of firefighters and other stakeholders involved in vegetation firefighting.

In summary, environmental protection and the aftermath of vegetation fires require a holistic approach that includes both preventive and reactive measures. The integration of knowledge from different disciplines and close cooperation between different stakeholders are fundamental prerequisites for effectively and sustainably managing the challenges associated with vegetation fires. In this sense, this chapter will not only present theoretical foundations and practical measures but also demonstrate ways to promote more effective cooperation between the various stakeholders.

15.1 Impact of fires on the environment

Vegetation fires have profound impacts on the environment, manifesting themselves in various dimensions. A central issue is the impairment of hydrology and soil properties. Fires often lead to an increase in surface runoff and increased soil erosion, which has been documented in several studies (Modeling, 2020, p. 1). These processes can significantly deteriorate water quality and increase the risk of flooding. The loss of vegetation and changes in soil properties also lead to



to increased surface runoff, which increases the risk of landslides (Modeling, 2020, pp. 2-3).

Another significant aspect is the release of greenhouse gases. During a fire, large amounts of CO₂, methane, and other greenhouse gases are released into the atmosphere, contributing to global warming. In addition, deforestation caused by fires reduces the ability of affected areas to absorb CO₂ from the atmosphere, exacerbating the climate impacts (Does more deadwood in the forest increase the risk of forest fires, 2024, pp. 1-2).

Biodiversity also suffers greatly from vegetation fires. This destruction of habitats leads to the loss of numerous plant and animal species.

Specific species that depend on old and dense forest structures could even disappear forever. The decline in biodiversity has negative consequences for the ecosystem, as the complex interactions between different species are disrupted (Does more deadwood in the forest increase the risk of forest fires, 2024, p. 1).

The flora itself also changes. Areas exposed to frequent fires tend to be dominated by plant species adapted to fire. These plants regrow more quickly and recolonize the area, while less fire-resistant species gradually disappear. This leads to a homogenization of the flora and a loss of ecological complexity (Does more deadwood in the forest increase the risk of wildfire, 2024, p. 2).

The floor itself also suffers massive damage. The heat of the flames leads to the burning of organic matter in the soil, which significantly reduces its fertility. The loss of organic matter reduces the soil's ability to retain water and provide nutrients, which in the long term impairs the productivity of the entire ecosystem (Modeling, 2020, pp. 2-3). Studies also show that soil sealing caused by fire reduces water infiltration, thus further increasing surface runoff (Modeling, 2020, pp. 3-4).

Another relevant aspect is the change in the chemical composition of the soil. Fires can alter the concentrations of nitrogen, phosphorus, and potassium in the soil, which negatively impacts the availability of nutrients for plant growth. In the long term, this has implications for the recolonization and regeneration of vegetation, as the altered nutrient balance can favor or inhibit the growth of certain plant species (Does more deadwood in the forest increase the risk of forest fires?, 2024, pp. 2-3).



Air quality also suffers significantly from vegetation fires. During a fire, large amounts of smoke and particulate matter are released into the atmosphere, worsening air quality in surrounding regions. These particulate matter and other pollutants can cause health problems among the population, including respiratory diseases and cardiovascular problems (2022 FORESTFIRE SEASON REPORT, p. 1). In the long term, these particles can also contribute to the acidification of soils and water bodies.

The socioeconomic consequences are also significant. Vegetation fires cause significant economic damage through the destruction of forestry, infrastructure, and property. The costs of firefighting, reforestation, and rehabilitation of affected areas can be enormous (Forest Roads and Operational Wildfire Response Planning, 2021, pp. 2-3). In many regions, frequent and intense fires can have long-term impacts on local economies and the livelihoods of the population.

In summary, these aspects demonstrate that the impacts of vegetation fires are complex and far-reaching. This requires comprehensive and coordinated management to effectively address the ecological and socioeconomic damage. The development of effective prevention and aftermath strategies is crucial to strengthen the resilience of affected ecosystems and minimize future risks.

15.2 Reforestation and ecological management

Reforestation after wildfires plays a crucial role in restoring damaged ecosystems and ensuring their long-term stabilization. It is important to consider the specific conditions of each site, as these strongly influence the selection of suitable tree species and the forestry methods to be applied (Wohlgemuth et al., 2010, p. 1).

A key aspect of reforestation is supporting natural regeneration processes. While some tree species, such as oak or beech, have a high capacity for self-establishment, others require more intensive support through targeted planting. Research shows that these species can establish themselves independently if sufficient seed trees are available and germination conditions are favorable (Maringer et al., 2020, p. 7). Targeted ecological management promotes these natural processes by incorporating measures to stabilize soil and promote biodiversity (Wohlgemuth et al., 2010, p. 15).



In reforestation, it is crucial to create landscape structures that are both ecologically valuable and resilient to future changes. Pioneer species play an important role here, as they are often the first to establish themselves after a fire, thus improving the microclimatic environment for subsequent species (Wohlgemuth et al., 2010, p. 9). One example is the silver birch, which promotes rapid growth and provides shade in suitable locations. This mitigates erosion effects and creates the conditions for the development of later successional stages (Maringer et al., 2020, p. 7).

Another important point is the creation of habitat structures for flora and fauna. It is increasingly recognized that structurally rich forests, containing both young and old trees as well as deadwood, not only exhibit higher biodiversity but are also more resilient to environmental changes (Frey & Adelmann, 2024, p. 1). Measures such as leaving deadwood as a nutrient source and habitat, as well as the targeted creation of clearings and wetlands, are central elements of ecological management (Frey & Adelmann, 2024, p. 2).

In addition, the use of forest fire protection strips and firebreaks plays an important role in preventive ecological management in reforested areas. These strips not only serve as fire protection, but can also contribute to biodiversity through targeted planting with resistant species, creating ecological corridors and expanding the habitat of many species (Wohlgemuth et al., 2010, p. 2).

A successful example of ecological management and reforestation is the "INCENDI" program in Switzerland. This program aims to implement preventive measures by analyzing weather data and identifying risk situations. Infrastructure is also being improved in at-risk areas, for example, by creating new water intake points (Wohlgemuth et al., 2010, p. 14).

Overall, it appears that an integrative approach that includes both preventive and curative measures is most effective. It is important to incorporate climate change and its impacts on water availability and vegetation cycles into planning (Maringer et al., 2020, p. 13). Only through such a holistic approach can reforestation be implemented sustainably and successfully.



15.3 Prevention of future vegetation fires

Vegetation fire prevention is a crucial aspect of environmental management, aiming to ensure the sustainable management and protection of forest and vegetation areas. Various scientific approaches and practical measures play a role in minimizing the risk of fire events and reducing their ecological impacts.

An effective approach to fire prevention is early fire detection. This involves the use of various technological systems, including thermal sensors and satellite-based monitoring systems. Schneider's dissertation (2017, p. 46) demonstrates that systems such as Firewatch can identify smoke development within minutes, enabling a rapid response to fires. Furthermore, Schneider (2017, p. 121) highlights the potential of UAVs (unmanned aerial vehicles), which can be used for early fire detection and monitoring larger areas.

Another preventive approach to reducing fire risk is adapting forest management, particularly the targeted management of deadwood. Various studies show that deadwood can pose a fire hazard when it is dry and fine, and also acts as a moisture reservoir, thereby reducing fire risk (Frey & Adelman, 2024, p. 3). The creation of deadwood islands and preventive silviculture that promotes biodiversity and a balanced age structure can help prevent the spread of fires (Frey & Adelman, 2024, p. 2).

Regular fire prevention inspections and the development of specific forest fire protection plans represent another important preventive approach. These plans should be adapted to local conditions and risks and include preventive measures such as firebreaks and protection corridors. The Expert Commission on Forest Fires in Saxony emphasizes the importance of such measures and recommends the establishment of comprehensive forest fire protection concepts (SZG, 2023, p. 53). Additionally, it recommends changing the forest structure to ensure a higher proportion of deciduous trees, as they are more fire-resistant than conifers (SZG, 2023, p. 11).

Integrating meteorological data into fire protection planning can also provide preventive benefits. The German Weather Service (DWD) provides weather and climate information that is helpful in assessing fire risk.

Factors such as wind, temperature, and humidity play a crucial role (Rudolph, 2024, p. 3). The introduction of the "30-30-30 rule," which states that wind speeds above 30 km/h, temperatures above 30°C, and humidity below 30% are high fire risks, can improve operational



Significantly improve planning and awareness among emergency services (Rudolph, 2024, p. 69).

In addition to technological and forestry measures, raising public awareness is a key aspect of preventive strategies. Since many forest fires are triggered by human behavior, it is crucial to raise awareness of the risks and consequences of vegetation fires through education and outreach (SZG, 2023, p. 65). This includes imparting knowledge and rules of conduct in schools, as well as public campaigns.

In summary, preventing future vegetation fires requires a multifactorial approach that integrates technological innovations, forestry practices, and community engagement. Through coordinated and consistent implementation of these measures, fire risk can be significantly reduced and ecosystem resilience strengthened.





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